Appendix H:
Hydrology and Water Quality Supporting Information

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H.1 - Hydrology and Water Quality Memorandum

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MEMO

To: Liza Baskir - First Carbon Solutions
From: Johnny Caspers, Scott Brown
Date: August 23, 2022

Subject: City of Berkeley Solid Waste and Recycling Transfer Station Project CEQA Hydrology and Water Quality Memo

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The City of Berkeley is seeking authorization to redevelop the City's Solid Waste and Recycling Transfer Station (SW&RTS, proposed project) located on a 7.73-acre site at the intersection of Gilman Street and Second Street in the northwest corner of the City (see **Figure 1**). The existing Transfer Station was constructed in 1983 with the sole purpose of receiving the City's solid waste. Over the course of the next fifteen years the facility was expanded to include other services related to waste handling, recycling, and disposal (Zero Waste, 2019). The proposed design, outlined in the Solid Waste and Recycling Center Transfer Station Feasibility Study Final Report ('Feasibility Study') prepared by Zero Waste Collaborative in 2019, will provide the City of Berkeley with an integrated facility that can accept a variety of materials and ensure environmentally compliant methods will be implemented for the handling and processing of refuse, fiber and container recyclables, and compostable materials not only for today but for many decades to follow (Zero Waste, 2019).

The objectives of this report are to outline the existing regulatory hydrologic framework that governs the development of the project site, identify the potential impacts of the project on environmental quality as it pertains to hydrology and water quality, and address potential methods to mitigate any identified impacts to a less than significant level. It is our understanding that this report will be incorporated with other CEQA permitting documentation; therefore, we have formatted it in a consistent manner.

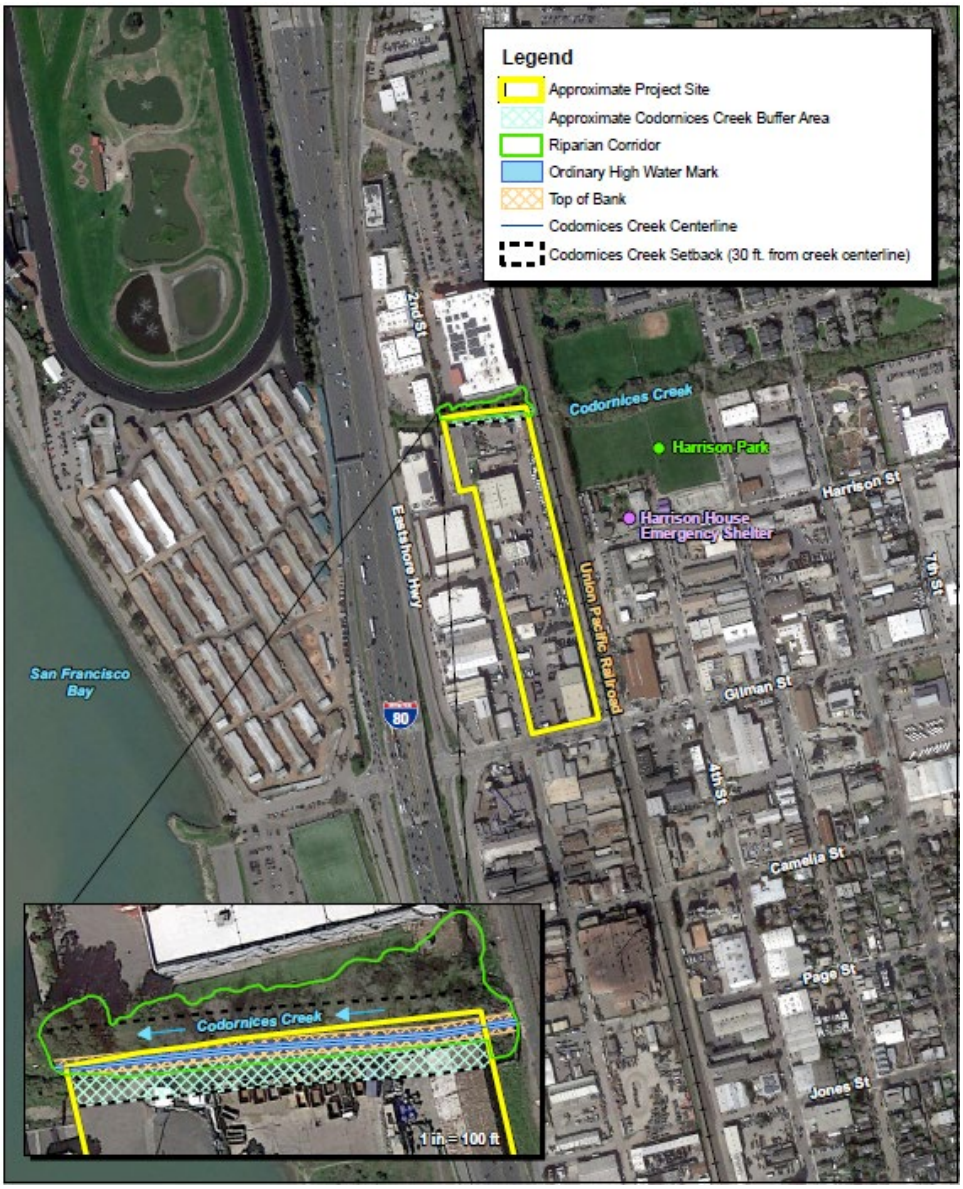


Figure 1. ***Local Vicinity Map.*** *Prepared by First Carbon Solutions.*

1.1 REGULATORY SETTING

This section describes the federal, state, and local regulatory context to be considered for the proposed project, and addresses hydrology and water quality concerns, including development strategies, stormwater pollution prevention plans, and stormwater management practices.

1.1.1 Federal and State Regulations

1.1.1.1 Section 402 of the Clean Water Act

The Clean Water Act (CWA) authorizes the U.S. Environmental Protection Agency to regulate water quality in California by controlling the discharge of pollutants to water bodies from point and non-point sources through the National Pollution Discharge Elimination System (NPDES).

In Alameda County, as with the rest of the Bay Area, NPDES permits are administered by the San Francisco Bay Regional Water Quality Control Board (RWQCB), a division of the State Water Resources Control Board (SWRCB). Phase I of the NPDES program covered discharges from industrial sites, construction sites larger than five acres, and municipal separate storm sewer systems (MS4s) serving populations of more than 100,000 people. The Phase II expansion of the MS4 program in 1999 expanded its coverage to include "Small" MS4s that serve:

- Urbanized areas as defined by the U.S. Census;
- Areas of high growth or high growth potential;
- Areas that discharge to sensitive water bodies or another regulated MS4; and,
- Areas that make a significant contribution to pollution of waters of the U.S.

Stormwater discharges in the City of Berkeley, as with all of Alameda County, are permitted under the NPDES requirements for large MS4s (Phase I). The SWRCB has established a Municipal Regional Stormwater permit (Regional Municipal Permit) for the Bay Area Counties (and permittees with the Counties) of Alameda, Contra Costa, Santa Clara, and San Mateo, along with the Fairfield-Suisun and Vallejo permitees (Order No. R2-2015-0049, NPDES General Permit No. CAS612008) to regulate stormwater discharges in these areas. All eligible MS4s are required to comply with its provisions, which include construction- and post-construction-phase stormwater runoff controls and water quality best management practices (BMPs) for new and re-development.

MS4 owners/operators must reduce pollutants in storm water to the maximum extent practicable (MEP) to protect water quality. The regulations specify that compliance with the MEP requirement can be attained by developing a storm water management plan that addresses the six minimum control measures described in the storm water regulations (Berkeley, 2016).

The Regional Municipal Permit provisions include post-construction provisions to prevent non-storm water discharges, minimize the discharge of pollutants in storm water runoff, and prevent a development-caused worsening of stream channel erosion and sediment deposition resulting from hydromodification of a watershed. To minimize pollutant discharges, projects must provide the capacity to either infiltrate, evapotranspire, or otherwise treat all runoff generated by the 85th percentile storm event.[1] Typically this is completed using low impact development (LID) design

[1] The 85th percentile storm represents a specific intensity for which all storms of lesser or equal intensity generate 85 percent of the total annual rainfall, while more intense storms only generate 15 percent of the total.

principles that seek to minimize the amount of land covered by impervious surfaces and maximize opportunities for infiltration. In instances where this cannot be completed solely using LID features, treatment measures must be provided to remove or otherwise neutralize identified pollutants. High flow rates that cannot be sufficiently reduced by site design strategies must be controlled through the provision of detention storage or through stabilization of downstream conveyances that would be adversely affected by development. Because these treatment and detention facilities will require maintenance for the life of a development project, developers must also establish a dedicated funding responsibility for either the future owners of the land or a designated public entity. In its role as an MS4 operator and permit holder, Alameda County is required to enforce these site design and water quality protection measures for all new and redevelopment projects within its jurisdiction. The City of Berkeley participates in the Alameda Countywide Clean Water Program, which is a joint effort with the other municipalities in the County to prevent stormwater pollution, protect and enhance water quality in creeks and wetlands, preserve beneficial uses of local waterways, and comply with State and Federal regulations.

Construction Sites

The SWRCB administers the NPDES General Permit for Discharges of Stormwater Runoff Associated with Construction Activity (General Construction Permit). In order to cover a construction project disturbing one acre or more of land under the General Construction Permit, a facility must submit a Notice of Intent to the SWRCB prior to the beginning of construction. Effective July 1, 2010, all dischargers are required to obtain coverage under the Construction General Permit Order 2009-0009-DWQ adopted on September 2, 2009. The General Construction Permit requires that projects develop and implement a Stormwater Pollution Prevention Plan (SWPPP), identifying potential sources of pollution and specifying runoff controls during construction for the purpose of minimizing the discharge of pollutants in stormwater from the construction area. The SWPPP should contain a site map which shows the construction site perimeter, existing and proposed buildings, lots, roadways, storm water collection and discharge points, general topography both before and after construction, and drainage patterns across the project (see Figure 2 in **Appendix A**). The SWPPP must list BMPs the discharger will use to protect storm water runoff and the placement of those BMPs. Additionally, the SWPPP must contain a visual monitoring program, a chemical monitoring program for "non-visible" pollutants to be implemented if there is a failure of BMPs, and a sediment monitoring plan if the site discharges directly to a water body listed on the 303(d) list for sediment.

The permit also includes post-construction standards with the requirement for all construction sites to match pre-project hydrology to ensure that the physical and biological integrity of aquatic ecosystems is maintained. This “runoff reduction” approach is analogous in principle to LID and serves to protect related watersheds and water bodies from both hydrologic-based and pollution impacts associated with the post-construction landscape.

Consequently, treatment facilities designed to accommodate runoff from the 85th percentile storm have sufficient capacity to treat 85 percent of total annual runoff.

1.1.1.1 Section 401 of the *Clean Water Act*

Under the auspices of the CWA, the United States Army Corps of Engineers (USACE) administers permitting programs that authorize impacts to "waters of the United States" including "wetlands" and "other waters." Such impacts may not be permitted until the SWRCB, acting through its regional boards, certifies that the activities covered by the permit will not violate water quality standards. Certification must be consistent with the requirements of the federal CWA, CEQA and California Endangered Species Act, and with the SWRCB's mandate to protect beneficial uses of waters of the state.

The San Francisco Bay RWQCB has adopted the USACE policy that there shall be "no net loss" of wetlands. Thus, prior to waiving or certifying water quality, the RWQCB requires a proposed project to ensure there are no impacts on existing wetlands, or, if such impacts are unavoidable, that they are fully mitigated.

1.1.1.2 California Porter-Cologne Act

The Porter-Cologne Act requires "any person discharging waste, or proposing to discharge waste, within any region that could affect the waters of the State (any surface water or groundwater, including saline waters) to file a report of discharge" with the local RWQCB by applying for waste discharge. The RWQCB determines if a project should be regulated pursuant to this act based on the likelihood that it would pose any "threat" to water quality. The San Francisco Bay RWQCB considers the placement of clean fill in waters of the State to constitute "pollution," because it can potentially alter existing water quality, which may adversely affect its beneficial uses.

Areas that drain to the San Francisco Bay are regulated by the Region 2 (San Francisco Bay) Basin Plan which was approved in 1995 and most recently updated in 2019. Beneficial water uses are designated in the Basin Plan for local aquifers, streams, marshes, and rivers, as well as water-quality objectives that must be met to protect these uses. Identified beneficial uses for Codornices Creek include cold- and warm- freshwater habitat, fish migration, preservation of rare and endangered species, fish spawning and wildlife habitat, and contact and non-contact recreational uses. In addition, the Basin Plan establishes water quality management objectives for various contaminants and pollutants (CA RWQCB, 2019).

1.1.1.3 California Fish and Game Code

Existing stream channels in California are protected under sections 1600-1603 of the State Fish and Game Code. These regulations specify that it is a landowner's responsibility to obtain a state permit before undertaking any modifications within an existing stream channel up to the top of bank. Stream channels are defined by the California Department of Fish and Wildlife (CDFW) as exhibiting evidence of scour, having a definable bank, or having or being capable of supporting riparian vegetation.

1.1.1.4 Sustainable Groundwater Management Act

In 2014, a new law was signed intended to create a framework for sustainable management of groundwater resources in California. The law, called the Sustainable Groundwater Management

Act (SGMA), requires governments and water agencies with management responsivities in medium- and high-priority subbasins to halt groundwater overdraft through development of a Groundwater Sustainability Plan (GSP). East Bay Municipal Utility District (EBMUD) and the City of Hayward have submitted a Draft Groundwater Sustainability Plan for the East Bay Plain groundwater subbasin, which includes the area underlying the City of Berkeley (Luhdorff & Scalmanini, 2021). The Draft GSP establishes the sustainability goals, measurable objectives, minimum thresholds, and potential undesirable results if the goals of the GSP are not met. Projects and management actions geared toward achieving the sustainability goals are also included in the report.

1.1.2 Local Regulations

1.1.2.1 Alameda County Storm Water Program

As noted above, the operators of municipal storm sewer systems must comply with the permitting provisions of the NPDES. The Alameda County Clean Water Program (ACCWP) is a joint effort of Alameda County, the cities of Alameda, Albany, Berkeley, Dublin, Emeryville, Fremont, Hayward, Livermore, Newark, Oakland, Piedmont, Pleasanton, San Leandro, Union city, the Alameda County Flood Control and Water Conservation District (ACFCD), and the Zone 7 Water Agency for managing conditions of the NPDES permit and addressing potential stormwater quality concerns. Though the entities of the ACCWP carry out their own individual stormwater pollution prevention programs, the ACCWP provides for the coordination and consistency of approaches between the individual participants and documents their efforts in annual reports.

ACCWP has developed several resources as guidance for projects toward various requirements of the NPDES permit. Of key importance are the documents outlining requirements and practices for new development within the county (ACCWP, 2021), which include construction-phase BMPs and technical guidance for post-construction runoff management. An important aspect of the post-construction runoff management outlined by the ACCWP is hydromodification management; the technical guidance states that all new projects must mitigate peak flows from 10 percent of the 2-year rainfall event up to the 10-year event to protect downstream waterways from degradation. To facilitate compliance with the hydromodification standard, the guidance encourages the use of the Bay Area Hydrology Model (BAHM), a continuous simulation modeling software created by Clear Creek Solutions to help applicants size specialized detention facilities (ACCWP, 2021).

1.1.2.2 Alameda County Flood Control District

Current practice and standards associated with flood control for Alameda County are outlined in the Alameda County Flood Control and Water Conservation District's (ACFCD) Hydrology and Hydraulics Manual. The manual stipulates peak flow design requirements for stormwater holding facilities to mitigate downstream flooding associated with the increase in impervious area of new and re-development. As it pertains to this project, ACFCD guidance stipulates that post-

development peak runoff must not exceed the pre-development peak runoff for the 5-, 15-, 25-, and 100-year recurrence interval storm events[2] (ACFCD, 2018).

1.1.2.3 City of Berkeley Department of Planning and Development

Additional standards related to hydrology and water quality management are included in the City of Berkeley General Plan (General Plan), published by the Department of Planning and Development in 2003. Within the General Plan, two main sections – referred to in the text as elements - highlight the City's current policy regarding hydrology and water quality. The two elements are Environmental Management and Disaster Preparedness and Safety.

1.1.2.3.a Environmental Management Element

Within the Environmental Management Element of the General Plan, the City outlines five policies – EM-23 through 27 - that are designed to improve water quality within City creeks and the San Francisco Bay. While some of the points emphasize adherence to existing legislation, they also expand upon state and county level regulations. For example, policies stipulate a need to eliminate sanitary and storm sewer cross connections; establish 30-foot setbacks from open and culverted creeks;[3] and improve community awareness of water quality management.

1.1.2.3.b Disaster Safety and Preparedness Element

Under the sub-heading Flood Hazards within the Disaster Safety and Preparedness Element, the General Plan outlines three policies – S-26 through 28 – that define measures to address flood hazards within the City. In summation, the element seeks to reduce the threat from existing flood hazards, prevent additional flood hazards from future development, and make flood insurance costs less prohibitive for property owners in the City.

1.2 ENVIRONMENTAL SETTING

1.2.1 Regional Hydrology

Codornices Creek, which collects runoff from roughly 10% of the City of Berkeley's area, is present along the northern boundary of the project site (**Figure 2)**. This creek is notable within the region for several reasons. Primarily, it is one of the least culverted creeks in the East Bay and as a result, supports a substantial salmonid population (Hagar, 2005). Furthermore, the creek has seen multiple restoration efforts in the past twenty years that have focused on improving the section of creek between San Pablo Avenue and the Union Pacific Railroad (UPRR), just upstream of the project site. Downstream of the project site and Second Street, the creek is culverted under I-80 before it discharges into the Bay north of Golden Gate Fields (see Section 1.2.3 for a discussion of the I-80 culvert and localized flooding).

The Gilman Street subwatershed ('Gilman Basin') is located to the south of the project site. The Gilman Basin is the second smallest watershed in the city, servicing roughly 249 acres of heavily

[2] All recurrence interval storm events are modeled using the 24-hour design storm.
[3] In 2004 the City of Berkeley adopted new regulations under Municipal Code 17.08, Preservation and Restoration of Natural Watercourses (Creeks Ordinance). The ordinance outlines a number of measures to protect and improve both open and culverted creeks from adjacent development.

urbanized area within the lower Bay Plain area of the City of Berkeley. The basin is entirely culverted and outfalls into the Bay at the western end of Gilman Street (Balance, 2011).

Average annual rainfall in the Codornices Creek and Gilman Basin watersheds are approximately 24 and 20 inches, respectively; higher rainfall amounts fall on the Berkeley Hills where the Codornices Creek watershed originates, as opposed to the low-lying western portion of the city that the Gilman Basin services. Nearly 95 percent of this precipitation falls during the winter rainy season, October through April, with the heaviest rainfall typically occurring in December, January, and February.

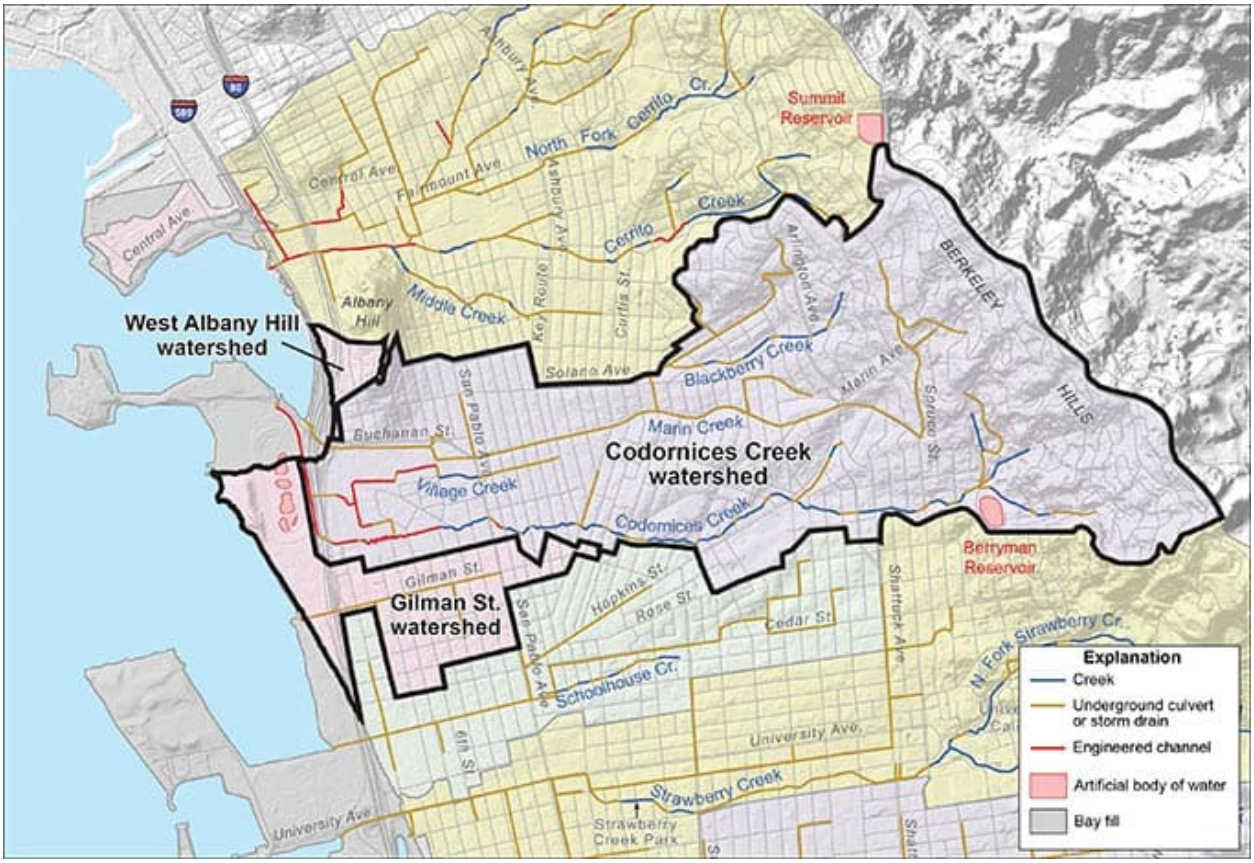


Figure 2. ***Map of the Codornices and Gilman Street watersheds.*** *The project site (shaded red) straddles the watershed boundary within the lower portion of both watersheds. Figure adapted from Fugro, 2014.*

1.2.2 Project Site Hydrology

As discussed above, the project site straddles two watersheds within the City of Berkeley. As shown in **Figure 3**, the northern portion (less than one-fifth) of the project site drains to Codornices Creek. This area is currently occupied by a natural gas refilling station and a sorting/storage area. The remaining portion of the project site (just over 6 acres) drains to the culverted Gilman Basin. This south-draining areas includes the main transfer station with a

tipping floor building; "Berkeley Recycling" operated by a City vendor that includes a the recycle materials drop-off area, material recovery sort buildings and a California Redemption Container buy-back center; a vehicle maintenance building and wash bay; the Ecology Center; residential curbside collection vendor operations yard and other facilities.

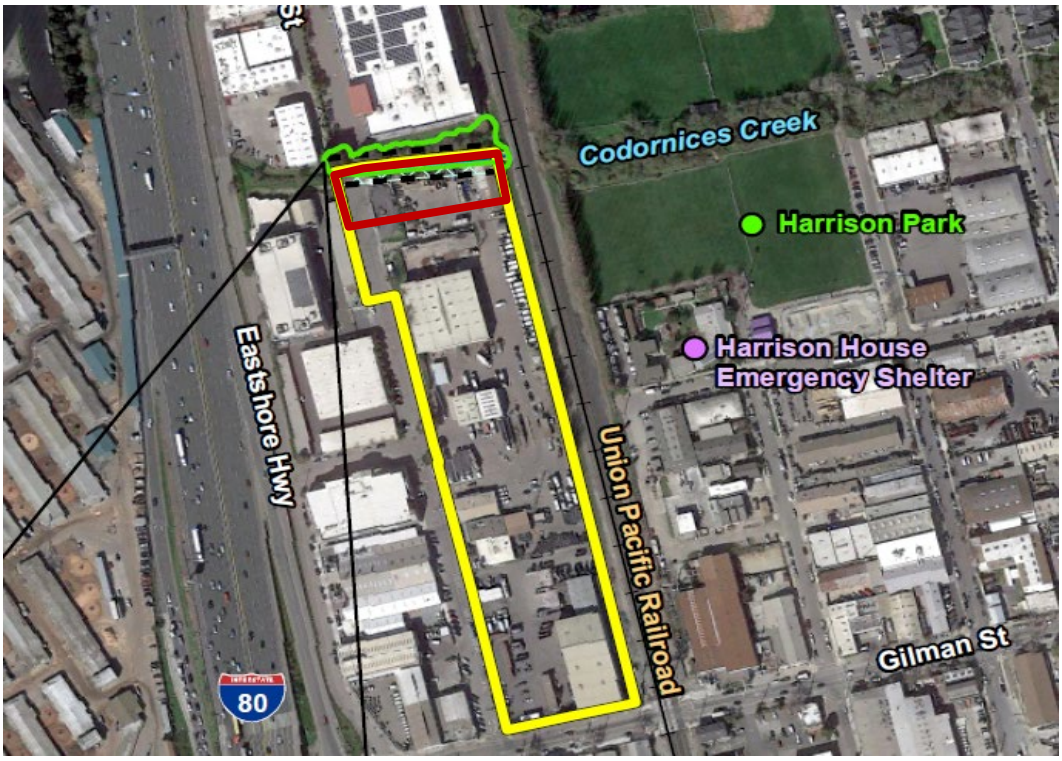


Figure 3. ***Aerial photograph of project site.*** *The northern portion of the project site (outlined in red) drains to the Codornices watershed, while the remainder drains south to the Gilman Basin. It should be noted that the area that drains to Codornices Creek is smaller than the area subject to flooding or run-on from the creek. Figure adapted from First Carbon Solutions, 2022.*

As a result of the current stormwater control measures on-site, a large portion of the site's runoff is deemed impacted for water quality and is prevented from entering the storm drain system. Runoff from impacted areas is diverted to EBMUD's sanitary sewer line for treatment. A detailed description of this diverted runoff is presented in the most recent Stormwater Pollution Prevention Plan (SWPPP), completed by Wood Environment and Infrastructure Solutions, Inc. (Wood) in 2018 (see **Appendix A**). The SWPPP states that runoff from the covered refuse and organic temporary storage (tipping floor), collection trucks parking, and construction & demolition materials tipping floor areas in the transfer station are directed to a 21,000-gallon retention tank. The runoff from the Berkeley Recycling's glass sorting and commingled container storage areas, operated by the City's vendor (Community Conservation Center, Inc.), is directed to two (2) 4,000-gallon holding tanks, respectively (see **Figure 4** for excerpts of SWPPP Figure 2 that show the location of the stormwater holding tanks). After the tests are provided to EBMUD, all three tanks are pumped to the EBMUD sanitary sewer for treatment. The un-

impacted runoff from the site flows to multiple catch basins on Second Street where it is conveyed primarily south to Gilman Street. The small portion that flows northwest into Codornices Creek enters the creek by way of a catch basin at the northern end of Second Street[4] (refer to Figure 2 in **Appendix A** for an overview of the catch basin locations).

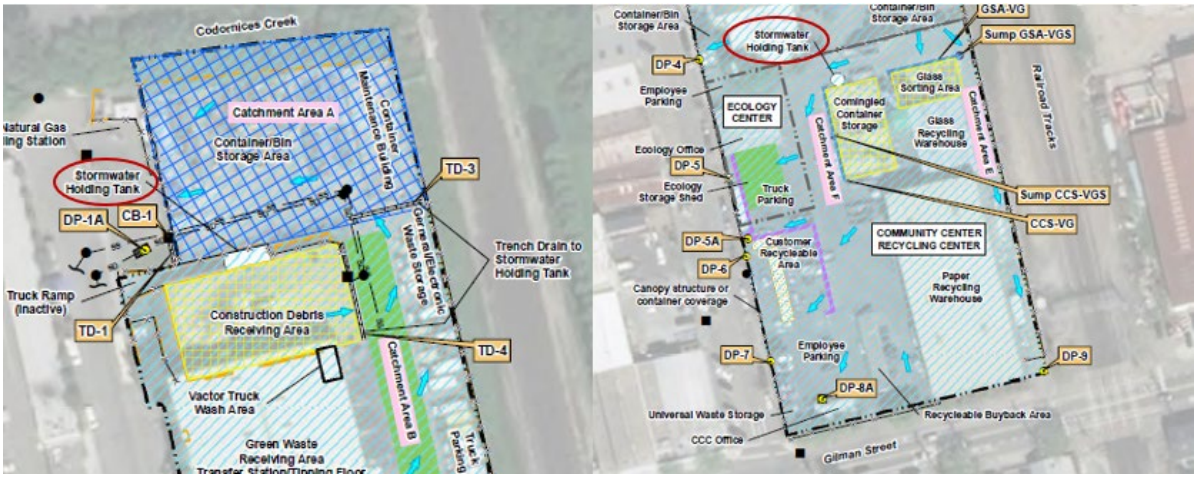


Figure 4. ***Locations of stormwater holding tanks.*** *Original callouts from the SWPPP figure are circled in red. It should be noted that the figure of the CCS Facility - at right - omits a second 4,000-gallion holding tank that was added after completion of the 2018 SWPPP.*

1.2.3 Downstream Hydrology

Runoff from the site that drains to the Gilman Basin is collected on Second Street and conveyed by pipe a quarter mile to the system's outfall at the western end of Gilman Street. This infrastructure downstream of the project site is undersized for the 10-year peak flow event (Balance, 2011). Under current conditions, flows exceeding the 10-year event will back-up within the storm drain system and cause ponding on surface streets and other low-lying areas. Due to the Gilman Basin's low time-of-concentration for stormwater runoff (a result of the small size and high imperviousness of the watershed) flooding is most prominent during short-duration, very high intensity rainfall events.

To the north of the project, Codornices Creek is culverted below I-80 prior to discharging into the Bay. The I-80 culvert has been the subject of significant study given that it constricts flow in the Creek, inducing flooding on Second Street adjacent to the project site.[5] Flooding has been further exacerbated since the conveyance of the UPRR culvert was increased when it was replaced in 1999 (Klein, 2009); a larger culvert upstream allows for more water to pass under the

[4] During a site visit by Balance Hydrologics' staff, rainwater harvesting cisterns were observed adjacent to buildings located in the Community Conservation Center; however, it is unclear what percentage of site runoff is currently collected and retained on-site within the cisterns.

[5] According to a report prepared for Caltrans by Randy Klein in 2009, Codornices Creek will overtop its banks upstream of I-80 when flowrates exceed 190 cfs. Furthermore, the Storm Drain Hydrology and Hydraulics Report completed by Balance Hydrologics for the City of Berkeley in 2011 outlines modeling that shows flooding occurring at Second Street during the 10-year 6-hour design storm event.

UPRR and which permits increasing peak flow of the Creek downstream of UPRR and has the potential to further overwhelm the I-80 culvert.

1.2.4 Project soils and groundwater

Near surface soils underlying the project site are classified as urban fill and clay loam, both of which have a low infiltration capacity (SoilWeb, 2021). More broadly, the project site is in a region dominated by alluvial fans – ranging in depth from 100 to 200 feet below ground surface (bgs) - deposited on a west-sloping bedrock surface (Groundwater Committee, 1999). Due to the proximity to the Bay, groundwater levels at the project site range seasonally from 5 to 10 feet bgs (Luhdorff and Scalmanini, 2021). Given the history of industrial work in this part of Berkeley, there are multiple active contamination cleanup programs within a half-mile of the project site, the most notable being an actively managed hexavalent chromium plume associated with the WRE/Colortech Facility located on 6th Street. Further contaminants of concern include petroleum by-products, tetrachloroethylene, lead, and other heavy metals (Geotracker, 2021).

1.3 PROJECT IMPACTS

1.3.1 Thresholds of Significance

Appendix G to the CEQA Guidelines includes questions for determining whether impacts to hydrology and water quality are significant. These questions reflect the input of planning and environmental professionals at the Governor's Office of Planning and Research and the California Natural Resources Agency, based on input from stakeholder groups and experts in various other governmental agencies, nonprofits, and leading environmental consulting firms. They also reflect the regulatory requirements of laws other than CEQA that protect hydrology and water quality (e.g., the federal Clean Water Act, the Porter-Cologne Water Quality Control Act). As a result, many lead agencies derive their significance criteria from the questions posed in Appendix G. The City has chosen to do this for this project.

Accordingly, the impact of the proposed project on hydrology and water quality would be considered significant if it would:

(a) violate any water quality standards or waste discharge requirements or otherwise substantially degrade surface or groundwater quality;

(b) substantially decrease groundwater supplies or interfere substantially with groundwater recharge such that the project may impede sustainable groundwater management of the basin;

(c) substantially alter the existing drainage pattern of the site or area, including through the alteration of the course of a stream or river or through the addition of impervious surfaces, in a manner that would:

    (i) result in substantial erosion or siltation on- or off-site

    (ii) substantially increase the rate or amount of surface runoff in a manner that would result in flooding on or off-site;

(iii) create or contribute runoff water that would exceed the capacity of existing or planned stormwater drainage systems, or provide substantial additional sources of polluted runoff; or

(iv) impede or redirect flood flows

(d) in flood hazard, tsunami, or seiche zones, risk release of pollutants due to project inundation;

(e) conflict with or obstruct implementation of a water quality control plan or sustainable groundwater management plan.

1.3.2 Water Quality Standards, Surface, and Groundwater Quality

Stormwater contaminants from the site may be present in the runoff that is discharged from both the construction and post-construction phase of the proposed project. The potential for site runoff to affect water quality and/or hydromodification is evaluated below.

1.3.2.1 Surface Water Quality

Construction Phase

During construction, clearing, grading and other activities would increase the potential for on-site erosion, potentially leading to increased turbidity and sedimentation in Codornices Creek and the Gilman storm drain. Sedimentation in Codornices Creek may degrade in-stream habitat and reduce flow capacity at downstream culverts and open channels, potentially inducing or exacerbating flooding. Other pollutants that might impact surface water quality during project construction include petroleum products (gasoline, diesel, kerosene, oil, and grease), contaminants from construction debris, hydrocarbons from asphalt paving, paints, solvents, and litter.

Because the project would disturb more than one acre of land, the applicant is required to prepare a SWPPP, per NPDES general construction permit requirements through the SWRCB. The SWPPP would address potential erosion and sedimentation issues through a project-specific erosion control plan, as well as other BMPs to reduce the potential for increased off-site sedimentation, spills, and other contamination from on-site construction activities. Appropriate measures for control of sediment and other pollutants from construction sites are included in the "Construction Handbook of Best Management Practices" (CASQA, 2019). The project SWPPP is likely to include, but is not limited to, the following BMPs related to construction water-quality impacts:

- If the entire site is not graded in a single operation, leave existing vegetated areas undisturbed until construction of improvements on each portion of the development site is ready to begin;
- Immediately re-vegetate or otherwise protect all disturbed areas from both wind and water erosion upon completion of grading;

- Collect storm water runoff into stable drainage channels and/or small drainage basins to prevent the buildup of large, potentially erosive storm water flows;
- Direct runoff away from all areas disturbed by construction;
- Use sediment ponds or siltation basins to trap eroded soils before runoff is discharged into on-site or off-site drainage culverts and channels;
- Install straw rolls, hay bales or other approved materials below all disturbed areas adjacent to drainages and drainage inlets to prevent eroded soils from entering the stream channel. Maintain these control measures until all disturbed upslope areas are fully stabilized;
- To the extent possible, schedule major site development work involving excavation and earthmoving for construction during the dry season;
- Develop and implement a program for the handling, storage, use, and disposal of fuels and hazardous materials. The program shall also include a contingency plan covering accidental hazardous material spills;
- Avoid cleaning, fueling, or maintaining vehicles on-site, except in an area designated to contain and treat runoff; and
- After construction is completed, inspect all drainage facilities immediately downstream of the grading site for accumulated sediment, and clear these facilities of debris and sediment as necessary.

In addition, the project site has the potential to experience stormwater run-on during large storm events when the capacity of the adjacent Codornices Creek channel is exceeded. The project SWPPP will need to include provisions to account for these flows during construction. Potential items not included in the list above may include:

- Staging work on the northern end of the project site (the area adjacent to Codornices Creek) early in the project so that the new flood protection berm is in place while the remainder of the construction proceeds.
- Phasing and scheduling the project to complete demolition and major earthwork during the dry season when run-on would not occur.
- Use of temporary retaining walls or other methods to reduce the risk of run-on to the project site during the construction period.

In compliance with the requirements of the statewide NPDES Construction General Permit, the developer will prepare a SWPPP for submittal with a Notice of Intent to the RWQCB prior to the start of construction. The SWPPP would specify BMPs to be employed during the construction phase to control sediment loads and pollutants. The SWPPP shall include the minimum BMPs required for the identified Risk Level of the project and receiving waters. BMP implementation shall be consistent with the BMP requirements in the most recent version of the California

Stormwater Quality Association Stormwater Best Management Handbook-Construction or the Caltrans Stormwater Quality Handbook Construction Site BMPs Manual. Proper implementation of the project-specific SWPPP would reduce the potential construction-related water quality impacts to a ***less-than-significant*** level. No further mitigation is required.

Operational Phase

Based on two preliminary design concepts that include a vegetated set-back from Codornices Creek and a focus on landscaped area throughout the site, the proposed project would decrease the site's total impervious percentage from 98 to approximately 89 percent.[6] Despite this decrease in impervious surface, it is possible that more runoff to the storm drain system will be generated on-site. Currently, waste material sorting and temporary storage is done in uncovered areas and runoff that falls on these areas is diverted to holding tanks prior to being sent to an EBMUD sanitary sewer line under special permit.[7] With the implementation of the proposed design, rainfall will no longer be able to interact with waste material as all material handling will be done in covered areas. One result of this water quality improvement, assuming that special discharge is no longer required, is that a larger portion of runoff will be generated within the Gilman Basin watershed. While runoff from the newly roofed areas is no longer expected to contact waste debris, it may still contain pollutants typical of urban area impervious surfaces. As required by the NPDES permit, runoff from these areas (and all other impervious surfaces on the project site) will still require appropriately designed stormwater quality treatment measures. As described in the Feasibility Study, stormwater BMPs are expected to include media filter treatment vaults, and rainwater harvesting and re-use cisterns, as methods of treating or preventing stormwater runoff from impervious surfaces (Zero Waste, 2019). With the implementation of the proposed suite of stormwater BMPs, the potential for the project to have an operational phase impact on stormwater quality is ***less than significant***.

Codornices Creek Watershed

Regarding the northern portion of the site that drains to Codornices Creek, the proposed design calls for a vegetated set-back from the creek that will replace a portion of the existing natural gas fueling station and container storage area. The remainder of the area will serve as the entryway for the main public scale[8] (Zero Waste, 2019). All stormwater discharge from impervious surfaces in this area will be treated in facilities to comply with ACCWP water quality treatment requirements. Facilities will be incorporated into landscaped areas surrounding the proposed buildings, parking areas, roads, and related impervious infrastructure. Proper design, sizing, and implementation of treatment measures, as outlined in ACCWP's C.3 Stormwater Technical Guidance, will effectively address both water quality and hydromodification concerns; because the Codornices Creek watershed is a zone of special consideration, as defined by the ACCWP in Section 7.2 of the C.3. Guidance, it is subject to hydromodification requirements. Given that the

[6] At this stage in the design, both site design concepts have the same landscaping plan despite differing facility layouts.
[7] Refer to Figure 4 in Appendix A for a layout and further description of the diversion to the EBMUD sanitary sewer line.
[8] The Feasibility Study outlines two conceptual designs for the Transfer Station; however, both designs propose the same plans for this portion of the site.

proposed project will include the addition of a vegetated, landscaped buffer along Codornices Creek that would treat runoff, and that the portion of the project draining toward Codornices Creek will be less impervious and of generally less intensive industrial use relative to existing conditions, stormwater runoff that drains to Codornices Creek will likely improve under post-project conditions; therefore, the impact related to stormwater quality will be ***less than significant***.

Gilman Basin

As stated above, by moving all the waste sorting and recycling processes indoors there will be a net increase in runoff from the project site to the Gilman Basin. To ensure that the increase in runoff does not impair water quality downstream of the project, LID and treatment vaults employed onsite will be designed to meet water quality standards outlined by the ACCWP's C.3. Stormwater Technical Guidance. Because water is conveyed solely by pipe downstream of the project site in the Gilman Basin, the portion of the project located in Gilman Basin is not located in an area susceptible to hydromodification as defined in Section 7.2 of the CCCWP C.3. Guidance (ACCWP, 2021).

Through the adherence to design standards outlined above and the implementation of a stormwater control plan, in compliance with the Regional Municipal NPDES Permit, stormwater quality and hydromodification impacts will be reduced to a ***less-than-significant*** level in both watersheds. No mitigation is required.

1.3.2.2 Groundwater Quality

The stormwater treatment BMPs discussed above are likely to be constructed as media filter treatment vaults, given the relatively shallow groundwater conditions at the project site. The BMPs are expected to meet water quality treatment standards as required under Provision C.3 of the Regional Municipal NPDES Permit.

Even if final designs incorporate vegetated swales or flow-through planters as stormwater treatment measures, underdrains[9] would be required, given the low soil infiltration rates and high groundwater conditions. In addition, with a known plume of groundwater contamination in the vicinity of the site, the use of underdrains for these features is preferred to avoid the potential increased mobilization of the existing plume. **Mitigation Measure HYD-1** requires that any proposed stormwater treatment features included as part of the final design incorporate underdrains to reduce potential impacts to groundwater quality to a ***less-than-significant*** *level*.

1.3.3 Groundwater supplies and recharge

Water supply for the proposed project is obtained from EBMUD. The proposed project would not pump groundwater at the project site or otherwise use groundwater for supply, and therefore would not deplete local groundwater supplies.

[9] An underdrain consists of one or multiple small diameter perforated pipes installed at the bottom of a stormwater treatment BMP. In situations where infiltration into the underlying soil is either not feasible or not wanted, it conveys the treated runoff from the BMP filter media to downstream stormwater drainage infrastructure.

The existing site surface area is currently 98% impervious. The proposed project would decrease the overall imperviousness to approximately 89% by creating a vegetated set-back from Codornices Creek and implementing landscape-based LID design features. Due to the proximity of the project to the Bay, and the high groundwater table (see Section 1.2.4), there is no major opportunity for the use of infiltration when selecting these features or stormwater treatment BMPs. Furthermore, given the existing groundwater contamination mentioned above, measures will likely be implemented to avoid significant groundwater recharge – see **Mitigation Measure HYD-1**. As a result, the potential impact to groundwater recharge is considered ***less than significant***.

1.3.4 Altered Drainage Patterns

1.3.4.1 Erosion or Siltation On- or Off-Site

Construction-phase impacts relating to erosion or siltation both on and off-site are discussed in Section 1.1.2.1, above. Implementation of the project specific SWPPP in compliance with County regulations would reduce erosion and siltation during project construction. The resulting impact would be ***less than significant***.

Compliance with the NPDES Phase II Permit through the ACCWP, and the addition of a vegetated set-back to the portion of the site that drains to Codornices Creek, will limit post-construction hydromodification effects that could induce in-stream erosion and cause siltation in Codornices Creek. As the Gilman Basin storm drain is a direct, piped outfall to the Bay, there is no concern for hydromodification.[10] As such, the impact related to potential hydromodification effects would be ***less than significant***.

1.3.4.2 Flooding On- or Off-Site

The proposed project would decrease the overall imperviousness of the project site surface area from 98 percent to approximately 89 percent, primarily through the incorporation of a vegetated set-back from Codornices Creek and through implementation of landscape-based LID design features. In general, this decrease would be expected to slightly reduce the overall runoff from the project site relative to existing conditions, especially for the Codornices watershed where the majority of the imperviousness will be removed. Despite the decrease in imperviousness, the additional stormwater runoff from newly covered areas (water that was diverted to EBMUD under existing conditions, see Section 1.2.2.1) has the potential to cause or exacerbate flooding downstream of the project site. This would represent a potentially significant impact.

Preliminary hydrologic modeling was completed by Balance Hydrologics using the project description and site concepts outlined in the Feasibility Study to determine how the change in land use and stormwater conveyance covered in Section 1.2.2 would affect stormwater runoff from the project site. The modeling was performed using USACE's HEC-HMS modeling program and reflects the methodology outlined in ACFCD's guidance explained in Section 1.1.2.2. For the purposes of this analysis, the project site was modeled as a single drainage area.

[10] Per Chapter 7 of the Alameda County C.3 Technical Guidance Manual (ACCWP, 2017), hydromodification control requirements do not apply to projects that drain into channel segments that are hardened continuously downstream to their outfall in a tidal area.

In the post-project condition, the vegetation bordering Codornices Creek will sheet flow into the creek.[11]

Modeling of the pre-project condition includes the stormwater holding facilities shown in **Figure 4** and reflects the outline included in Figure 4 of the 2018 SWPPP (see **Appendix A**). Through correspondence with Zero Waste Division (ZWD) staff, it was determined that the stormwater holding capacity for the Community Conservation Center was doubled to comply with a recommendation outlined in the 2018 Wood SWPPP. In total, 30,000 gallons, or 0.089 acre-feet of storage are included in the pre-project modeling condition. Furthermore, conversations with ZWD staff and review of the requirements outlined in the EBMUD special discharge permit confirmed that after the capacity of the stormwater holding tanks is met, runoff is directed to the storm drain in Second Street. In contrast, the post-project condition omits any stormwater quality treatment measures and assumes no peak flow mitigation measures. The model was structured in this manner to present a conservative analysis, quantify the difference in peak flow and runoff volume, and inform the maximum amount of detention that may be required to avoid flooding impact. Results from the modeling are presented below in **Table 1**; the storm events are consistent with those required in Chapter 5 of ACFCD's Hydrology and Hydraulics Manual; all storm durations are 24 hours.

Recurrence Interval	Pre-Project with Detention		Post Project		Difference	
	Peak Flow *(cfs)*	Total Volume *(acre-feet)*	Peak Flow *(cfs)*	Total Volume *(acre-feet)*	Peak Flow *(cfs)*	Total Volume *(acre-feet)*
5 yr	6.5	1.9	6.2	1.8	-0.4	-0.1
15 yr	8.5	2.5	8.0	2.3	-0.5	-0.2
25 yr	9.5	2.8	8.8	2.6	-0.7	-0.2
100 yr	11.8	3.5	11.0	3.2	-0.8	-0.3

Table 1. Preliminary Hydrologic Modeling Results

The modeling results demonstrate that the 9-percent decrease in impervious area of the proposed design is enough to offset the existing stormwater holding capacity, and that peak flow and runoff volume is expected to decrease under post-project conditions. This result is to be expected considering how small the existing holding capacity is in comparison with the total runoff volume of the storm events. However, because alterations may occur in the final project designs and because flooding is of particular concern in this area, especially for the Codornices

[11] In compliance with the NPDES permit, the project proponent will be required to submit BAHM modeling to show that the design of the project site within the Codornices Creek watershed does not contribute to hydromodification within the watershed.

watershed, **Mitigation Measure HYD-2** is included to confirm that final project designs and stormwater plans do not result in increases to peak stormwater flows or volumes. With adherence to **HYD-2**, potential impacts related to increase in stormwater flows would be reduced to a ***less-than-significant*** level.

1.3.4.3 Exceedance of Storm Drain Capacity

Both the Codornices and Gilman Street watersheds have stormwater capacity constraints downstream of the project site for flows with a 10-year or greater recurrence (see Section 1.2.3). As discussed in the previous section, stormwater runoff from the site is expected to decrease under post-project conditions. Detailed stormwater routing has not yet been designed for post-project conditions, though the site drainage is expected to remain generally the same. Per discussion with ZWD staff, the final design of either Concept A or B are not anticipated to result in the rerouting of current drainage patterns but if portions of the project site that previously drained to one watershed are diverted to the other, exceedance of storm drain capacity may occur despite the overall decrease in stormwater runoff. To address this potentially significant impact, **Mitigation Measure HYD-3** provides guidance for final design of the stormwater infrastructure. Compliance with **Mitigation Measure HYD-3** would confirm that the project does not exacerbate storm drain capacity constraints and would reduce the potential impact to a ***less-than-significant*** level.

1.3.4.4 Hydrologic Impacts to Flood Flows

During significant storm events the highest likelihood of flooding in Codornices Creek occurs adjacent to and on the northern portion of the project site where Second Street acts as a point of release for the creek (see Section 1.2.2). The capacity of the culvert beneath I-80 lacks the necessary conveyance for events equal to or greater than the 10-year recurrence interval. This currently results in flooding adjacent to and on the project site at Second Street. Because the proposed project includes a setback in compliance with the City of Berkeley Creek Ordinance that will be graded at or below the existing site grade, the overbank storage volume will be improved compared to existing conditions. For the purposes of impacts to flood flows, CEQA directs lead agencies to evaluate whether a proposed project would cause or risk exacerbating existing flooding conditions by bringing development or people into an affected area. Because the proposed project would improve the existing flooding conditions on the project site and within Second Street, impacts to flood flows would be ***less than significant***.

While the City does not identify a threshold requiring evaluation of climate change on flooding and CEQA does not require analysis of the existing environment's effect on future users or residents of a project unless the project itself exacerbates an existing risk or hazard, this provides a discussion of potential flooding impacts related to climate change for informational purposes. The additional vegetated buffer along the southern boundary of Codornices Creek would provide minor additional storage that is unlikely to reduce the breakout of flood flows to Second Street; however, the setback does provide space for a future response to climate change when the incorporation of future adaptive measures is feasible and necessary. For example, a flood wall could be constructed as sea level rise further exacerbates existing flooding conditions.

1.3.5 Risk of Pollutant Release Due to Inundation

Due to the proximity of the project site to San Francisco Bay, it is subject to potential inundation by tsunami.[12] Maps provided by the California Geological Survey (CGS)[13] show the potential tsunami inundation zone extending from San Francisco Bay to 5th Street, east of the project site. The proposed project would replace existing infrastructure, but the overall elevation and usage of the site would generally remain the same. The proposed project would not cause the site to be more susceptible to tsunami inundation, and thus the risk of resultant pollutant release would be ***less than significant*** when compared to existing conditions.

Federal Emergency Management Agency maps for the Berkeley/Albany area show the presence of Special Flood Hazard Area (Zone AO; the 1-percent chance annual flood or what has traditionally been called the '100-year flood' hazard zone) for Codornices Creek adjacent to the northern boundary of the project site. The project site itself is within the 0.2-percent annual chance Flood Hazard Area ('500-year flood') that covers the East Bay Plain in this area between Highway 80 and 4th Street. Though rare, the current site is subject to potential flooding from the adjacent Codornices Creek, which would (if water quality treatment storage capacity is exceeded) result in release pollutants to downstream receiving waters. However, the proposed project (under both Concept A and Concept B) would include a 30-foot creek buffer and berm wall that would serve to reduce the risk of run-on flooding relative to existing conditions. As such, the potential impact related to release of pollutants due to flood inundation is ***less than significant***.

1.3.6 Water Quality Control Plan and Sustainable Groundwater Management Plan Consistency

As discussed in Sections 1.1.1.2 and 1.1.1.4, the proposed project falls within the jurisdiction of the San Francisco Bay Water Quality Control Plan (CA RWQCB, 2019) and East Bay Plan Groundwater Management Plan (EBMUD and Hayward, 2021). In both cases, the proposed project would continue existing uses on a similar footprint to existing conditions and would not have significant impacts related to water quality or groundwater (see Sections 1.3.2 and 1.3.3). Proposed stormwater BMPs would be designed to meet NPDES standards and are consistent with the goals and objectives outlined in the Basin Plan. In some cases, project plans have specifically targeted ways to reduce the potential for generation of contaminated runoff (e.g., covering areas designated for material handling and sorting).

With respect to the Groundwater Management Plan, the project site is located in a heavily urbanized area of the East Bay Plain, with a long-standing history of light and heavy industrial uses. Because of these previous and existing industrial land uses, the underlying aquifer is not a regional source of water supply, nor is it anticipated to be developed as one in the future. The project would not intensify use at the site, nor would it significantly affect groundwater recharge to, or movement within, the underlying aquifer relative to existing conditions.

[12] Analysis by the USGS (1975) suggests that an earthquake-induced seiche within the San Francisco Bay is unlikely, and thus only tsunami and flood inundation risk are assessed for this project.
[13] https://www.conservation.ca.gov/cgs/tsunami/maps/alameda

Therefore, the proposed project is considered consistent with applicable Water Quality Control and Groundwater Management Plans, and the potential impact is ***less than significant***.

1.4 CUMULATIVE IMPACTS

The cumulative context to assess the project impacts include the current and historic land use of the project site and the broader region of Berkeley; the potential hydromodification impact to Codornices Creek; and the potential to worsen existing flooding near the project site. When analyzing cumulative impacts, the watershed is used as the geographic unit based on the concept that many water quality problems, like the accumulation of pollutants or nonpoint source pollution, are best addressed at the watershed level. In addition, California's regulatory framework for protection of water quality focuses on the watershed. Potential cumulative impacts specific to water quality, flooding, and groundwater are discussed below.

1.4.1 Surface Water Quality

The Codornices and Gilman watersheds are both highly urbanized and include past and existing heavy industrial facilities within the lower portions near the project site. In addition, much of the development in this area originally occurred prior to the passage of the Clean Water Act, and, though many sites have been re-developed and/or retrofitted since, legacy water quality impacts remain. Because of the high existing level of urbanization within these watersheds (especially the Gilman Street watershed), future development is likely to consist largely of infill and redevelopment projects and, similar to the proposed project, will be required to incorporate updated stormwater quality treatment measures in compliance with applicable federal, State, and local laws and regulations. With respect to cumulative surface water quality impacts, implementation of cumulative projects would not contribute to a considerable cumulative impact.

Due to the small percentage of the project site that falls within the Codornices Creek watershed and the planned setback from the creek, the proposed project would likely not have a large impact on the overall water quality of the watershed if treatment requirements are met. Conversely, in the Gilman Basin, there is potential for the project to improve water quality by incorporating additional and updated stormwater treatment measures (compared to existing conditions) that comply with current NPDES treatment requirements and all other applicable federal, State, and local laws and regulations. The existing facility had to retroactively implement requirements to meet standards after the passage of major environmental legislation listed in Section 1.1.1 and has struggled with upholding the requirements outlined for industrial facilities.[14] By designing the proposed facility to meet necessary treatment requirements, the potential for pollutants to enter receiving waters would be greatly reduced. Therefore, the proposed project would not make a cumulatively considerable contribution to the less than significant cumulative impact and the cumulative impact would be ***less than significant***.

[14] In April of 2016 the non-profit organization San Francisco Baykeeper filed a complaint against the City of Berkeley alleging that the existing facility was violating its Industrial Stormwater Permit regulated under Section 402 of the Clean Water Act.

1.4.2 Flooding

Both watersheds currently lack sufficient capacity to handle storm events larger than the 10-year event and face an existing impact from flooding. Codornices Creek is constrained by the culvert under I-80 and insufficient overbank storage between the UPRR and I-80 that results in overland release to Second Street. In the Gilman Basin, the storm drain system is undersized such that storm flows back up and flood low-lying areas along Gilman Street at events greater than a 10-year recurrence. Future cumulative development and redevelopment along Codornices Creek would continue to add additional overbank storage near the creek in compliance with the City of Berkeley Creek Ordinance. Given the existing high level of urbanization within both watersheds, large-scale cumulative flooding impacts are unlikely, especially when considering the increased emphasis on LID design and the opportunities to retrofit detention facilities. With respect to cumulative flooding impacts, implementation of cumulative projects would not contribute to a considerable cumulative impact.

The proposed project would reduce stormwater runoff to both watersheds (see Section 1.3.4.2) and would provide additional overbank storage along Codornices Creek (see Section 1.3.4.4). Therefore, the proposed project would not make a cumulatively considerable contribution to the less than significant cumulative impact and the cumulative impact would be considered ***less than significant***.

1.4.3 Groundwater

Based on the high degree of urbanization in West Berkeley and the generally low-infiltration soils in the area, there is minimal recharge to the underlying groundwater and most rainfall is conveyed as surface runoff to San Francisco Bay. Future projects in the area would be required to incorporate LID elements in accordance with applicable federal, State, and local laws, which encourage infiltration, where subsurface conditions allow. With respect to cumulative impacts to groundwater recharge, implementation of cumulative projects would not contribute to a considerable cumulative impact.

The proposed project would decrease impervious area by approximately 9 percent. As discussed in Section 1.3.3, because of the proximity of the project site to the Bay, and the high groundwater table, there is no major opportunity for the use of infiltration when selecting LID design features or stormwater treatment BMPs. However, this is an existing condition, and the proposed project is not expected to cumulatively decrease groundwater recharge below existing conditions. Therefore, the proposed project would not make a cumulatively considerable contribution to the less than significant cumulative impact and the cumulative impact would be considered ***less than significant***.

Groundwater quality within the West Berkeley area has been significantly impacted by past industrial activities. Redevelopment within this area, especially conversion of heavy industrial uses to lighter industrial, commercial, or residential uses, would likely result in general improvements in water quality under future cumulative conditions. With respect to cumulative impacts to groundwater quality, implementation of cumulative projects would not contribute to a considerable cumulative impact.

The project itself would maintain a similar land use to existing conditions and is not expected to degrade groundwater quality (see Section 1.3.2.2). Therefore, the proposed project would not make a cumulatively considerable contribution to the less than significant cumulative impact and the cumulative impact would be considered ***less than significant***.

1.5 MITIGATION MEASURES

Impact HYD-1: The proposed project could potentially mobilize existing contaminant plumes located near the project site if infiltration-based LID measures are used to treat stormwater runoff in compliance with NPDES requirements.

Mitigation Measure HYD-1: Infiltration will not be used in LID elements to avoid the possibility of mobilizing existing contaminant plumes. LID features will be fitted with underdrains that collect and convey treated stormwater to the existing storm drain infrastructure located on Second Street.

Impact HYD-2: The proposed project could potentially alter the existing drainage pattern of the site or area in a manner that would significantly increase the rate or amount of surface runoff resulting in flooding off-site.

Mitigation Measure HYD-2: In compliance with ACFCD requirements, prior to approval of the final project design by the City of Berkeley, the project proponent shall complete a hydrologic study to show that, at a minimum, peak flow from the project site does not increase for the 5-, 15-, 25-, and 100-year storm events. If appropriate detention capacity cannot be provided to meet this requirement, then the site design shall be altered to reduce impervious area (or otherwise alter the drainage), such that the flood control requirements can be met.

Impact HYD-3: Per discussions with ZWD staff the proposed project is not anticipated to alter existing drainage patterns, however, the proposed project has the potential to alter site drainage, shifting drainage from one sub-watershed to another. This could exacerbate existing storm drain capacity issues even if the overall peak runoff from the site is reduced. This would be a significant impact related to storm drain capacity.

Mitigation Measure HYD-3: Prior to approval of the final project design by the City of Berkeley, the project proponent shall provide documentation that the final stormwater plan maintains the existing proportion of runoff between the Codornices and Gilman Street watersheds, or that peak flows are not increased in each subwatershed for the 5-, 15-, 25-, and 100-year storm events. If increases in peak flow are documented at that time, the project shall provide additional detention or plans shall be altered to re-apportion flows such that no peak-flow increases shall occur to either watershed.

References

(ACCCWP, 2017) Alameda County Clean Water Program, 2017, C.3. Stormwater Technical Guidance: A handbook for developers, builders, and project applicants, v. 6, 169 p.

(ACFCD, 2018) Alameda County Flood Control & Water Conservation District, 2018, Alameda County Hydrology and Hydraulics Manual, p. 44-50.

(Balance, 2011) Ballman, E.D., Guensch, J., Thompson, K., 2011, Storm Drain Hydrology and Hydraulics Report: Balance Hydrologics consulting report prepared for the City of Berkeley, 40 p.

(Berkeley, 2016) City of Berkeley Planning and Development Department, 2016, Stormwater Requirements Overview, 2 p.

(Berkeley, 2003) City of Berkeley City Council and Planning Commission, 2003, City of Berkeley General Plan: A Guide for Public-Decision Making.

(CA RWQCB, 2019) California Regional Water Quality Control Board, San Francisco Bay Region, 2019, Water Quality Control Plan (Basin Plan). 456p.

(Fugro, 2014) Fugro Consultants, Inc, 2014, Creek & Watershed Map of Western Alameda. Prepared in collaboration with the Alameda County Flood Control and Water Conservation District and the Alameda County Resource Conservation District.

(Geotracker, 2021) State Water Resources Control Board Geotracker Report for WRE/Colortech Facility, accessed on March 18, 2021.

(Groundwater Committee, 1999) California Regional Ground Water Quality Control Board San Francisco Bay Region Ground Water Committee, 1999, East Bay Plain Groundwater Basin Beneficial Use Evaluation Report Alameda and Contra Costa Counties, California, p. 37-43.

(Hagar, 2005) Hagar Environmental Science, 2005, Codornices Creek – Restoration Activities between 2nd Street and 5th Street Fish Removal Activities: Hagar Environmental Science consulting report prepared for the City of Albany, California, 11 p.

(Klein, 2009) Klein, R., 2009, Capacities of Codornices and Village Creeks Culverts at East Shore Highway, Albany and Berkeley, California: consulting report prepared for Caltrans, 22 p.

(Luhdorff and Scalmanini, 2021) Luhdorff & Scalmanini Consulting Engineers and others, 2021, Draft East Bay Plain Subbasin Groundwater Sustainability Plan: Luhdorff and Scalmanini consulting report prepared for East Bay Municipal Utility District GSA and City of Hayward GSA, p. 27-29, app. 2.A.

(SoilWeb, 2021) National Resource Conservation Service Soil Web Report for 669 Gilman Street, Berkeley, California, Accessed on November 3, 2021.

(USGS, 1975) United States Geological Survey, 1975, Studies for Seismic Zonation of the San Francisco Bay Region, USGS Professional Paper 941-A, 116p.

(Wood, 2018) Wood Environmental & Infrastructure Solutions, Inc., 2018, Storm Water Pollution Prevention Plan City of Berkeley Transfer Station and Recycling Center 1201 Gilman Street and 669 Gilman Street Berkeley, California, p. 6-8, fig. 2-4.

(Zero Waste, 2019) Zero Waste Collaborative, Inc., 2019, Solid Waste & Recycling Transfer Station Feasibility Study Final Report: Zero Waste Collaborative consulting report prepared for the City of Berkeley, p. 37-38.

H.2 - Stormwater Pollution Prevention Plan

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STORM WATER POLLUTION PREVENTION PLAN

City of Berkeley Transfer Station and Recycling Center
1201 Second Street and 669 Gilman Street
Berkeley, California

Prepared for:

City of Berkeley
Berkeley, California

Prepared by:

WSP USA Environment & Infrastructure Inc.
555 12th Street, Suite 215
Oakland, California 94612

June 2026

Project No. 5025220024.06.****

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REVISION SHEET

All revisions to the Stormwater Pollution Prevention Plan must be documented. Presented below is a listing, by date, of the sections that have been revised.

Revision 1

Date:	November 20, 2015
Section(s) Revised:	M. MONITORING IMPLEMENTATION PLAN (MIP)
Purpose of Revision:	Pollutant assessment added copper
Revised By:	Jim Honniball
Title:	Associate Scientist

Revision 2

Date:	November 30, 2015
Section(s) Revised:	Table of Contents, Figures 2 & 3, MIP sections M.4.A, M.4.B, M.5.A, M.5.G
Purpose of Revision:	Place holders for inspection/observation forms, weather tracking and BMPs on figures
Revised By:	Jim Honniball
Title:	Associate Scientist

Revision 3

Date:	October 28, 2016
Section(s) Revised:	The majority of the SWPPP
Purpose of Revision:	New BMP implementation
Revised By:	Jim Honniball
Title:	Associate Scientist

Revision 4

Date:	December 21, 2017
Section(s) Revised:	
Purpose of Revision:	Additional BMP implementation and analytical sampling suite
Revised By:	Jim Honniball
Title:	Associate Scientist

Revision 5

Date:	August 7, 2018
Section(s) Revised:	M.5. G
Purpose of Revision:	Update ERA Level
Revised By:	Joy Brown
Title:	Senior Management Analyst

Revision 6

Date:	June 15, 2021
Section(s) Revised:	B, C, E, F.1, F.2, F.4, G, H.1, J.1, L (H.4a), M.2, M.5.7, Site Map Figure 2
Purpose of Revision:	Update pollution team, industrial areas, minimum BMPs, exposure BMPs, sample reduction and QISP information
Revised By:	Jim Honniball
Title:	Associate Scientist TofR QISP, QSD/QSP

Revision 7

Date:	May 26, 2022
Section(s) Revised:	C.2, C.3, F.2-4, 8, 10, 11, 12, 13, 14, F.3-4, F.7-3, 6, G, J.2, L, M.5.3, Figures, 2, 3 and 4, Appendix C and D
Purpose of Revision:	Annual review
Revised By:	Jim Honniball
Title:	Associate Scientist TofR QISP, QSD/QSP

Revision 8

Date:	March 31, 2023
Section(s) Revised:	A.2, B.1-B.3, F.1.3, F.1.7, F.1.11, F.1.14 F.2.7, F.2.11, F.2.14, F.7.1, F.7.3 – 5, F.7.8, F.7.12, G, Sections X.H.1.a.ii X.H.1.a.iv, X.H.1.b.ii, X.H.1.b.iii, X.H.1.b.iv, X.H.1.c.i, X.H.1.c.ii, X.H.1.c.iii, X.H.1.d.v, J.2, J.3, L, M.2, M.7.
Purpose of Revision:	Update BMP information and request from Water Board
Revised By:	Jim Honniball

Title: Assistant-Vice President, Environmental Scientist, CPESC, TofR IGP, CGP, QISP, QSD,QSP

Revision 9

Date: October 20, 2023

Section(s) Revised: F.1-3, F.2-1, F.2-3, F.2-7, F.2-11, F.2-13, F.2-14, Minimum BMP table, L table (BMP summary), M.4.2, M.5.1, M.5.1.1, M.5.3,

Purpose of Revision: Facility Changes

Revised By: Jim Honniball

Title: Assistant Vice-President, Environmental Scientist, CPESC, TofR IGP, CGP, QISP, QSD,QSP

Revision 10

Date: June 16, 2026

Section(s) Revised: A.2 Regulatory Background, A.3 Existing Facility Plans and Plan Compliance Modifications, B.1.Legally Responsible Person, B.2 Duly Authorized Representative, B.3 Stormwater Pollution Prevention Team Personnel, C.1 Facility Information, C.2 Facility Description, C.3 Site Stormwater Drainage, C.4.1 Describe Neighboring Operations, C.4.2 Run-On, D. Required Site Maps(S) Information, E. List Of Industrial Materials, F.1 Industrial Processes, F.2 Materials Handling And Storage, F.3 Dust & Particulate Generating Activities, F.4 Potential For Significant Spills And Leaks, F.6 Authorized Non-Stormwater Discharge, F.7 Unauthorized Non-Stormwater Discharge, F.8 Erodible Surfaces, F.9 Assessment of Potential Pollutant Sources, G Minimum BMPs, H Employee Training Program, J.1 Exposure Minimization, J.3 Treatment Control BMPs, L. Bmp Summary Table,

	M.1 Team Members, M.2 Discharge Locations, M.3 Representative Sampling Reduction Or Qualified Combined Samples, M.5.2 Sample Collection, M.5.3 Sample Analysis, M.6 Annual Comprehensive Facility Compliance Evaluation, Removal of Exhibits, Figures, 2, 3, 4, and 5.
Purpose of Revision:	Facility Changes
Revised By:	Jim Honniball
Title:	Assistant Vice-President, Environmental Scientist, CPESC, TofR IGP, CGP, QISP, QSD,QSP

STORMWATER POLLUTION PREVENTION PLAN
City of Berkeley Transfer Station and Recycling Center
1201 Second Street and 669 Gilman Street
Berkeley, California

A. INTRODUCTION

This document is the Stormwater Pollution Prevention Plan (SWPPP) for the City of Berkeley (City) Transfer Station and Recycling Center located at 1201 Second Street and 669 Gilman Street, Berkeley, California, respectively. The Recycling Center is owned by the City but operated by the Community Conservation Center (CCC). The SWPPP was prepared in accordance with the requirements of the California State Water Resources Control Board (State Board), National Pollutant Discharge Elimination System (NPDES) General Permit for Stormwater Discharges Associated with Industrial Activities dated April 1, 2014, as amended by Order WQ 2015-0122-DWQ and Order WQ 2018-0028-DWQ (General Permit).

This SWPPP sets forth specific practices, procedures, monitoring, and reporting that are the responsibility of designated Facility personnel to implement.

A.1 PLAN OBJECTIVES

This SWPPP is designed to:

- Identify and evaluate sources of pollutants associated with industrial activities that may affect the quality of stormwater discharges and authorized non-stormwater discharges (NSWDs) from the Facility;
- Identify and describe the minimum Best Management Practices (BMPs) (General Permit Section X.H.1) and any advanced BMPs (Section X.H.2) implemented to reduce or prevent pollutants in industrial stormwater discharges and authorized NSWDs; and
- Identify and describe conditions or circumstances which may require future revisions to this SWPPP.

The SWPPP provides Facility staff with guidance for achieving the stated objectives and maintaining compliance with the requirements of the General Permit. A copy of this SWPPP is maintained at the Facility.

A.2 Regulatory Background

The United States Environmental Protection Agency (U.S. EPA) developed the stormwater regulatory program through the authority of the Clean Water Act amendments of 1987. The U.S. EPA's goal was to reduce discharges of contaminated stormwater from industrial facilities. The U.S. EPA, through the NPDES permitting program, regulates discharges of potentially contaminated wastewater and stormwater into Waters of the United States. California has been delegated NPDES general permitting authority by the U.S. EPA. On April 1, 2014, the State Water Resources Control Board (SWRCB) adopted Order 2014-0057-DWQ. This General Permit became effective on July 1, 2015. The General Permit was subsequently amended by Order WQ 2015-0122-DWQ and Order WQ 2018-0028-DWQ. This SWPPP was written pursuant to the requirements set forth in the General Permit.

A.3 Existing Facility Plans and Plan Compliance Modifications (General Permit Section X.D.2)

This section lists permits and other governing documents that may be relevant to water quality-related issues. In addition to the General Permit, the following documents have been taken into account while preparing this SWPPP:

1. 303(d)-list of impaired water bodies;
2. Total maximum daily loads (TMDLs);

Additional documents that may affect practices employed to control stormwater discharges associated with industrial activities include the following:

3. Spill prevention control and countermeasures plan, which describes petroleum storage, handling, and spill response activities at the Facility (updated 10 November 2021); and
4. Hazardous material business plan, which maintains an inventory of hazardous chemicals, hazardous materials storage volumes, and hazardous materials storage locations (updated 2019).

This SWPPP will be amended and revised as needed. SWPPP revisions will be completed when:

- BMPs require significant changes;
- There is a physical or operational change at the Facility which may affect stormwater discharges;
- Otherwise deemed necessary by the Pollution Prevention Team members based on visual observations (General Permit XI.A.3);
- Upon entering Level 1 or Level 2 status (General Permit XII.1.C);

- Additional pollutant source is identified;
- Modification is made to the Monitoring Implementation Plan (MIP);
- Change is needed to prevent Numeric Action Level (NAL) exceedances;
- Annual Comprehensive Facility Compliance Evaluation (ACFCE) identifies necessary changes;
- Modifications required to be consistent with applicable municipal, state, and federal requirements pertaining to the General Permit;
- Directed to do so by the Regional Water Quality Control Board (RWQCB); and
- There are changes to alternative discharge location identification (General Permit XI.C.3.b), Representative Sampling Reduction justifications (General Permit XI.C.4.c), or Qualified Combined Samples justification (General Permit XI.C.5.c).

Significant SWPPP Revisions are required to be certified and submitted via SMARTS within 30 days of the significant revision. Other revisions (other than typographical errors) are required to be submitted within three (3) months. The pollution prevention team will assess what is significant. If no changes are made to the SWPPP, there are no resubmittal requirements. A revision sheet has been provided to track revisions made within this SWPPP.

Per Section XV of the General permit, the Discharger shall revise the SWPPP, as appropriate, and implement the revisions within 90 days of the ACFCE.

A.4 Plan Availability and Records Retention (General Permit Section XXI.J and Section XXI.H)

Paper or electronic records of documents required by this SWPPP will be retained for a minimum of five (5) years from the date generated or date submitted, whichever is later, for the following items:

- Employee Training Records;
- BMP Implementation Records;
- Spill and Clean-up Related Records;
- Records of Monitoring Information;
- Records of Visual Observations;
- Response to the observations including identification of SWPPP revisions, if needed;
- Level 1 Exceedance Response Action (ERA) Reports;

- Level 2 ERA Action Plan;
- Level 2 ERA Technical Report; and
- Annual Reports from SMARTS (checklist and any explanations).

Upon request by the relevant agency, the Facility will provide information to determine compliance with this General Permit to the SWRCB, RWQCB, U.S. EPA, or local Municipal Separate Storm Sewer System (MS4) within a reasonable amount of time. The Facility will furnish, upon request by the relevant agency, copies of records that are required to be kept by this General Permit. In addition, the electronic version of the SWPPP will be uploaded to SMARTS.

A.5 Permit Registration Documents & Signed Certification (General Permit Section II.A)

Permit Registration Documents (PRDs) are to be submitted through the Stormwater Multiple Application and Report Tracking System (SMARTS) by the Legally Responsible Person (LRP), or authorized personnel (i.e., Duly Authorized Representative) under the direction of the LRP. PRDs include:

1. Notice of Intent (NOI);
2. Signed Certification Statement (LRP Certification is provided electronically with SMARTS PRD submittal);
3. Site Map(s);
4. SWPPP; and
5. Annual Fee.

The individual designated as the LRP and other authorized personnel are identified in Section B. SMARTS submittal documentation, including a copy of NOI and Signed Certification Statement, is attached in PRDs, Appendix A of this SWPPP. The Facility Site Maps are included as Figures 1 through 4.

A.6 Temporary Suspension of Activities (General Permit Section X.H.3)

The General Permit includes a provision for facilities that temporarily suspend industrial activities for ten (10) or more consecutive calendar days during a reporting year. In the event the Facility's operations are temporarily suspended, the SWPPP will be revised to include stabilization BMPs. In addition, the date the Facility is fully stabilized and projected start date

will be documented. If monitoring is infeasible, justification for not monitoring will be provided. The revised SWPPP will be uploaded via SMARTS within seven calendar days prior to suspension of activities.

B. POLLUTION PREVENTION TEAM (GENERAL PERMIT SECTION X.D.1)

The LRP, DAR, and Pollution Prevention Team designated for this Facility are summarized below. In the event a regularly assigned Pollution Prevention Team member is temporarily unavailable, a new member will be assigned the role by the LRP or DAR. Team members are cross-trained to assist with various activities, as necessary, to implement this SWPPP in the event staff become unavailable due to vacation, illness, or other reasons.

The procedures to identify alternate team members to implement the SWPPP and conduct required monitoring when the regularly assigned team members are temporarily unavailable are listed below.

Transfer Station, Ecology Center and CCC SWPPP BMP staff are all trained. In the event one person is not available, other trained staff are available to perform the duties. Only trained staff implement the SWPPP.

Monitoring, sampling, reporting and General Permit administrative tasks are contracted out to consulting firms. These consulting firms have personnel that are available to implement the General Permit for the City.

The Facility training program is detailed in Section H of this SWPPP.

B.1. LEGALLY RESPONSIBLE PERSON (LRP)

Name	Title
Wahid Amiri	Acting Public Works Director

B.2 DULY AUTHORIZED REPRESENTATIVE

Name	Title
Robert Hernandez	Environmental Compliance Specialist

B.3 STORMWATER POLLUTION PREVENTION TEAM PERSONNEL

Name	Title	Responsibilities and Duties
Leticia Jauregui	Solid Waste and Recycling Manager	Reviews program performance, supports regulatory compliance efforts, and coordinates with City leadership on significant compliance, operational, or policy matters.
Robert Hernandez	Environmental Compliance Specialist	Compliance, Annual Reporting, SWPPP development, training, paperwork/records, delegation of SWPPP implementation duties, and record keeping.
Dave Johnson	CCC Manager	Oversight of stormwater program and implementation, delegation of BMP implementation duties, and training.
Daniel Maher	Ecology Center Recycling Manager	Oversight of stormwater program and implementation, delegation of BMP implementation duties, and training.
Victtor Lopez	Operations Manager	Oversees transfer station operations.
Joy Brown	Interim Deputy Public Works Director	Coordination of project contracts and subcontractors. Supports facility improvement projects. supports regulatory compliance efforts, and coordinates with City leadership on significant compliance, operational, or policy matters.
Don Colbert	Senior Equipment Supervisor	Oversees equipment maintenance, daily activities of equipment maintenance, and implementation of stormwater BMPs.
Gary Triplett	Senior Supervisor	Supervisor at transfer station, oversees daily operations and stormwater BMPs.
Jeff Belchamber	General Manager, CCC	Oversight of stormwater program and implementation, delegation of BMP implementation duties.

A Qualified Industrial SWPPP Practitioner (QISP) is required if the Facility changes from baseline status to Level 1 or Level 2. This SWPPP will be updated upon entering Level 1 or Level 2 for a new parameter.

C. FACILITY INFORMATION

The Facility information is described in the sections below.

C.1 FACILITY INFORMATION

Facility Name:	City of Berkeley Transfer Station & Recycling Center
Street Address:	1201 Second Street & 669 Gilman Street
Telephone Number:	510-981-6362
WDID Number:	201I009237
Facility Contact Name:	Leticia Jauregui
Contact Title:	Zero Waste Manager
Latitude/Longitude:	37.87978/-122.30588
Total Facility Area (Acres):	7.4
Total % Site Imperviousness (Percent):	98
Total Area of Industrial Activities and/or Materials Exposed to Precipitation (Acres):	4.5
Primary SIC Code(s):	5093, 4212
Receiving Water:	Central San Francisco Bay
Hours of Operation:	The Transfer Station facility is open to the Public Monday through Saturday 8:00 a.m. – 4:30 p.m. The Recycling Center facility is open to the Public Monday through Saturday 8:30 – 4:00 p.m.

C.2 FACILITY DESCRIPTION

The facility consists of two operations: the Transfer Station in the northern and central portion and the Recycling Center in the southern portion of the site.

The Transfer Station encompasses 5.0 acres and is operated by the City of Berkeley. Facilities on this portion of the site include the Administration Building, Transfer Station/Tipping Floor, Equipment Maintenance Building, Used Oil Depot, Scale House, Temporary Mobile Diesel Fueling Tank, Wash Rack, Container Maintenance Building, Container/Bin Storage Areas, General Storage Area, Green Waste Receiving Area, General/Electronic Waste Storage, and Parking.

The Transfer Station is owned and operated by the City to accept municipal, commercial and private solid waste and transfer it to a sanitary landfill. The Transfer Station does not accept hazardous waste or infectious materials. It accepts scrap metals, construction debris, refrigerated appliances, electronic waste, mattresses and box springs, used tires, propane cylinders, and green waste.

Equipment is fueled on site with a mobile diesel fueling tank. Equipment is repaired on site in either the Equipment Maintenance Building or the Container Maintenance Building. Equipment, vehicles, and refuse bins are cleaned at the wash rack.

The Recycling Center borders the Transfer Station to the south and encompasses 2.4 acres. This portion of the site includes the CCC and Ecology offices, parking, Paper and Glass Recycling Warehouses, Glass Sorting Area, Commingled Container Storage, Universal Waste Storage, Customer Recyclable Area, Recyclable Buyback Area, Ecology Administrative Shed, and Parking.

The Recycling Center is owned by the City and currently operated by the CCC. The Recycling Center accepts recyclable materials from the pickup of recyclables by the Ecology Center and from the general public. Minor equipment maintenance is performed on site and fueling takes place at the Transfer Station by the mobile diesel fueling tank.

C.3 SITE STORMWATER DRAINAGE

The site is located in the Gilman Street watershed, which is a highly urban and industrial section of the City. Site drainage is conveyed through the City storm drain system (south on 2nd Street and west on Gilman Street) and drains into the San Francisco Bay at the Gilman Street outfall.

The site encompasses 7.4 acres of land that slopes towards the Bay. Approximately 98 percent of the site is paved with asphalt or concrete, buildings or other structural facilities.

The facility includes 11 Drainage areas (Drainage Areas 1 through 11) of which one is non-industrial (Drainage Area 11). Drainage Areas 1-11 are depicted on Figure 3. A description of the Drainage Areas is provided below.

Drainage Area 1

This Drainage Area contains Container Bin Storage, and a Container Maintenance Building which also stores tires. Additionally, garbage trucks and heavy equipment are temporarily stored in this area while awaiting scheduled maintenance.

Stormwater in this area flows to two trench drains and a catch basin CB-1 then is directed to a Contech filter for treatment, after which it is discharged into the City's stormwater conveyance

system in Second Street. Stormwater samples are collected from the Contech filter's discharge point, at DP-1A.

Drainage Area 10 may overflow into Drainage Area 1 during large storm events that exceed the available storage capacity (21,000-gallon or 42,000-gallon storage volume when an extra aboveground storage tank [AST] is brought on site). Additionally, while no off-site run-on from Codornices Creek has been observed entering Drainage Area 1, Second Street may flood with municipal stormwater during large storm events at the end of the cul-de-sac. When flooding occurs, it may impede the flow within the storm drain lines in Drainage Area 1 resulting in on-site pooled water.

Drainage Area 2

This Drainage Area contains the Scale House, a portion of the Equipment Maintenance Building, General Waste Storage, Truck Parking, Mattress Area, the Wash Rack which drains to the East Bay Municipal Utility District (EBMUD) under permit, vehicle/customer access areas, a portion of the non-industrial Employee Parking Area, and a non-industrial Administration Building.

Stormwater in Drainage Area 2 flows to two trench drains (TD-2 and TD-2A) and a catch basin (CB-2) (Figure 3). Stormwater from TD-2 flows to an underground conveyance system and to the Contech filter. Stormwater from TD-2A and CB-2 flows to a manhole and then to a Contech filter. Stormwater from catch basin and trench drains passes through the Contech filter and is sampled at the effluent side of the vault (DP-2A).

Drainage Area 3

This Drainage Area contains a portion of General Waste Storage, General Storage, Temporary Mobile Diesel Fueling, a portion of the Equipment Maintenance Building, Used Oil Depot, and Drum/Diesel Exhaust Fuel Storage.

Stormwater flows through a strip drain depression on the north side of the Equipment Maintenance Building and is directed to catch basin CB-3A which flows to CB-3. CB-3 also receives overland flow from the area north of the Wash Rack. Stormwater on the eastern portion of the Drainage Area flows to catch basin CB-4 which flows to a manhole and then

connects to underground piping to a Contech filter. Stormwater from CB-3 and CB-4 passes through the Contech filter and is sampled at the effluent side of the vault (DP-3A).

Drainage Area 4

This Drainage Area contains industrial areas including Truck Parking, and a portion of the Container Bin Storage Area, and non-industrial areas including a portion of the Administration Building and a portion of the Employee Parking area.

Sheet flow leaves the facility along the western portion of this Drainage Area. Therefore, a Representative Sampling Reduction (RSR) sampling point, DP-4 was designated as described in Section M.3. Stormwater that discharges from Drainage Area 4 flows into the City's stormwater conveyance system on Second Street (see Figure 3).

Drainage Area 5

This Drainage Area contains The Ecology Center's office, Storage Shed, Truck Parking, and maintenance. Portions of this area are bermed which directs sheet flow to leave the facility along the western portion of this Drainage Area. Therefore, a RSR sampling point, DP-5A has been designated as described in Section M.3. Stormwater that discharges Drainage Area 5 flows into the City's stormwater conveyance system on Second Street (see Figure 3).

Drainage Area 6

This Drainage Area contains Recycling Bins, Customer Recyclable Area, Universal Waste and Fats Oil and Grease Storage. Materials are unloaded and placed within canopy covered bins. Water in this drainage area flows northwest towards Second Street. Sheet flow leaves the facility along the western portion of this Drainage Area. Therefore, a RSR sampling point, DP-6 has been designated as described in Section M.3.

Drainage Area 7

Water within the Glass Sorting Area and the Commingled Container Storage Area is contained with berms, sumps, and holding tanks. A valley trench (GSA-VG) is located in front of the Glass Sorting area which directs water towards Sump GSA-VGS. A valley trench is also located in front of the Commingled Container Storage area (CCS-VG) which directs water towards Sump CCS-VGS. These sumps have a high liquid level float switch. When the float switch detects a

high liquid level, pumps are turned on and flow is pumped to two, 4,000-gallon interconnected tanks. A third tank, with a volume of 5,400 gallons, is located next to the existing tanks but is not connected to them. If the water level in the interconnected tanks is high, stormwater from one of those tanks is pumped into the third tank for additional storage. Water within the tanks is discharged to EBMUD sanitary sewer after 24 hours of dry weather under permit via a 4-inch pipe with a flow meter and ball valve. Water from within the Commingled Container Storage area may overtop the berm and flow into Drainage Area 5 towards discharge point DP-5A. Additional tanks may be brought in as-needed in advance of heavy rain events to provide additional storage.

When the liquid in the stormwater holding tanks rises to a high level, the float switch sensors send an electrical signal that turn off the sump pumps. The pumps remain deactivated until the liquid drops back down to a low level. During large storm events that exceed the tank storage capacity, water within the Glass Sorting Area may overtop the berm and flow into Drainage Area 9 towards discharge point DP-9 as described in Section M.3.

Drainage Area 8

This Drainage Area contains operations by the Recycling Center including industrial activities from the Customer Recyclable and Recyclable Buyback Areas and a portion of the Paper Recycling Warehouse. Non-industrial activities within this Drainage Area include the Employee Parking and lunch area. DP-8A is a catch basin that is connected to the City's stormwater conveyance system on Gilman Street. Stormwater is collected as it enters the catch basin.

Drainage Area 9

This Drainage Area includes a portion of the Container Bin Storage Area, the eastern portion of the Glass Recycling Warehouse, and the eastern portion of the Paper Recycling Warehouse. Stormwater from this Drainage Area flows down the eastern alley of the Paper Recycling Warehouse and discharges as sheet flow to RSR sampling point DP-9 towards Gilman Street as described in Section M.3. Water from within the Glass Sorting Area may overtop the berm and flow into Drainage Area 9 towards discharge point DP-9 during large storm events. During large storm events, run-on from the off-site adjacent railroad can overtop the curb located along the eastern boundary of the Drainage Area and can comingle with on-site industrial water, Figure 3.

Drainage Area 10

This Drainage Area contains the Transfer Station building, Construction Debris, General/Electronic Waste Storage and Garbage Truck Parking areas. Within the Transfer Station building is a Vactor Truck wash area. Customers bring their materials for disposal in this Drainage Area.

Flow in Drainage Area 10 is routed to trench drains in front of the Construction Debris receiving area (TD-4) and near the General/Electronic Waste Storage Area (TD-3) (Figure 3).

TD-4 directs flow to a manhole with two sluice gates. The sluice gate directing flow to the stormwater conveyance system (gate closest to TD-4) is maintained in the open position while the sanitary sewer sluice gate is maintained in the closed position. Flow is directed to the main stormwater conveyance system, where it is joined by flow from trench drain TD-3. Flow proceeds via gravity flow to a second manhole equipped with a weir to deflect flow from going toward catch basin CB-1. Flow is instead directed to the stormwater holding tank(s) via a 2-inch pipe inlet by gravity flow. The stormwater holding tank is a 21,000-gallon aboveground storage tank. At least 24-hours after a rain event, the Environmental Compliance Specialist drains the 21,000-gallon tank to a sump pump with a digital flowmeter that pumps water to the Sanitary Sewer System. A second 21,000-gallon tank is present for additional storage, but is not directly plumbed to the stormwater system. Additional tanks are brought on-site for larger events to prevent discharge. Stormwater within the tank(s) is discharged into East Bay Municipal Utility District (EBMUD) sanitary sewer after 24 hours of dry weather under permit. During high flow events, stormwater that bypasses the weir flows to the municipal stormwater system and is sampled at location DP-1A after passing through a Contech filter vault.

The Vactor Truck wash area, located inside the Transfer Station Building, is bermed and has a trench drain. Wash water flows into the trench drain to the sanitary sewer (Figure 3). Behind the Vactor Truck wash area wall, an additional trench drain exists that connects to the same sanitary sewer line.

Drainage Area 11 – Non-Industrial

Drainage Area 11 is non-industrial and includes the western portion of the Green Waste Receiving Area roof that discharges to landscaped areas and Second Street.

C.4 Neighboring Operations and Stormwater Run-On

C.4.1 Neighboring Operations and Run-On

North: To the north of the Facility is Codornices Creek. There have been no recorded incidents of the creek flooding into Drainage Area 1. The northern area of Second Street is adjacent to the Facility and may flood with municipal stormwater during extreme rain events. When flooding occurs, it may impede the flow Facility drainage to the municipal storm drain system, but no overland run-on to the Facility has been recorded.

South: To the south of the Facility is Gilman Street. There is no potential run-on from this area as Gilman Street is at a lower elevation than the Facility.

East: To the east of the Facility are railroad tracks. There is run-on from this area during large storm events to Drainage Area 9.

West: To the west of the Facility is Second Street. This street floods during large back to back storm events, which can impede Facility drainage to the municipal storm drain system, but no overland run-on to the Facility has been recorded.

D. REQUIRED SITE MAPS(S) INFORMATION (GENERAL PERMIT SECTION X.E)

The Facility's Site Maps are included within this SWPPP as Figures 1 through 4.

Included on Site Map(s)? Yes/No/NA	SWPPP Site Map Requirements
Yes	The Facility boundary (Section X.E.3.a)
Yes	Stormwater drainage areas within the Facility boundary (Section X.E.3.a)
Yes	Portions of any drainage area impacted by discharges from surrounding areas
Yes	Flow direction of each drainage area (Section X.E.3.a)
NA	On-Facility surface water bodies (Section X.E.3.a)
NA	Areas of soil erosion (Section X.E.3.a)
Yes	Location(s) of nearby water bodies such as rivers, lakes, wetlands (Section X.E.3.a)
Yes	Location(s) of municipal storm drain inlets that may receive the Facility's industrial stormwater discharges and authorized NSWDs (Section X.E.3.a)
Yes	Locations of stormwater collection and conveyance systems and associated points of discharge, sample locations, and direction of flow (Section X.E.3.b)
Yes	Any structural control measures (that affect industrial stormwater discharges, authorized NSWDs, and run-on) (Section X.E.3.c)
Yes	All impervious areas of the Facility, including paved areas, buildings, covered storage areas, or other roofed structures (Section X.E.3.d)

Included on Site Map(s)? Yes/No/NA	SWPPP Site Map Requirements
Yes	Locations where materials are directly exposed to precipitation (Section X.E.3.e)
NA	Locations where significant spills or leaks (Section X.G.1.d of the General Permit) have occurred (Section X.E.3.e)
Yes	Areas of industrial activity subject to the General Permit (Section X.E.3.f)
Yes	All storage areas and storage tanks (Section X.E.3.f)
Yes	Shipping and receiving areas (Section X.E.3.f)
Yes	Fueling areas (Section X.E.3.f)
Yes	Vehicle and equipment storage/maintenance areas (Section X.E.3.f)
Yes	Material handling and processing areas (Section X.E.3.f)
Yes	Waste treatment and disposal areas (Section X.E.3.f)
Yes	Dust or particulate generating areas (Section X.E.3.f)
Yes	Cleaning and material reuse areas (Section X.E.3.f)
Yes	Any other areas of industrial activity which may have potential pollutant sources (Section X.E.3.f)

E. LIST OF INDUSTRIAL MATERIALS (GENERAL PERMIT SECTION X.F)

Presented below is a list of industrial materials that are handled and stored at the Facility. Industrial materials include raw materials, intermediate products, final or finished products, recycled materials, and waste or disposed materials.

Material	Physical Characteristics	Storage Location(s); Typical Quantity Stored; and Typical Frequency of Storage	Receiving Location(s); Typical Quantity Received; and Typical Frequency of Receiving	Shipping Location(s); Typical Quantity Shipped; and Typical Frequency of Shipping	Handling Location(s); Typical Quantity Handled; and Typical Frequency of Handling
Ethylene Glycol (anti-freeze)	Liquid	Site Map Locations: Equipment Maintenance / Ecology Storage Shed Typical Quantity: Approximately 100 gallons / 30 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance / Ecology Storage Shed Typical Quantity Received: 100 gallons / 5 gallons Typical Frequency of Receiving: Quarterly	Shipping Location(s): Equipment Maintenance Typical Quantity Shipped: 55-75 gallons Typical Frequency of Shipping: Quarterly	Handling Location(s): Equipment Maintenance / Ecology Storage Shed Typical Quantity Handled: 10-15 gallons Typical Frequency of Handling: Daily
Motor and Gear Oil	Liquid	Site Map Locations: Equipment Maintenance / Ecology Storage Shed, Container Maintenance Building Typical Quantity: Approximately 240 gallons / 30 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance / Ecology Storage Shed, Container Maintenance Building Typical Quantity Received: 100 gallons / 10 gallons Typical Frequency of Receiving: Weekly	N/A – See Used Oil for Shipping information.	Handling Location(s): Equipment Maintenance / Ecology Storage Shed, Container Maintenance Building Typical Quantity Handled: 10 gallons Typical Frequency of Handling: Daily
Lead Acid Batteries	Solid	Site Map Locations: Equipment Maintenance Typical Quantity: Approximately 400 pounds based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance Typical Quantity Received: 250 pounds Typical Frequency of Receiving: Monthly	Shipping Location(s): Equipment Maintenance Typical Quantity Shipped: 250 pounds Typical Frequency of Shipping: Monthly	Handling Location(s): Equipment Maintenance Typical Quantity Handled: 25 pounds Typical Frequency of Handling: Daily
Hydraulic Fluid	Liquid	Site Map Locations: Equipment Maintenance Typical Quantity: Approximately 250 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance Typical Quantity Received: 75-100 gallons Typical Frequency of Receiving: Weekly	N/A	Handling Location(s): Equipment Maintenance Typical Quantity Handled: 10-15 gallons Typical Frequency of Handling: Daily
Grease	Semi-Solid	Site Map Locations: Equipment Maintenance Typical Quantity: Approximately 55 gallons (400 lbs.) based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance Typical Quantity Received: 55 gallons (400 lb.) Typical Frequency of Receiving: Quarterly	N/A	Handling Location(s): Equipment Maintenance Typical Quantity Handled: 3-5 lbs. Typical Frequency of Handling: Daily
Transmission Fluid	Liquid	Site Map Locations: Equipment Maintenance Typical Quantity: Approximately 400 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance Typical Quantity Received: 50 gallons Typical Frequency of Receiving: Weekly	N/A	Handling Location(s): Equipment Maintenance Typical Quantity Handled: 5-10 gallons Typical Frequency of Handling: Daily
Assorted Flammables, Solvents, and Paints	Liquid, Semi-Solid	Site Map Locations: Equipment Maintenance, Container Maintenance Building Typical Quantity: Approximately 100 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance, Container Maintenance Building Typical Quantity Received: 5 gallons Typical Frequency of Receiving: Quarterly	N/A	Handling Location(s): Equipment Maintenance, Container Maintenance Building Typical Quantity Handled: .5 gallons Typical Frequency of Handling: Daily
Used Oil	Liquid	Site Map Locations: Equipment Maintenance, Used Oil Depot Typical Quantity: Approximately 1300 gallons based on demand Typical Frequency of Storage: Annually	N/A – Used oil is generated in the shop.	Shipping Location(s): Equipment Maintenance, Used Oil Depot Typical Quantity Shipped: 250 gallons, 600 gallons Typical Frequency of Shipping: every 6-8 weeks.	Handling Location(s): Equipment Maintenance, Used Oil Depot Typical Quantity Handled: 5 gallons Typical Frequency of Handling: Daily
Used Oil Filters	Solid	Site Map Locations: Equipment Maintenance, Used Oil Depot Typical Quantity: Approximately 3 – 55 gallon drums based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Used Oil Depot Typical Quantity Received: 5-10 filters Typical Frequency of Receiving: Daily	Shipping Location(s): Equipment Maintenance, Used Oil Depot Typical Quantity Shipped: 2-5 – 55 gallon drums Typical Frequency of Shipping: Quarterly	Handling Location(s): Equipment Maintenance, Used Oil Depot Typical Quantity Handled: 10-20 filters Typical Frequency of Handling: Daily
Sodium Metasilicate Cleaner (biodegradable)	Liquid	Site Map Locations: Wash Rack Typical Quantity: Approximately 100 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Wash Rack Typical Quantity Received: 25 gallons Typical Frequency of Receiving: Quarterly	N/A	Handling Location(s): Wash Rack Typical Quantity Handled: 1 gallon (mixed with water) Typical Frequency of Handling: Daily

Material	Physical Characteristics	Storage Location(s); Typical Quantity Stored; and Typical Frequency of Storage	Receiving Location(s); Typical Quantity Received; and Typical Frequency of Receiving	Shipping Location(s); Typical Quantity Shipped; and Typical Frequency of Shipping	Handling Location(s); Typical Quantity Handled; and Typical Frequency of Handling
Diesel	Liquid	Site Map Locations: Temporary Mobile Fueling Double Walled Tank near north west corner of maintenance building Typical Quantity: Approximately 512 gallons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Temporary Mobile Fueling Double Walled Tank near north west corner of maintenance building Typical Quantity Received: Varies Typical Frequency of Receiving: 2 – 3 times per week	N/A	Handling Location(s): Temporary Mobile Double Walled Fueling Tank near north west corner of maintenance building Typical Quantity Handled: 100 –200 gallons Typical Frequency of Handling: Daily
Diesel Exhaust Fuel (DEF)	Liquid	Site Map Locations: Drum and DEF Storage Area , east side of maintenance building Typical Quantity: Approximately two 275-gallons containers based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Drum and DEF Storage Area , east side of maintenance building Typical Quantity Received: Varies Typical Frequency of Receiving: Varies	N/A	Handling Location(s): Drum and DEF Storage Area , east side of maintenance building Typical Quantity Handled: two 275-gallon containers Typical Frequency of Handling: Daily
Absorbent/Absorbent Pads	Solid, Powder	Site Map Locations: Equipment Maintenance Typical Quantity: Approximately 400 pounds based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Equipment Maintenance Typical Quantity Received: 200 pounds Typical Frequency of Receiving: Quarterly	N/A	Handling Location(s): Equipment Maintenance Typical Quantity Handled: 40 pounds Typical Frequency of Handling: Weekly or as needed
Municipal Waste and Construction Debris	Solid, Non-Solid, Powder	Site Map Locations: Transfer Station Tipping Floor and Outdoor Construction Debris Receiving Area Typical Quantity: Approximately 400,000 pounds based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Transfer Station Tipping Floor and Outdoor Construction Debris Receiving Area Typical Quantity Received: 50 to 12,000 pounds depending on customer Typical Frequency of Receiving: Daily (except Sunday)	Shipping Location(s): Tipping Floor Truck Loading Area and Outdoor Construction Debris Receiving Area Typical Quantity Shipped: 80,000 pounds Typical Frequency of Shipping: Daily (except Sunday)	Handling Location(s): Transfer Station Tipping Floor and Outdoor Construction Debris Receiving Area Typical Quantity Handled: 1,000 pounds Typical Frequency of Handling: Daily (except Sundays)
Treated Wood Waste	Solid	Site Map Locations: Outdoor Construction Debris Receiving Area Typical Quantity: Approximately 4,000 pounds based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Outdoor Construction Debris Receiving Area Typical Quantity Received: 500 to 1,000 pounds depending on customer Typical Frequency of Receiving: Daily (except Sunday)	Shipping Location(s): Outdoor Construction Debris Receiving Area Typical Quantity Shipped: 70,000 pounds annually Typical Frequency of Shipping: Daily (except Sunday)	Handling Location(s): Outdoor Construction Debris Receiving Area Typical Quantity Handled: 500-1,000 pounds Typical Frequency of Handling: Daily (except Sundays)
Green Waste	Solid, Non-Solid	Site Map Locations: Transfer Station Tipping Floor Typical Quantity: Approximately 19,000 pounds based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Transfer Station Tipping Floor Typical Quantity Received: 50 to 12,000 pounds depending on customer Typical Frequency of Receiving: Daily (except Sunday)	Shipping Location(s): Green Waste Truck Loading Area Typical Quantity Shipped: 80,000 pounds Typical Frequency of Shipping: Daily (except Sunday)	Handling Location(s): Green Waste Typical Quantity Handled: 1,000 pounds Typical Frequency of Handling: Daily (except Sunday)
Universal and Electronic/General Waste	Solid	Site Map Locations: Universal Waste Storage (Recycling Center), Electronic/General Waste (Transfer Station) Typical Quantity: Varies based on customer deliveries Typical Frequency of Storage: Annually	Receiving Location(s): Universal Waste Storage (Recycling Center), Electronic/General Waste (Transfer Station) Typical Quantity Received: Varies based on customer deliveries Typical Frequency of Receiving: Varies	Shipping Location(s): Universal Waste Storage (Recycling Center), Electronic/General Waste (Transfer Station) Typical Quantity Shipped: varies Typical Frequency of Shipping: Weekly or as needed	Handling Location(s): Universal Waste Storage (Recycling Center), Electronic/General Waste (Transfer Station) Typical Quantity Handled: Varies Typical Frequency of Handling: Weekly or as needed
Bed Mattresses	Solid	Site Map Locations: Mattress Area (Transfer Station) Typical Quantity: Varies based on customer deliveries Typical Frequency of Storage: Annually	Receiving Location(s): Mattress Area (Transfer Station) Typical Quantity Received: Varies based on customer deliveries Typical Frequency of Receiving: Varies	Shipping Location(s): Mattress Area (Transfer Station) Typical Quantity Shipped: varies Typical Frequency of Shipping: Weekly or as needed	Handling Location(s): Mattress Area (Transfer Station) Typical Quantity Handled: Varies Typical Frequency of Handling: Weekly or as needed

Material	Physical Characteristics	Storage Location(s); Typical Quantity Stored; and Typical Frequency of Storage	Receiving Location(s); Typical Quantity Received; and Typical Frequency of Receiving	Shipping Location(s); Typical Quantity Shipped; and Typical Frequency of Shipping	Handling Location(s); Typical Quantity Handled; and Typical Frequency of Handling
Plastic Containers	Solid	Site Map Locations: Recycling Center Plastic Storage Bins Typical Quantity: Approximately 86 tons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Recycling Center Plastic Storage Bins Typical Quantity Received: Approximately 86 tons based on demand Typical Frequency of Receiving: Varies	Shipping Location(s): Recycling Center Plastic Storage Bins Typical Quantity Shipped: Approximately 86 tons based on demand Typical Frequency of Shipping: Weekly or as needed	Handling Location(s): Recycling Center Plastic Storage Bins Typical Quantity Handled: Approximately 86 tons based on demand Typical Frequency of Handling: Weekly or as needed
Glass Bottles and Containers (clear, green, brown, and mixed glass)	Solid	Site Map Locations: Recycling Center Glass Storage Bins Typical Quantity: Approximately 299 tons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Recycling Center Glass Storage Bins Typical Quantity Received: Approximately 299 tons based on demand Typical Frequency of Receiving: Varies	Shipping Location(s): Recycling Center Glass Storage Bins Typical Quantity Shipped: Approximately 299 tons based on demand Typical Frequency of Shipping: Weekly or as needed	Handling Location(s): Recycling Center Glass Storage Bins Typical Quantity Handled: Approximately 299 tons based on demand Typical Frequency of Handling: Weekly or as needed
Non-Ferrous Metal (Tin Cans and Scrap Aluminum and Metal)	Solid	Site Map Locations: Recycling Center Aluminum/Tin Can Storage Typical Quantity: Approximately 58 tons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Recycling Center Aluminum/Tin Can Storage Typical Quantity Received: Approximately 58 tons based on demand Typical Frequency of Receiving: Varies	Shipping Location(s): Recycling Center Aluminum/Tin Can Storage Typical Quantity Shipped: Approximately 58 tons based on demand Typical Frequency of Shipping: Weekly or as needed	Handling Location(s): Recycling Center Aluminum/Tin Can Storage Typical Quantity Handled: Approximately 58 tons based on demand Typical Frequency of Handling: Weekly or as needed
Paper	Non Solid	Site Map Locations: Recycling Center - Paper Recycling Area Typical Quantity: Approximately 760 tons based on demand Typical Frequency of Storage: Annually	Receiving Location(s): Recycling Center - Paper Recycling Area Typical Quantity Received: Approximately 760 tons based on demand Typical Frequency of Receiving: Varies	Shipping Location(s): Recycling Center - Paper Recycling Area Typical Quantity Shipped: Approximately 760 tons based on demand Typical Frequency of Shipping: Weekly or as needed	Handling Location(s): Recycling Center - Paper Recycling Area Typical Quantity Handled: Approximately 760 tons based on demand Typical Frequency of Handling: Weekly or as needed

F. POTENTIAL POLLUTANT SOURCES AND ASSESSMENT (GENERAL PERMIT SECTION X.G)

This section presents descriptions of the industrial potential pollutant sources at the Facility. The pollutants likely to be present in industrial stormwater discharge from these areas are included below.

F.1 INDUSTRIAL PROCESSES (GENERAL PERMIT SECTION X.G.1.A)

Presented below are descriptions of the Facility's industrial processes that have the potential for exposure to stormwater discharges.

1. *Location*: Equipment Maintenance Building

   Description of the Industrial Process: Equipment maintenance of trucks and vehicles.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Equipment maintenance is performed inside when feasible. Materials identified above are accumulated until they are transported for recycling or disposal.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: All above ground tanks are double walled and located indoors.

2. *Location*: Temporary Mobile Diesel Fueling Tank

   Description of the Industrial Process: Diesel Fueling of heavy equipment and garbage trucks. Diesel fuel is delivered two to three times weekly to the temporary mobile above ground double walled storage tank, and vehicles fill up daily from the dispensers.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Diesel is added to the tank two to three times weekly by a tanker truck.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The mobile diesel fueling tank is double walled, providing 512 gallons of containment capacity.

3. *Location*: Wash Rack

   Description of the Industrial Process: Biodegradable soap and pressure washers are used for washing the exterior and interior of roll off bins, containers, garbage trucks, street sweepers and other vehicles.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Wash water is contained on the wash pad until discharge to the sanitary sewer is permitted (during dry weather). During wet weather, a drain cover is put over the sanitary sewer grate to prevent discharges per EBMUD permit. The standard operating procedure for cleaning the wash rack area is power spraying water into the grated manhole. Wash water is then filtered through a Contech filter vault prior to discharge to the sanitary sewer. Accumulated debris is shoveled into trash bins.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The wash rack is walled and bermed on all sides and is graded to divert flow to a drain connected to a manhole that contains two valves, one that flows to the storm drain system and one that flows to the sanitary conveyance system. Due to confined space safety concerns, staff no longer enter the manhole to operate the gate valve, therefore the gate valve is always closed to the storm drain and open to the sanitary conveyance system. A spill cover is used to cover the manhole during rain events. When the spill cover is in place, the wash rack area has a containment capacity of 1,800 gallons. Water from the wash rack flows to a media filter treatment vault (Contech Filter) that discharges to the EBMUD sanitary drain after 24 hours of dry weather under permit. No washing occurs during storm events to reduce the potential for unauthorized non-stormwater discharges if the storage capacity is exceeded.

4. *Location*: Used Oil Depot

   Description of the Industrial Process: Public customers bring used oil, small propane tanks, and fire extinguishers to the used oil area for recycling.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Not applicable.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The above ground tank is double walled and stored within a concrete containment area with a roof over the entire area. The

containment area has a secondary containment capacity of 1,100 gallons. There are spill pallets as well to contain any drips or spills from customers.

5. *Location: General Storage Area*

   Description of the Industrial Process: Transfer station staff use the storage cabinets as general storage of supplies to assist in operations.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Materials and equipment consist of brooms, shovels, rakes, storm drain filter fabric, etc.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Materials and equipment are used at the transfer station.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Not applicable.

6. *Location*: Transfer Station/Tipping Floor and Vactor Truck Wash Area

   Description of the Industrial Process: Customers and City vehicles deposit municipal refuse including green and food waste. The wastes are mixed by a front loader and pushed into long haul trucks for each separate waste for final disposal at the Altamont Landfill. Materials dumped are visually inspected by personnel for hazardous or unauthorized materials and returned to the customer when possible.

   Storm drain and sewer cleanout material is emptied within the vactor truck wash area. The area is bermed and connected to sanitary sewer via trench drains in front and behind the vactor truck wash area.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Municipal waste is disposed at a landfill. Green waste is hauled off-site and recycled into compost.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The tipping floor building is designed to cover the refuse material pile and uses misters to provide dust suppression.

7. *Location*: Outdoor Construction Debris Receiving Area

   Description of the Industrial Process: The construction debris area receives debris from municipal customers. It is mixed by a front loader and pushed into long haul trucks for transport to a City-contracted recycling facility. Materials dumped are visually inspected by personnel for hazardous or unauthorized materials and returned to the customer when possible. A water cannon is used when necessary for dust suppression on construction debris. The water cannon is not used during rain events.

Treated wood, concrete and miscellaneous debris are stored in bins.

Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Construction debris is disposed of at a landfill or recycled.

If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The trench drain (TD-4) directs flow to a 21,000-gallon aboveground storage tank. Containment capacity in this area exceeds the 85th percentile 24-hour design storm as defined in Section X.H.6.a.i. A second 21,000-gallon tank is present for additional storage; water is transferred manually between the two tanks on an as-needed basis using a sump pump. An additional 21,000-gallon storage tank may be brought on-site as needed for large storm events. Wastewater from the tank(s) discharges into EBMUD sanitary sewer after 24 hours of dry weather under permit. The bins are covered with a tarp when not in use and during rain events.

8. *Location*: Universal Waste and General/Electronic Waste Storage Container

   Description of the Industrial Process: Universal Waste is brought by the Public for recycling by a vendor. It is stored to prevent breakage, but sometimes materials are already delivered damaged. Personnel assist the public to prevent breakage. Electronic/General Waste is brought in by customers at the Transfer Station and temporarily stored inside a 20' shipping container.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Universal waste consists of batteries and lamps. Electronic and Universal Waste include televisions, stereo equipment, computers, tires, mattresses, microwaves, refrigerators and other appliances. Estimated quantities are listed in Section E.

   Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Universal waste is hauled off-site for recycling.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Materials are stored in in 20' shipping containers and covered with tarps when not in use and during storm events.

9. *Location*: Container/Bin/Tire Storage Area/Container Maintenance Building/Truck Parking

   Description of the Industrial Process: The replacement containers for recycling and municipal waste, debris boxes, roll off containers and used tires are stored. As well as occasional truck and heavy equipment parking.

   Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Plastic garbage bins, debris boxes, large roll off containers

and tires are stored in the northern end (Drainage Area 1) and near the Administration Building of the Transfer Station. Additionally, truck and heavy equipment parking is located adjacent to the employee parking area.

Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Not applicable.

10. *Location*: Mattress Area

    Description of the Industrial Process: Used mattresses are brought in by customers for disposal and stored under canopy until transported offsite for final disposal.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Bed mattresses composing of cloth, wood and springs and cushioning.

    Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Not applicable.

11. *Location*: Drum Storage and Diesel Exhaust Fuel (DEF) Area

    Description of the Industrial Process: Empty 55-gallon drums and DEF area stored in this area under cover and within secondary containment. Drums containing liquids (motor oil and hydraulic oil) are stored on secondary containment structures under cover. The drums are used as needed along with DEF to fuel heavy equipment.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Refer to Section E.

    Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Not applicable.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The covered containment berm has a capacity of approximately 1,070 gallons.

12. *Location*: Recycling Center – Recycling Commingled Container Storage

    Description of the Industrial Process: Recycling trucks deliver loads of mixed plastic containers daily to the recycling center container storage area where they are loaded and sorted on conveyor belts. Front-end loaders assist in managing the stockpile.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Plastic containers that previously contained milk, juices, water, etc. Estimated quantities are listed in Section E.

    Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Material is sorted and recycled by outside vendor.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: A valley gutter in front of the bins contains a sump and pump, to capture stormwater. Stormwater is pumped into two 4,000-gallon tanks

when the water level activates a float switch in the sump. An additional 5,400 gallon tank is used to transfer stormwater from the 4,000 gallon tanks for storage. Containment capacity in this area exceeds the 85th percentile 24-hour design storm as defined in Section X.H.6.a.i. Additional storage tanks may be brought on-site as needed for large storm events. When the water level in the tanks is too high, float switches turn off the sump pump to prevent overfill. Should this occur, a 6-inch berm surrounding the Commingled Container Storage Area provides an additional 2,650 gallons of containment capacity. Collected water is discharged after 24 hours of dry weather into EBMUD sanitary sewer under permit.

13. *Location*: Recycling Center – Customer Recyclable Area

    Description of the Industrial Process: The public primarily brings plastics, glass, aluminum containers, aluminum foil/cans/scrap metal, steel scrap metal, pet food cans (bi-metal), cardboard, books and non-ferrous metal, glass bottles, for recycling through a buyback program. There are buckets provided for the Public to drain any liquids from their containers before having them loaded onto a conveyor belt for sorting and baling.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Plastic, glass bottles, aluminum and can containers that previously contained milk, juices, soft-drinks, beer, wine, liquor, processed foods, etc. Estimated quantities are listed in Section E.

    Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Material is sorted and recycled by outside vendor.

14. *Location*: Recycling Center – Paper Recycling Warehouse

    Description of the Industrial Process: Paper and cardboard for recycling are baled and stored for transfer to another recycler.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Paper and cardboard. Estimated quantities are listed in Section E.

    Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Material is sorted for recycling in a warehouse and recycled by outside vendor.

15. *Location*: Recycling Center – Glass Sorting Area

    Description of the Industrial Process: Glass is processed in the glass recycling warehouse and place on conveyor belts and sorted by color and dropped into bins and stored for transfer to another recycler.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Different colored glass (i.e., clear, brown, green, blue). Estimated quantities are listed in Section E.

Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Material is sorted for recycling by outside vendor.

If applicable, describe the areas protected by containment structures and the corresponding containment capacity: A valley gutter in front of the bins contains a sump and pump, to capture stormwater.

Stormwater is pumped into two 4,000-gallon interconnected tanks when the water level activates a float switch in the sump. An additional 5,400 gallon tank is used to transfer stormwater from the 4,000 gallon tanks for storage. Containment capacity in this area exceeds the 85th percentile 24-hour design storm as defined in Section X.H.6.a.i. Additional storage tanks may be brought on-site as needed for large storm events. When the water level in the tanks is too high, float switches turn off the sump pump to prevent overfill. Collected water is discharged after 24 hours of dry weather into EBMUD sanitary sewer under permit.

16. *Location*: Recycling Center – Universal Waste Storage and Fat Oil and Grease (FOG)

    Description of the Industrial Process: Universal Waste is brought by the Public for recycling by a vendor. It is stored to prevent breakage, but sometimes materials are already delivered damaged. Personnel assist the public to prevent breakage. Public customers bring used cooking grease, batteries (alkaline, automotive and lithium), lamp ballast, fluorescent light bulbs, and appliances for recycling.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Universal waste consists of small metal appliances include toasters, toaster ovens, blenders, fans, etc. (no microwaves), and large appliances (water heaters, stoves, washers, dryers, dishwashers and furnaces [no refrigerators]). The quantity of materials varies. Estimated quantities are listed in Section E.

    Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Materials are stored until they are recycled by outside vendor.

17. *Location*: Ecology Center – Truck Parking Area

    Description of the Industrial Process: The Ecology Center operates trucks that pick up cans, bottles, newsprint, mixed paper, plastic, and cardboard throughout the City of Berkeley. After pickup, the trucks deposit the materials at the Recycling Center. Trucks are parked in a designated area with minor maintenance performed by outside contractor Mobile Fleet Care in the Recycling Center. Major truck maintenance is performed offsite.

    Description of the type, characteristics, and quantity of industrial materials used in or resulting from the process: Oil, hydraulic and transmission fluids are changed onsite. Estimated quantities are listed in Section E.

Description of the manufacturing, cleaning, maintenance, recycling, disposal or other activities related to the process: Used fluids and filters generated from scheduled maintenance are temporarily stored in the cargo bin for scheduled pickup by outside vendor for proper disposal or removed offsite by Mobile Fleet Care after maintenance work. Large plastic containers are placed under the hydraulic lines of the trucks when parked to contain any leaks. Leaks and/or spills are cleaned up immediately and disposed of properly.

F.2 Materials Handling and Storage (General Permit Section X.G.1.b)

Presented below are descriptions of the Facility's material handling and storage areas that have the potential for exposure to stormwater.

1. *Location*: Equipment Maintenance Building

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Materials used in the Equipment Maintenance Building include liquids such as oil, solvents, antifreeze, grease, and hydraulic fluid. Adjacent to the northern side of the building is catch basin CB-3A that is connected to CB-3. Stormwater from these catch basins flow to the Contech filter. The approximate quantities of materials handled or stored at the Equipment Maintenance Building are included in Section E. Spills or leaks from equipment such as end loaders and ancillary mobile equipment may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Materials identified above are accumulated indoors in secondary containment until they are transported for recycling or disposal.

2. *Location*: Temporary Mobile Diesel Fueling Tank

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: The material handled at the temporary mobile fueling station is diesel. Garbage trucks and heavy equipment fuel at the area and vendor replenish the storage tank when the fuel is low. The approximate quantities of materials handled or stored at the Equipment Maintenance Building are included in Section E. Spills and leaks can occur during the fueling process. Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.2, above.

3. *Location*: Wash Rack

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Quantities of biodegradable soap are listed in Section E. Spills or leaks from the biodegradable soap above ground storage tank may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.3, above.

4. *Location*: Used Oil Depot

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Used oil is put into a double walled above ground tank and stored within a concrete containment area with a roof over the entire area. There are spill pallets as well to contain any drips or spills from customers. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.4, above.

5. *Location*: General Storage Area

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Transfer station staff use equipment (shovels, push brooms, rakes, etc.) to aid operations. There is low potential for spills or leaks during equipment handling, however vehicles and the sweeper are parked in this area. Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Materials are stored within a concrete containment area with an awning or in the general materials shed.

6. *Location*: Transfer Station /Tipping Floor and Vactor Truck Wash Area

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: There are no significant materials used in the receiving of municipal and green waste, however, there is sediment and debris that is generated from the process. On the tipping floor, customers and City vehicles dump municipal refuse where it is mixed by a front loader and pushed into long haul trucks for final disposal at the Altamont Landfill. Green waste is also dumped on the tipping floor and sent offsite for composting. Materials dumped are visually inspected by personnel for hazardous or

unauthorized materials and returned to the customer when possible. The vactor truck wash area is located at the north-east corner of the transfer station/tipping floor. The area is bermed and connected to sanitary sewer via trench drains in front and behind the vactor truck wash area. Material collected from vactor trucks are emptied within the bermed area. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill or leak response supplies are located at the Facility. Spill and leak response and prevention BMPs are described in Section G.

If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See section F.1.6, above.

7. *Location*: Outdoor Construction Debris Receiving Area

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Customers drop off construction debris in the receiving area. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill or leak response supplies are located at the Facility. Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.7, above.

8. *Location*: Universal Waste and General Electronic/Electronic General Waste Storage

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Universal Waste is brought by the Public for recycling by a vendor. It is stored to prevent breakage, but sometimes materials are already delivered damaged. These are stored in a cardboard box. Personnel assist the public to prevent breakage. The quantities vary. Spills or leaks may occur during material handling Spill and leak response and prevention BMPs are described in Section G.

   If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.8, above.

9. *Location*: Container / Bin Storage Area/Container Maintenance Building/Truck Parking

   Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: The replacement containers for recycling and municipal waste, debris boxes, roll off containers and used tires are stored. There are no hazardous materials or spill risk associated with this storage area. However, there could be leaks from trucks delivering the bins and roll off containers and parking and residual liquid in the bins/containers. Spill and leak response and prevention BMPs are described in Section G.

If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Some of the bins are stored in a warehouse. The bins stored outdoors are covered when not in use and during storm events.

10. *Location*: Mattress Area

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Spent and old mattresses are stacked up under canopy awaiting for final disposal. Quantities of bins are not applicable. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Not applicable.

11. *Location*: Drum Storage and Diesel Exhaust Fuel (DEF) Area

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Motor oil, hydraulic fluid and DEF are stored in this area to service trucks and heavy equipment. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See section F.1.11, above.

12. *Location*: Recycling Center – Recycling - Commingled Container Storage

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: The recycling trucks deliver loads of mixed containers daily to the recycling center container storage area where they are loaded and sorted on conveyor belts. The plastic containers at times contain residual liquids from soft drinks, water, milk, juices, etc. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.12, above.

13. *Location*: Recycling Center - Customer Recyclable Area

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: The Public primarily brings plastic, glass and aluminum containers for recycling through a buyback program. There are buckets provided for the Public to drain any liquids from their containers before having them loaded onto a conveyor belt for sorting and baling. The quantities are variable based on customer usage. Spills or

leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Materials other than containers such as scrap metal are stored in an open top metal container with a cover. These containers are stored under canopy. Furthermore, a berm has been installed at the Customer Recyclable Area to mitigate stormwater run-on originating from the central section of the Recycling Center.

14. *Location*: Recycling Center - Paper Recycling Warehouse

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Paper and cardboard for recycling are baled and staged for transfer to another recycler. The paper and cardboard are loaded onto a truck from a ramp in the alley way behind the warehouse for offsite recycling. Loose material falls on the ground in this area from loading activities and can create paper debris outdoors. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Paper recycling activities are done within a warehouse building.

15. *Location*: Recycling Center – Glass Sorting Area

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Glass is sorted and placed on conveyor belts based on color and dropped into bins. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: See Section F.1.15, above.

16. *Location*: Recycling Center – Universal Waste Storage and FOG

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Materials are brought to the area by customers and place into appropriate bins or on the ground. Appliances are put into a bin. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

If applicable, describe the areas protected by containment structures and the corresponding containment capacity: Not applicable.

17. *Location*: Ecology Center – Truck Parking Area

    Description of the type, characteristics, and quantity of materials handled or stored; the shipping, receiving, and loading procedures; and the spill or leak prevention and response procedures: Cans, bottles, newsprint, mixed paper, plastic, and cardboard are deposited at their designated area at the Recycling Center after pickup. Trucks are parked in a designated area with minor maintenance performed by outside contractor Mobile Fleet Care at the Ecology Center. All major truck maintenance are performed offsite. The approximate quantities of materials handled or stored are included in Section E. Spills or leaks may occur during material handling. Spill and leak response and prevention BMPs are described in Section G.

    If applicable, describe the areas protected by containment structures and the corresponding containment capacity: The Ecology Center storage shed (a cargo bin) has drums and containers to store used and unused oil, lubricants, hydraulic fluid, and absorbents. These materials are stored in quart size containers and 55 gallon drums. Used fluids and absorbents are disposed of offsite by vendor.

F.3 Dust & Particulate Generating Activities (General Permit Section X.G.1.c)

Presented below are descriptions of the Facility's dust and particulate generating activities that have the potential for exposure to stormwater or authorized non-stormwater discharges. BMPs to address dust and particulates are described in Section G.

1. *Industrial activity*: Tipping Floor Building

   Discharge location(s): Dust within the tipping floor is managed by an overhead misting system.

   Source Type: Municipal and green waste.

   Description of the physical characteristics of the dust and particulate pollutants: During offloading of municipal and green waste dust and particulates may be generated. In general, the particulates include sand, dirt and particulates from debris.

2. *Industrial activity*: Truck Parking Area and Paved Access Roads

   Discharge location(s): The primary discharge points would be DP-1A, DP-2A, DP-3A, and DP-4 from Drainage Areas 1, 2, 3, and 4. Dust from the paved access road in Drainage Area 10 would primarily be discharged to EBMUD sanitary sewer after 24 hours of dry weather, or discharge via DP-1A.

   Source Type: Dirt from vehicular traffic and construction debris

Description of the physical characteristics of the dust and particulate pollutants: Dust may be observed from trucks driving across the site and tracking from tipping floor and outside construction debris receiving area.

3. *Industrial activity*: Construction Debris Receiving Area

   Discharge location(s): The primary discharge point would be EBMUD sanitary sewer from trench drains in Drainage Area 10. If dust were to migrate, discharges could occur at DP-1A, DP-2A and DP-3A.

   Source Type: Dirt and dust

   Description of the physical characteristics of the dust and particulate pollutants: Dust and particulates may accumulate near the construction debris before it is loaded into the long-haul trucks.

4. *Industrial activity*: Glass Sorting Area

   Discharge location(s): DP-5A, DP-9 and EBMUD sanitary sewer

   Source Type: Glass dust from broken glass during sorting into storage bins

   Description of the physical characteristics of the dust and particulate pollutants: Dust and glass particles may accumulate near the glass sorting area due to broken glass and tracking of particulates from loading activities onto paved access roads.

F.4 Potential for Significant Spills and Leaks (General Permit Section X.G.1.d.i)

Presented below are descriptions of the Facility's areas where spills and leaks can occur.

Location Where Spills and Leaks Can Likely Occur	General Description of Potential for Spills and Leaks
Equipment Maintenance	Potential for spills of oil, solvents and antifreeze from leaks during material transfer and usage
Used Oil Above Ground Tank	Potential for spills during material transfer, spills, leaks, or tank rupture
Universal Waste and General/ Electronic Waste Storage and FOG	Potential for spills during material transfer, spills and leaks
Used Oil Depot Storage Area	Potential for spills during material transfer, spills and leaks
Employee Parking Areas	Leaks from vehicles, ruptured hydraulic fuel lines, drips from motors
Mattress Area	Leaks from vehicles, ruptured hydraulic fuel lines, drips from motors
Temporary Mobile Diesel Fueling Tank	Spills during material transfer and leaks during fueling
Drum and DEF Storage	Spills during material transfer and leaks during fueling
Equipment Maintenance Building	Leaks from vehicles, ruptured hydraulic fuel lines, drips from motors and overall spills
General Storage Area	Leaks from vehicles
Customer Buyback Area	Spills of residual fluids during transfer into buckets and recycling bins, automobile motor oil, hydraulic and anti-freeze leaks
Customer Recyclable Area	Spills of residual fluids during transfer of material into bins, automobile motor oil, hydraulic and anti-freeze leaks

Location Where Spills and Leaks Can Likely Occur	General Description of Potential for Spills and Leaks
Commingled Container Storage and Glass Sorting Areas	Leaks from residual fluids in plastic containers during material handing
Truck Parking Areas at the Transfer Station and Ecology Center	Leaks from residual fluids in green waste, ruptured hydraulic fuel lines, drips from motors
Container/Bin/Tire Storage Area and Container Maintenance Building	Leaks from residual fluids in bins if not properly cleaned, vehicle leaks from motors, anti-freeze and hydraulic hoses
Construction Debris Receiving and Transfer Station	Leaks from ruptured hydraulic fuel lines, drips from motors

F.5 SIGNIFICANT SPILLS AND LEAKS (GENERAL PERMIT SECTION X.G.1.D.II AND III)

No industrial materials have spilled or leaked in significant quantities and discharged from the Facility's stormwater conveyance system or had the potential to discharge within the previous five-year period. However, should spills or leaks identified in 40 Code of Federal Regulations section 302 that have discharged from the Facility's stormwater conveyance system as reported on U.S. EPA Form R, as well as oil and hazardous substances discharges in excess of reportable quantities (40 C.F.R §§110, 117, and 302) occur, the information would be included in the table below.

Describe Material leaked or spilled	Location of Spill or Leak & Corresponding Drainage Area	Describe whether the spill or leak discharged or had the potential to discharge from Facility	Describe spill (Include characteristics, quantity spilled/leaked, and quantity discharged as appropriate)	Describe cleanup or remedial actions	Describe preventative measures taken to avoid reoccurrence
None	None	None	None	None	None

F.6 AUTHORIZED NON-STORMWATER DISCHARGE(S) (GENERAL PERMIT SECTION X.G.1.E.I AND II, AND IV)

Presented below is a summary of potential authorized NSWD at the Facility. If there is no potential for authorized non-stormwater discharges, identify "none" under the first line below.

1. *Description of non-stormwater discharge*: Fire Hydrant Testing

   Frequency of Discharge: Annually during fire hydrant flow testing.

   Source(s) of non-stormwater discharge: Fire hydrant water near Paper Recycling Building near DP-9 and TD-3 and TD-4.

Quantity of non-stormwater discharge (per typical event, annually, etc.): Fire hydrants are rated for 1,000 gallons per minute flow rate. Flow tests typically last between 2 to 4 minutes, totaling 2,000-4,000 gallons of flow.

Characteristics of discharge: Potable water.

Associated drainage area and/or discharge point(s): Drainage Areas 9 and 10.

Authorized by General Permit? Yes

Description of BMPs to prevent or reduce contact of non-stormwater discharges with significant materials or equipment: Refer to Section G.

F.7 Unauthorized Non-Stormwater Discharge(s) (General Permit Section X.G.1.e.iii)

Presented below is a summary of potential unauthorized NSWDs at the Facility. BMPs to eliminate potential unauthorized NSWDs are described in Sections G and L.

1. Area*: Recycling Center* – Commingled Container Storage Area

   Potential Unauthorized Discharge: Liquid from recycling trucks, stockpiled plastic and potential equipment leaks from front-end loader.
2. *Area*: Recycling Center – Customer Recyclable Area

   Potential Unauthorized Discharge: Liquid from waste storage bins.
3. *Area*: Recycling Center – Glass Sorting Area

   Potential Unauthorized Discharge: Residual liquid from broken glass.
4. *Area*: Construction Debris Area

   Potential Unauthorized Discharge: Dust Control Water
5. *Area*: Temporary Mobile Diesel Fueling Area

   Potential Unauthorized Discharge: Diesel fuel.
6. *Area*: Used Oil Depot (Transfer Station) and FOG Storage Area (Recycling Center)

   Potential Unauthorized Discharge: Motor Oil and FOG
7. *Area*: Equipment Maintenance Building

   Potential Unauthorized Discharge: Oils, solvents, greases, antifreeze, transmission fluids, hydraulic fluid, and batteries.
8. *Area*: Wash Rack

   Potential Unauthorized Discharge: Rinsate and biodegradable soap from washing vehicles.
9. *Area*: Truck Parking Areas

   Potential Unauthorized Discharge: Drips from engines and green waste in haul trucks.

10. *Area*: Transfer Station/Tipping Floor and Vactor Truck Wash Area

    Potential Unauthorized Discharge: Dust control water, vactor truck material, and green waste liquids.

11. *Area*: Container/Bin Storage Areas/Container Maintenance Building/Truck Parking

    Potential Unauthorized Discharge: Residual liquids and oil and hydraulic leaks from parked trucks.

F.8 Erodible Surfaces (General Permit Section X.G.1.F)

Presented below are descriptions of the Facility locations where erosion may be caused by industrial activity, contact with stormwater, authorized and unauthorized NSWDs, or run-on from areas surrounding the Facility.

12. *Description of Area*: A small unpaved area measuring approximately 50 by 50 feet is present in Drainage Area 1 where the paving has deteriorated. The area is relatively flat and has low potential for erosion. This area is a low-traffic area that is used for container storage.

    Description of Potential Erosion: No erosion observed to date.

    Associated Drainage Area(s): Drainage Area 1.

    Describe Sediment or Erosion Control BMP(s): Stormwater in this area flows to trench drain TD-1 and CB-1, which are protected by filter fabric and sediment control wattles.

F.9 Assessment of Potential Pollutant Sources (General Permit Section X.G.2)

The General Permit requires a narrative assessment of all areas of industrial activity with potential industrial pollutant sources. The narrative source assessment with items that must be considered is included below or the applicable section of the SWPPP is referenced within the table.

Assessment Requirement	Narrative Description of Assessment
Areas of the Facility with likely sources of pollutants in industrial stormwater discharges and authorized NSWDs (Section X.G.2.a.i)	Areas of the Facility with likely sources of pollutants are described within Section F (Potential Pollutant Source) above.
Pollutants likely to be present in industrial stormwater discharges and authorized NSWDs (Section X.G.2.a.ii)	Industrial materials that may come in contact with stormwater are presented in Section E of this SWPPP. BMPs implemented at the Facility are presented in Sections G, H, I, J, and L. These BMPs are implemented at the Facility to minimize the potential for off-site discharge of the pollutants identified in Sections E and F.

Assessment Requirement	Narrative Description of Assessment
Approximate quantity, physical characteristics (e.g., liquid, powder, solid, etc.) and locations of each industrial material handled, produced, stored, recycled, or disposed (Section X.G.2.a.iii)	Quantities, locations, and physical characteristics for locations of industrial materials handled, produced, stored, recycled, or disposed are included within this SWPPP under Section E, List of Industrial Materials.
The degree to which the pollutants associated with those materials may be exposed to, and mobilized by contact with, stormwater (Section X.G.2.a.iv)	Pollutants associated with industrial materials and processes are included in Sections E and F. BMPs to minimize exposure and/or mobilization of pollutants are included in Sections G, H, I, J, and L of this SWPPP.
The direct and indirect pathways by which pollutants may be exposed to stormwater or authorized NSWDs (Section X.G.2.a.v)	The pathways to which pollutants associated with industrial materials may be exposed to, and mobilized by contact with stormwater is identified in Sections F.1 to F.8.
All sampling, visual observation, and inspection records (Section X.G.2.a.vi)	The MIP, Section M, describes the assessment of sampling, visual observation, and inspection records. Additional pollutant sources observed will be included in the SWPPP.
The effectiveness of existing BMPs to reduce or prevent pollutants in industrial stormwater discharges and authorized NSWDs (Section X.G.2.a.vii)	The existing BMPs implemented at the Facility to reduce or prevent pollutants in industrial stormwater discharges and authorized NSWDs are presented in Sections G, H, I, J, and L of this SWPPP. The BMPs described in this SWPPP are intended to minimize the potential for off-site discharge of pollutants associated with the industrial activities. The MIP, Section M of this SWPPP, describes the assessment of the BMPs. If additional BMPs are necessary based on visual observations or analytical results, additional BMPs will be implemented and the SWPPP will be revised as necessary.
The estimated effectiveness of implementing, to the extent feasible, minimum BMPs to reduce or prevent pollutants in industrial stormwater discharges and authorized NSWDs (Section X.G.2.a.viii)	The estimated effectiveness of implementing, to the extent feasible, minimum BMPs to reduce or prevent pollutants in industrial stormwater discharges and authorized NSWDs is presented in Sections G, H, I, and J of this SWPPP. Section H includes an employee training program with initial and periodic refresher trainings for Facility employees. The training program is intended to support consistent and effective implementation of the BMPs.
The identification of industrial pollutants related to the receiving waters with 303(d) listed impairments identified in Appendix 3 or approved TMDLs that may be causing or contributing to an exceedance of a water quality standard in the receiving waters (Section X.G.2.a.ix)	The Facility directly discharges to Central San Francisco Bay via the City's MS4 and is 303(d) listed as impaired for the following parameters. As described in Appendix 3 of the General Permit, invasive species, mercury, and Polycyclic Aromatic Hydrocarbons (PAHs) are described as typically not associated with industrial stormwater. • Chlordane • DDT • Dieldrin • Dioxins • Furans • Invasive Species • Mercury • PCBs and includes dioxin like compounds • Polybrominated diphenyl ethers (PBDEs) • Selenium • Mercury (sediment) • PAHs (sediment) In addition, the 10-digit Hydrologic Unit Code (HUC 10) watershed downstream of the Facility was evaluated based on email correspondence with the Water Board. These constituents are:

Assessment Requirement	Narrative Description of Assessment
	• Dacthal • Diazinon • E. Coli and Enterococcus • Copper • Zinc • Temperature Based on our review of industrial materials at the Transfer Station, copper has been added to the sampling list summarized in Section M.5.C based on this pollutant source assessment. Zinc is also monitored per the requirements for the Facility SIC code.
Identify any areas where minimum BMPs described in General Permit Subsection H.1 will not adequately reduce or prevent pollutants in stormwater discharges in compliance with section V.A. Identify advanced BMPs for those areas. (Section X.G.2.b)	Minimum BMPs are included in Sections G, H, I, and J of this SWPPP. Advanced BMPs are included in this SWPPP under Section J.
Identify drainage areas with no exposure to industrial activities and materials in accordance with the definitions in General Permit Section XVII (Section X.G.2.c)	Not Applicable
Identify additional parameters that indicate the presence of pollutants in industrial stormwater discharges (Section X.G.2.d)	No additional indicator parameters were added based on this assessment as shown in Section M, MIP.

G. MINIMUM BMPS (GENERAL PERMIT SECTION X.H)

The Facility is required to implement and maintain the minimum BMPs described in Section X.H.1 of the General Permit to the extent feasible. The extent feasible requirement reflects best industry practice considering technological availability and economic practicability and achievability. Presented below are the minimum BMP requirements, areas where the minimum BMP requirement is applicable, and site-specific BMP description used to comply with the minimum BMP.

Minimum BMP Requirement (Reference to Section in General Permit)	Area(s) Implemented	Site Specific BMP Description	Does Minimum or Alternate BMP Reflect Best Industry Practice? (Yes/No)	Actions Performed in lieu of BMP (Requirement from Section X.H.4.c)
Observe all outdoor areas associated with industrial activity. Any identified debris, waste, spills, tracked materials, or leaked materials shall be cleaned and disposed of properly (Section X.H.1.a.i).	Entire Facility	• Employees are trained to monitor their areas for spills and leaks from equipment and industrial areas that may result in spills or leaks. • The Facility either stores under cover or removes from the Facility all abandoned or broken equipment or materials no longer considered for future use that have the potential to contribute to pollutant loading. • Inspections are performed in accordance with the BMP Summary Table (Section L) and MIP (Section M). • Absorbent materials are available in event of a spill or leak.	Yes	Not Applicable
Minimize or prevent material tracking (Section X.H.1.a.ii).	Entire Facility	• Dust suppression water is used as necessary. • Trucks are washed daily in the designated Wash Rack. • Sweeping is performed and daily sweeping logs are kept.	Yes	Not Applicable
Minimize dust generated from industrial materials or activities (Section X.H.1.a.iii).	Tipping Floor / Transfer Station, Green Waste Area, Glass Sorting Area	• Dust suppression water is used as necessary in areas that discharge to the sanitary sewer. • Sweeping is performed daily.	Yes	Not Applicable
Ensure that all facility areas impacted by rinse/wash waters are cleaned as soon as possible (Section X.H.1.a.iv). Prevent disposal of any rinse/wash waters or industrial materials into the stormwater conveyance system (Section X.H.1.a.vii).	Wash Rack Area, Vactor Truck Wash Area	• The wash rack is walled and bermed on all sides and is graded to divert flow to a drain. During dry weather, the drain flows to the sanitary sewer. The gate valve is always closed to the storm and open to the sanitary conveyance system. Due to confined space safety concerns, staff no longer enter the manhole to operate the gate valve. Instead, a spill cover is used to cover the manhole during rain events. Water from the wash rack flows to a media filter treatment vault that discharges to the sanitary drain. No washing occurs during storm events to prevent wash water from accumulating within the bermed area to reduce the potential for non-stormwater discharges. The standard operating procedure for cleaning the wash rack area is power spraying water into the grated manhole with trash picked up and deposited in a bin. • Wash water flows to a drain that directs stormwater to a Contech filter vault and then discharges to the sanitary sewer. During storm events, no washing is performed in the wash rack. • Vactor truck waste material is emptied within a berm area and connected to sanitary sewer with a trench drain. • Spilled vactor truck material outside the berm is cleaned up immediately and deposited within the bermed area. • Garbage truck parking in the vicinity of the wash rack has been moved to Drainage Area 10 to prevent potential washed trucks from dripping wash water, oil and garbage residual liquids into catch basin CB-3 and CB-3A.	Yes	Not Applicable
Cover all stored industrial materials that can be readily mobilized by contact with stormwater (Section X.H.1.a.v).	Transfer Station/ Tipping floor (Green Waste), Area, Equipment Maintenance Building, Mattress Area, Paper Recycling Warehouse, General Storage, Used Oil Depot, Universal Waste, Electronic Waste, and Customer Recyclable Area, Bin Storage Areas (Active Bins Only)	• The Facility either stores under cover or removes from the Facility all abandoned or broken equipment or materials no longer considered for future use that have the potential to contribute to pollutant loading. • Materials are stored indoors, under canopy, in covered containers, and under overhangs when feasible, otherwise materials are stored or covered with tarps if they are to remain outdoors.	Yes	Outdoor construction debris receiving area (Transfer Station) and glass sorting and comingle storage areas (Recycling Center) are not covered. A portion of stormwater is contained in these areas with trench drains TD-3 and TD-4 for construction debris receiving area and valley trenches at the glass sorting and commingled container storage areas. Stormwater collected from these areas are stored in above ground tanks for discharge into EBMUD sanitary sewer after 24 hours of dry weather. Discharge to sanitary sewer is not allowed during rain per EBMUD permit.

Minimum BMP Requirement (Reference to Section in General Permit)	Area(s) Implemented	Site Specific BMP Description	Does Minimum or Alternate BMP Reflect Best Industry Practice? (Yes/No)	Actions Performed in lieu of BMP (Requirement from Section X.H.4.c)
Contain all stored non-solid industrial materials or wastes (e.g., particulates, powders, shredded paper, etc.) that can be transported or dispersed by wind or contact with stormwater (Section X.H.1.a.vi).	Paper Recycling Warehouse, Transfer Station/Tipping Floor, Green Waste Receiving Area, Universal Waste	• Materials are stored indoors/undercover when practicable.	Yes	Not Applicable
Minimize stormwater discharges from non-industrial areas (e.g., parking lots) that contact industrial areas of the facility (Section X.H.1.a.viii).	Employee Parking Areas, Administrative Building, Drainage Area 11	• These areas do not come in contact with industrial areas of the facility.	Yes	Not Applicable
Minimize authorized NSWDs from non-industrial areas that contact industrial areas of the facility (Section X.H.1.a.ix).	Drainage Areas 9 and 10	• The location of the fire hydrant near DP-9 perimeter of building and not within industrial area. The fire hydrant in Drainage Area 10 is located in an area with a clear flow path to trench drain TD-3 that traverses a paved access road. Fire hydrant water does not contact industrial materials prior to entering the trench drain.	Yes	Not Applicable
Identify all equipment and systems used outdoors that may spill or leak pollutants (Section X.H.1.b.i).	Used Oil Depot, General Storage, Fueling Area, Vehicle Parking Areas, Vehicle Maintenance Building, Wash Rack, Commingled Container Storage, Glass Sorting Area, Container Bin Storage Areas, Customer Recyclable Area, Stormwater Storage Tanks and Associated Equipment	• Areas with the potential for spills or leaks are identified in Section F.4 of this SWPPP.	Yes	Not Applicable
Observe the identified equipment and systems to detect leaks, or identify conditions that may result in the development of leaks (Section X.H.1.b.ii).	Used Oil Depot, General Storage, Fueling Area, Vehicle Parking Areas, Vehicle Maintenance Building, Wash Rack, Commingled Container Storage, Glass Sorting Area, Container Bin Storage Areas, Customer Recyclable Area, Stormwater Storage Tanks and Associated Equipment	• Employees are trained to monitor their areas for spills and leaks from equipment and vehicles areas that may result in spills or leaks. • Inspections are performed in accordance with the BMP Summary Table (Section L) and MIP (Section M). • Above ground tanks and other equipment that may leak are inspected at least monthly to detect leaks and to identify conditions that may result in leaks. • Stormwater storage tanks are inspected after rain events to determine adequate freeboard. Weather is monitored to evaluate storm rainfall amounts and probability. Stormwater in the tanks are discharged to sanitary sewer after 24 hours of dry weather. Stormwater is not allowed to be discharged during rain events per EBMUD permit. Stormwater discharge logs are kept at the TS and CCC.	Yes	Not Applicable

<table>
<tr><th>Minimum BMP Requirement (Reference to Section in General Permit)</th><th>Area(s) Implemented</th><th>Site Specific BMP Description</th><th>Does Minimum or Alternate BMP Reflect Best Industry Practice? (Yes/No)</th><th>Actions Performed in lieu of BMP (Requirement from Section X.H.4.c)</th></tr>
<tr><td>Establish an appropriate schedule for maintenance of identified equipment and systems (Section X.H.1.b.iii).</td><td>Used Oil Depot, Fueling Area, General Storage, Vehicle Maintenance Building, Wash Rack, Vactor Truck Wash Area, Conveyor Belt and Bailer System, Universal Waste, Stormwater Storage Tanks and Associated Equipment, Contech Filters</td><td>• The schedule for equipment and heavy equipment maintenance is based on the manufacturer's specifications. Preventative maintenance is performed at the maintenance building on-site.
• Hazardous materials and used oil recycling follow Transfer Station Spill Prevention Control Countermeasure Plan and or Hazardous Materials Business Plan protocols.
• Stormwater holding tanks sewage pump (Transfer Station) and sumps pumps for the Valley Gutter Drains (Recycling Center) are checked during the rainy season for operability. Trench drains (TD-3 and TD-4) and valley gutters (GSA-VG and CCS-VG) are regularly inspected and maintained.
• A contractor has been hired to perform BMP maintenance of the storm and trench drains and maintain Silt Socks and hydrocarbon wattles.
• Effluent stormwater samples are collected at DP-1A, DP-2A and DP-3A during QSEs with regular inspections at the catch basins (CB-1, CB-2, CB-3, CB-3A, and CB-4) and trench drains (TD-1, TD-2 and TD-2A) to determine filter (wattles and sediment filter fabric) efficiencies and necessity of Contech media filter media change-out.</td><td>Yes</td><td>Not Applicable</td></tr>
<tr><td>Establish procedures for prompt maintenance and repair of equipment, and maintenance of systems when conditions exist that may result in the development of spills or leaks (Section X.H.1.b.iv).</td><td>Used Oil Depot, Fueling Area, General Storage, Vehicle Parking Areas, Vehicle Maintenance Building, Wash Rack, Vactor Truck Wash Area, Conveyor Belt and Bailer System,
Stormwater Holding Tanks and Associated Equipment and Contech Filters</td><td>• Inspections are performed as described in the MIP Section of this SWPPP and BMP Summary Table Section L to identify conditions that may result in spills or leaks. If these conditions are observed, the Facility will have the equipment repaired at the maintenance shop, if feasible, and as soon as practicable.
• The Facility uses preventative maintenance to maintain equipment and to avoid spills and leaks to the extent practicable.
• Vactor truck wash area berm is inspected for openings and failures that would cause breaches during washout.
• Piping for the stormwater holding tanks are checked during the rainy season for leaks.
• Stormwater in the holding tanks are discharged into sanitary sewer after 24 hours of dry weather to prevent conditions that may cause overflow in addition to weather monitoring. Stormwater is not allowed to be discharged during rain events per EBMUD permit. Discharge logs are kept at the TS and the CCC.
• Contech Filters are inspected every six months. Change out of the media in the cartridges is based on percent capacity remaining by visual observations by the vendor. During each inspection, the vaults are cleaned of sediment. Contech Filter vault specifications are shown below. See Appendix D for more details.
<table>
<tr><th>Discharge Location</th><th>Cartridge Size (inches)</th><th>Number of Cartridges</th><th>Media Type</th></tr>
<tr><td>DP-1A</td><td>18</td><td>13</td><td>ZPG</td></tr>
<tr><td>DP-2A</td><td>18</td><td>13</td><td>ZPG</td></tr>
<tr><td>DP-3A</td><td>12</td><td>3</td><td>Perlite</td></tr>
<tr><td>EBMUD Filter from Truck Wash Rack</td><td>18</td><td>4</td><td>ZPG</td></tr>
</table></td><td>Yes</td><td>Not Applicable</td></tr>
<tr><td>Establish procedures and/or controls to minimize spills and leaks (Section X.H.1.c.i).</td><td>Used Oil Depot, General Storage, Vehicle Parking Areas, Fueling Area, Wash Area, Vactor Truck Wash Area, Vehicle Maintenance Building, Customer Recycle Area, Commingled Container Storage, Glass Storage Area, Customer Recyclable Area, Stormwater Holding Tanks and Associated Equipment</td><td>• Petroleum related above ground tanks are double-walled or located on spill containment pallets and covered by an awning.
• Above ground tanks and equipment identified in Section F.4 are inspected in accordance with the MIP, Section M of this SWPPP to identify areas that might result in a spill or leak.
• Employees are trained in accordance with Section H on BMPs used at the site to minimize spills and leaks.
• Vehicle maintenance is performed inside the Vehicle Maintenance Building when feasible and drip pans are used to prevent spills and large absorbent sheets under the vehicles at the EC truck parking area.
• Vactor truck waste material is emptied within a bermed area and connected to sanitary sewer.</td><td>Yes</td><td>Not Applicable</td></tr>
</table>

Minimum BMP Requirement (Reference to Section in General Permit)	Area(s) Implemented	Site Specific BMP Description	Does Minimum or Alternate BMP Reflect Best Industry Practice? (Yes/No)	Actions Performed in lieu of BMP (Requirement from Section X.H.4.c)
Develop and implement spill and leak response procedures to prevent industrial materials from discharging through the stormwater conveyance system. Spilled or leaked industrial materials shall be cleaned promptly and disposed properly (Section X.H.1.c.ii).	Facility Wide	• General spill response procedures include notifications to the Transfer Station Manager and affected employees; orderly evacuation if applicable; containment of spill; using spill kits (i.e., absorbent materials to cleanup up the spill/leak, and proper disposal of the used absorbent and other cleanup materials. • Stormwater in holding tanks are inspected and discharged to sanitary sewer after 24 hours of dry weather to prevent conditions that may cause overflow in addition to weather monitoring. Stormwater is not allowed to be discharge during rain events per EBMUD permit. Stormwater discharge logs are kept at the TS and CCC. • Absorbent sheet BMPs installed under bins at the Customer Recycle Area to capture residual liquids. • Large absorbent sheet BMPs installed under vehicles at the Ecology Center truck parking area.	Yes	Not Applicable
Identify and describe all necessary and appropriate spill and leak response equipment, location(s) of spill and leak response equipment, and spill or leak response equipment maintenance procedures (Section X.H.1.c.iii).	Used Oil Depot, Vehicle Maintenance Building, Vehicle Parking Areas, General Storage, Fueling Area, Commingled Container Area, Green Waste Area, Transfer Station/Tipping Floor, Customer Recyclable Area, Stormwater Holding Tanks and Associated Equipment	• Absorbent materials (hydrocarbon wipes, absorption material and kitty litter) and spill kits are located in strategic locations for proper response. In addition, all City Trucks have spill kits on them in case of spills or leaks. Ecology Center and CCC have spill kits. • To avoid spills/overflow, stormwater in holding tanks are inspected and discharged to sanitary sewer after 24 hours of dry weather to maintain freeboard for the next rain event(s) in addition to weather monitoring. Stormwater is not discharged during rain events. Stormwater discharge logs are kept at the TS and CCC. • Containers/bins in the Customer Recycling Area are regularly inspected for leaks and repaired or replaced as needed. • Large absorbent sheet BMPs installed under vehicles at the Ecology Center truck parking area.	Yes	Not Applicable
Identify and train appropriate spill and leak response personnel (Section X.H.1.c.iv).	Facility Wide	• Employees responsible for spill and leak cleanup are trained in accordance with Section H of this SWPPP.	Yes	Not Applicable
Prevent or minimize handling of industrial materials or wastes that can be readily mobilized by contact with stormwater during a storm event (Section X.H.1.d.i).	Construction Debris and Green Waste Receiving Areas, Transfer Station/Tipping Floor, Vactor Truck Wash Area, Glass Sorting, Commingled Container Storage and Customer Recyclable Areas	• The green waste and vactor truck wash area operations are conducted inside the transfer station/ tipping floor building. • Vactor wash area is bermed and connected to sanitary sewer. • Customer recyclable area is under canopy structure and a berm is installed to prevent run-on. Large absorbent sheets are placed under bins to capture residual liquids. • Large absorbent sheets are placed under the garbage trucks at the Ecology Center to capture leaks and spills.	Yes	Trench drains at the construction debris area and valley gutters at the glass sorting and commingled container storage areas collect stormwater into holding tanks for discharge to sanitary sewer after 24 hours of dry weather. Discharge to sanitary sewer is not allowed during rain per EBMUD permit.
Contain all stored non-solid industrial materials or wastes that can be transported or dispersed by the wind or contact with stormwater (Section X.H.1.d.ii).	Transfer Station/ Tipping floor, Construction Debris Receiving Area, Equipment Maintenance Building, Paper Recycling Warehouse, Glass Bin Storage Area, Universal Waste,	• Materials noted in Section X.H.1.a.v are containerized as appropriate. • Dust control is performed at the transfer station/tipping floor and construction debris receiving area. • Materials are stored indoors or an appropriate structure/container at the transfer station/tipping floor, equipment maintenance building, paper recycling warehouse, universal waste	Yes	Trench drains at the construction debris area and a valley gutter at the glass sorting area collect stormwater into a holding tank for discharge to sanitary sewer after 24 hours of dry weather. Discharge to sanitary sewer is not allowed during rain per EBMUD permit.

Minimum BMP Requirement (Reference to Section in General Permit)	Area(s) Implemented	Site Specific BMP Description	Does Minimum or Alternate BMP Reflect Best Industry Practice? (Yes/No)	Actions Performed in lieu of BMP (Requirement from Section X.H.4.c)
Cover industrial waste disposal containers and industrial material storage containers that contain industrial materials when not in use (Section X.H.1.d.iii).	Used Oil Depot Equipment Maintenance Building, Universal and Electronic Waste, Ecology Center Cargo Bin, Construction Debris, Customer Recyclable, Glass Sorting and Comingle Container Storage Areas	• Used Oil Depot is covered with a roof. • Universal Waste stored in shed. • Electronic Waste stored in cargo bin. • Greases and fluids used for equipment maintenance stored in containers in Equipment Maintenance Building • Used truck fluids stored in Ecology Center cargo bin. • At the Recycling Center, canopy structures are at the customer recyclable, universal waste storage/FOG, recycle bins areas	Yes	Stockpiled construction debris, glass sorting and comingle container areas are too large to cover. Instead trenches and valley gutters have been installed to capture stormwater or spills into stormwater holding tanks. Stormwater from the holding tanks are discharged to sanitary sewer after 24 hours of dry weather. Discharge to sanitary sewer is not allowed during rain per EBMUD permit.
Divert run-on and stormwater generate from within the facility away from all stockpiled materials (Section X.H.1.d.iv).	Customer Recyclable Area, Construction Debris, Glass Sorting and Comingle Container Storage Areas	• Berms have been installed to prevent run-on of stormwater onto the customer recyclable area, commingled container storage area, and the glass sorting area.	Yes	Stockpiled construction debris, glass sorting and comingle container areas are too large to cover. Instead trenches and valley gutters have been installed to capture stormwater or spills into stormwater holding tanks to prevent run-on to other areas of the Transfer Station, Ecology Center and Recycling Center. Stormwater from the holding tanks are discharged to sanitary sewer after 24 hours of dry weather. Discharge to sanitary sewer is not allowed during rain per EBMUD permit.
Clean all spills of industrial materials or wastes that occur during handling in accordance with the spill response procedures (Section X.H.1.d.v).	Use Oil Depot, General Storage, Equipment Maintenance Building, Customer Recyclable Area, Construction Debris, Commingled Container Storage, Stormwater Holding Tanks and Associated Equipment	• Spill or leak prevention response procedures are located at the Facility. • Trench drains and valley gutter drains installed at construction debris, commingled container storage and glass sorting areas to capture spills and stormwater into holding tanks. • To avoid spills/overflow, stormwater in holding tanks are inspected and discharged to sanitary sewer during dry days after rain events to ensure adequate freeboard for the next rain event(s) in addition to weather monitoring. Stored stormwater is discharged after 24 hours of dry weather. Discharge to sanitary sewer is not allowed during rain per EBMUD permit. Stormwater discharge logs are kept at the TS and CCC.	Yes	Not Applicable
Observe and clean as appropriate, any outdoor material or waste handling equipment or containers that can be contaminated by contact with industrial materials or wastes (Section X.H.1.d.vi).	Used Oil Depot, General Storage, Universal Waste, Vactor Truck Wash Area, Stormwater Holding Tanks and Associated Equipment	• Employees are trained to monitor their areas for equipment that may have spills or leaks. • Vactor truck wash area is contained with berms and discharges to sanitary sewer via a trench drain. • Inspections are performed in accordance with the BMP Summary Table (Section L) and MIP (Section M).	Yes	Not Applicable
Implement effective wind erosion controls for areas listed in SWPPP Section F.8 (Section X.H.1.e.i).	Not Applicable.	• The small unpaved area noted in Section F.8 is located on a flat surface in a low traffic area and frequently covered by storage containers. Wind erosion in this area is not anticipated.	Not Applicable	Not Applicable

Minimum BMP Requirement (Reference to Section in General Permit)	Area(s) Implemented	Site Specific BMP Description	Does Minimum or Alternate BMP Reflect Best Industry Practice? (Yes/No)	Actions Performed in lieu of BMP (Requirement from Section X.H.4.c)
Provide effective stabilization for inactive areas, finished slopes, and other erodible areas prior to forecast storm events (Section X.H.1.e.ii).	Not Applicable.	• The small unpaved area noted in Section F.8 is located on a flat surface in a low traffic area at a lower grade than the surrounding pavement. Erosion of this area has not been observed to date. Perimeter sediment controls are in place in this Drainage Area as described below.	Not Applicable	Not Applicable
Maintain effective perimeter controls and stabilize all site entrances and exits to sufficiently control discharges of erodible materials from discharging or being tracked off the site (Section X. H.1.e.iii).	Only a small portion of the northern Facility (Drainage Area 1) has erodible surfaces, however this area is flat and has low traffic activity and is used primarily for storage.	• The northern Facility perimeter has trench drain TD-1 has sediment wattle, BioChar and oil absorbing sock that discharge stormwater to treatment vault DP-1A. The central Facility trench drains TD-2 and TD-2A are lined with filter fabric, that discharge stormwater to treatment vault DP-2A. • The eastern perimeter of the facility is graded away from the perimeter and the eastern perimeter of the facility is curbed. • Where trench drains are not installed, Filtrex wattles are used to trap erodible materials prior to leaving the facility. Sampling locations use BioChar, sediment wattles and oil absorbent sock because there are industrial activities in those areas.	Yes	Not Applicable
Divert run-on and stormwater generated from within the facility away from all erodible materials (Section X.H.1.e.iv).	Not Applicable	• Not Applicable	Not Applicable	Not Applicable
If sediment basins are implemented, ensure compliance with the design storm standards (Section X.H.1.e.v).	No sediment basins at the Facility	• Not Applicable	Not Applicable	Not Applicable

H. EMPLOYEE TRAINING PROGRAM (GENERAL PERMIT SECTION X.H.1.F)

Presented below is the Facility's training program which demonstrates compliance with the minimum BMP requirements listed in General Permit Section X.H.1.f. The table below identifies the Facility personnel required for the training and provides a general description of the trainings provided to personnel.

Facility Title	Responsibilities and Duties	Type of Training	Source of Appropriate Training Manuals and Materials	Training Schedule	Describe how training is documented
Pollution Prevention Team	• BMP Implementation • BMP effectiveness evaluation • Visual Observations • Monitoring Activities	Staff Meetings	• Presentation materials developed internally and/or by consultant	Annual Training	Documents for all completed training classes for personnel are maintained with the hard copy SWPPP files in Appendix B. An example training log is included within the electronic SWPPP in Appendix B.
Operations Manager at the Transfer Station	• Oversee various employee job descriptions, including implementing BMPs to minimize materials from contacting stormwater • Maintaining BMPs that filter stormwater, record keeping • Sweeping and trash pickup program, • Implementing minimum BMPs as described in the SWPPP	Annual training and tailgates	• Environmental Compliance Specialist develops training materials.	Routine and new employees participate in the BMP training program within thirty (30) days of their hiring date or as soon as practicable.	Training records are kept with the supervisors and with the Environmental Compliance Specialist.
Facility staff responsible for BMPs, material handling, and spill cleanup	• BMP Implementation • Material handling • Spill cleanup • Completing BMP inspection forms	Annual training and tailgates	• Environmental Compliance Specialist develops training materials.	Routine and new employees participate in the BMP training program within thirty (30) days of their hiring date or as soon as practicable.	Training records are kept with the supervisors and with the Environmental Compliance Specialist.

I. QUALITY ASSURANCE AND RECORD KEEPING (GENERAL PERMIT SECTION X.H.1.G)

Presented below is the Facility's Quality Assurance and Record Keeping Policies. For reference, the table includes the minimum BMP requirements.

Minimum BMP Requirement (Reference to Section in General Permit)	Facility Specific Quality Assurance and Record Keeping Policy
Develop and implement management procedures to ensure that appropriate staff implements all elements of the SWPPP, including the MIP (Section X.H.1.g.i).	The Facility implements a training program for the pollution prevention team to implement the SWPPP. Additional trainings are described in Section H. Training documents are kept with the SWPPP.
Develop a method of tracking and recording the implementation of BMPs identified in the SWPPP (Section X.H.1.g.ii).	Monthly visual observation and sampling event visual observation forms are completed as required. These forms document BMP deficiencies and actions taken to correct. Inspection BMP forms are completed at different frequencies (i.e. prior to rain events, annually, monthly, weekly and daily).
Maintain the BMP Implementation records, training records, and records related to any spills and clean-up related response activities for a minimum of five (5) years (Section X.H.1.g.iii).	The required records are kept at City's Corporation Yard with the SWPPP files. Records will be maintained for a minimum of five (5) years

J. ADVANCED BMPS (GENERAL PERMIT SECTION X.H.2)

Advanced BMPs include Exposure Minimization, Stormwater Containment and Discharge Reduction, Treatment Control, and Other Advanced BMPs. Exposure minimization BMPs include storm resistant shelters to prevent the contact of stormwater with industrial activities and material. Stormwater Containment and Discharge Reduction BMPs include BMPs that divert, reuse, contain, or reduce the volume of stormwater runoff. Treatment control BMPs include one or more mechanical, chemical, biologic, physical, or any other treatment process technology and is sized to meet the treatment control design storm standard. Advanced BMPs, if any, are summarized below. Advanced BMPs are not required. Advanced BMPs may be evaluated and implemented during Exceedance Response Action (ERA) plan if necessary.

Advanced BMPs that have been installed at the Transfer Station and Recycling Center are described in Section L. Figure 5 shows the flow process diagram of Treatment Control, Stormwater Containment and Discharge Reduction BMPs at the Transfer Station and Recycling Center.

J.1 Exposure Minimization (Section X.H.2.b.i)

See Table L for list of Exposure Minimization BMPs.

J.2 Stormwater Containment and Discharge Reduction BMPs (Section X.H.2.b.ii)

The stormwater containment and discharge reduction BMPs for the Construction Debris Area, Glass Sorting and Commingled Container Storage areas are described in Table L. Contained water is discharged to EBMUD after 24 hours of dry weather under permit; a copy of the EBMUD discharge permit is located in Appendix C. See Section F.1 for details on containment capacities.

J.3 Treatment Control BMPs (Section X.H.2.b.iii)

See Table L for list of Treatment Control BMPs and details in Appendix D.

J.4 Other Advanced BMPs (Section X.H.2.b.iv)

See Table L for list of Other Advanced BMPs.

K. TEMPORARY SUSPENSION OF ACTIVITIES (GENERAL PERMIT SECTION X.H.3)

Presented below are the BMPs associated with temporarily suspending activities for ten (10) or more consecutive calendar days during a reporting year.

K.1 Describe Industrial Activities Temporarily Suspended

Describe Reason for Temporary Suspension	List Facility Stabilization BMPs	Area(s) Implemented	Date Facility is Fully Stabilized	Projected Date when Industrial Activities will Resume
Not applicable at this time	Not applicable at this time	Not applicable at this time	Not applicable at this time	Not applicable at this time

K.2 Include justification for why monitoring is infeasible.

Not applicable at this time.

L. BMP SUMMARY TABLE (GENERAL PERMIT SECTION X.H.4 AND X.H.5)

Presented below is a description of all stormwater BMPs implemented at the Facility for each potential pollutant source. This table meets the General Permit requirement of providing procedures/maintenance instructions for BMP implementation (Section X.H.4.a.v). Operational and maintenance manuals are also referenced by Facility personnel for select equipment.

Potential Pollution Source	Potential Pollutant(s) Reduced by BMP (Section X.H.4.a.i)	BMPs	Frequency of BMP implementation (Section X.H.4.a.ii)	Location of BMP (Section X.H.4.a.iii)	Equipment and Tools for BMP Implementation (Section X.H.4.a.vi)	Frequency for BMP Inspection (Section X.H.4.a.vii)	Individual and/or Position Responsible for Implementing the BMP (Section X.H.4.a.iv)
Transfer Station / Tipping Floor, Vactor Wash, Mattress Areas, Equipment/Vehicle Maintenance Building, Construction Debris Area, Temporary Mobile Diesel Fueling Area, Wash Rack, Truck Parking, General Waste and Electronic Waste Storage, Container Bin Storage Area and Container Maintenance Building	Suspended Solids, Oil and Grease, Chemical Oxygen Demand, metals, pH	1. A regular sweeping program is implemented by a mechanical sweeper. 2. Hand sweeping plus two industrial (Billy Goat) vacuums are used in areas where the mechanical sweeper cannot access. 3. Trucks are regularly washed to prevent tracking. 4. Pavement cleaning include pressure washing with full recovery and regenerative air sweeping to remove solids that remain after the mechanical sweeper has passed. 5. Documented inspections are performed regularly, including monthly observation as part of the General Permit. 6. Routine inspections are performed by employees. If a spill or leak is observed, it is cleaned up as soon as practicable. 7. Trash is picked up. 8. Employees responsible for spill response and BMP implementation are trained (refer to Section H for additional details). 9. The Facility either stores under cover or removes from the Facility all abandoned or broken equipment or materials no longer considered for future use that have the potential to serve as a source of pollutants (Exposure Minimization BMP). 10. Bins and containers are checked for holes and leaks and repaired as needed.	Sweeping program is implemented daily by the mechanical sweeper Hand sweeping is completed monthly, prior to rain events, and as needed. Trucks are washed daily Pavement cleaning including pressure washing is conducted at least annually. Documented inspections of BMPs are completed at least weekly. Inspections of leaks and spills are daily and ongoing. Trash is picked up daily Employees are trained annually, at minimum. Bins and containers are inspected and repaired once a month as needed.	Facility-wide within The Transfer Station	Mechanical sweeper Shovels and Brooms Digital and physical sweeping logs Pressure washer Spill response materials	Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Condition of pavement is inspected annually. Daily inspections occur for operational equipment. Bins and containers ae inspected once a month	1. Solid Waste Supervisor and Solid Waste Workers 2. Solid Waste Supervisor and Solid Waste Workers 3. Solid Waste Driver, Solid Waste Supervisor (Collections), any Supervisory staff present 4. Senior Supervisor and Solid Waste Supervisor 5. Solid Waste Supervisor, Environmental Compliance Specialist, Environmental Consultant (Monthly) (WSP) 6. Solid Waste Worker, Solid Waste Driver, Mechanic 7. Solid Waste Worker 8. Environmental Consultant, WSP (Annual IGP Training), Environmental Health and Safety Officer and Environmental Compliance Specialist (Annual Hazardous Materials and Spill Response Training) 9. Solid Waste Supervisors, Senior Solid Waste Supervisors 10. Soid Wate Supervisors, Senior Solid Waste Supervisors
Recycling Center - Customer Recyclable Area, Recycling Center – Commingled Container Storage Area, Ecology Center – Truck Parking Area, Recycling Center – Glass Bin Area, Paper Recycling Warehouse and Rear Alleyway	Suspended Solids, Oil and Grease, Chemical Oxygen Demand, metals, pH	1. A sweeping program is implemented by the mechanical sweeper. 2. Hand sweeping plus two industrial (Billy Goat) vacuums are used in areas where the mechanical sweeper cannot access, such as under the ramp. 3. Trucks are regularly washed to prevent tracking. 4. Pavement cleaning includes pressure washing with full recovery and regenerative air sweeping to remove solids that remain after the mechanical sweeper has passed. 5. Documented inspections are performed regularly, including monthly observation as part of the General Permit. 6. Routine inspections are performed by employees. If a spill or leak is observed, it is cleaned up as soon as practicable. 7. Trash is picked up. 8. Employees responsible for spill response and BMP implementation are trained (refer to Section H for additional details).	Sweeping program is implemented daily by the mechanical sweeper Hand sweeping is completed monthly, prior to rain events, and as needed. Trucks are washed daily Pavement cleaning including pressure washing is conducted at least annually. Documented inspections of BMPs are completed at least weekly.	Facility-wide within the Community Center Recycling Center	Mechanical sweeper Brooms Digital and physical sweeping logs Pressure washer Spill response materials	Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Condition of pavement is inspected annually. Daily inspections occur for operational equipment. Bins and containers ae inspected once a month	1. Recycling Operations Manager, Recycling Supervisors 2. Recycling Workers 3. Recycling Operations Manager (CCC and EC) 4. Recycling Supervisor (CCC), Recycling Operations Manager (EC) 5. Recycling Supervisors (CCC), Operations Manager (EC) Compliance Specialist, Environmental Consultant (Monthly)

Potential Pollution Source	Potential Pollutant(s) Reduced by BMP (Section X.H.4.a.i)	BMPs	Frequency of BMP implementation (Section X.H.4.a.ii)	Location of BMP (Section X.H.4.a.iii)	Equipment and Tools for BMP Implementation (Section X.H.4.a.vi)	Frequency for BMP Inspection (Section X.H.4.a.vii)	Individual and/or Position Responsible for Implementing the BMP (Section X.H.4.a.iv)
		9. The Facility either stores under cover or removes from the Facility all abandoned or broken equipment or materials no longer considered for future use that have the potential to serve as a source of pollutants (Exposure Minimization BMP). 10. Bins and containers are checked for holes and leaks and repaired as needed.	Inspections of leaks and spills are daily and ongoing. Trash is picked up daily Employees are trained annually, at minimum. Bins and containers are inspected and repaired once a month as needed.				(WSP) 6. Recycling Center Workers, Recycling Supervisors, Curbside Drivers 7. Recycling Center Workers, Recycling Supervisors 8. General Manager, Operations Manager, Environmental Consultant, WSP (Annual IGP Training) 9. General Manager and Operations Manager (CCC), Operations Manager (EC) 10. Operations Manager and Recycling Supervisors
Transfer Station / Tipping Floor, Green Waste Receiving Area, Vactor Wash, Truck Parking, General/Electronic Waste Storage, Construction Debris Receiving Area	Suspended Solids, Oil and Grease, Chemical Oxygen Demand, metals, pH	1. Misters are utilized inside the Tipping Floor building for dust suppression. The rate of misting is monitored and controlled such that misting water does not exit the Tipping Floor Building. 2. Green waste receiving area is located under cover (Exposure Minimization BMP). 3. The green waste area receives food waste and yard waste from municipal customers. Food waste may have some liquid runoff which is mixed in with the dry green waste by the loader, or the runoff will be contained by a floor drain within the roofed Transfer Station that is connected to the sanitary sewer. 4. General/Electronic Waste Storage is located under cover (Exposure Minimization BMP). 5. Stormwater from Drainage Area 10 drains into trench drains TD-3 and TD-4. There is a manhole near the construction area with two sluice gates that direct flow to the sanitary sewer or the storm drain. The sluice gate to sanitary sewer is closed to direct stormwater to the storm drain (blue handle up, black handle down) for discharge into a 21,000-gallon stormwater holding tank for discharge to EBMUD sanitary sewer (Stormwater Containment and Discharge Reduction BMP). 6. The manhole that contains the weir to bypass the stormwater storage tanks during high flow events is opened regularly and cleaned with a vactor truck to remove accumulated sediments to that the 2-inch inlet to the tanks is not clogged. 7. The manhole that contains the two sluice gates is opened, inspected and maintained on a monthly basis, and the sluice gates are exercised on an annual basis. 8. Trench drains (TD-3 and TD-4) are protected with sediment filter fabric. 9. Dust control water is applied manually at the construction debris receiving area by means of a water cannon. Dust control water is only applied during dry weather. Prior to applying dust control water, facility personnel verify sufficient storage capacity in the 21,000-gallon stormwater holding tank. If there is not sufficient capacity in the 21,000-gallon tank, water is transferred manually to a backup 21,000-gallon tank by means of a sump pump to create the needed capacity. Dust control water application rate is controlled to minimize flow to trench drain TD-4.	Dust control misters operate daily and are inspected daily during business hours. Trench drains are maintained at least weekly during September-May and prior to rain events. Filter fabric within trench drains are replaced as needed. The manhole that contains the weir with bypass to the stormwater storage tanks and the manhole containing two sluice gates are opened and cleaned of sediment on a weekly basis. Dust control water is applied to the construction debris receiving area on as-needed basis during dry weather. Observations of vactor wash area during washing activities. Weekly removal of accumulated sediments from the manhole containing the sluice gates, annual exercising of sluice gates	Drainage Area 10 Filter fabric are implemented at trench drains (TD-3 and TD-4)	Water hose and misters for dust suppression. Sediment filter fabric, sediment/hydrocarbon/ wattles Stormwater holding tank, sewage pump or vactor truck and associated piping used to collect stormwater for discharge to EBMUD sanitary sewer.	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Misters inspected daily for working condition. Observations of vactor wash area during washing activities. Post rain inspections are performed for the stormwater holding tank for discharge to sanitary sewer during dry days, Discharges to occur 24 hours after rain event, no discharges during rain events per EBMUD permit, discharge logs. Weekly inspections of trench drains and weir manhole, monthly inspection of sluice gate manhole	1. Supervisor, Solid Waste Supervisors and Solid Waste Workers 2. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 3. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 4. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 5. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 6. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 7. Solid Waste Senior Supervisor, Solid Waste Supervisors, Solid Waste Worker and Corporation Yard Worker 8. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 9. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid

Potential Pollution Source	Potential Pollutant(s) Reduced by BMP (Section X.H.4.a.i)	BMPs	Frequency of BMP implementation (Section X.H.4.a.ii)	Location of BMP (Section X.H.4.a.iii)	Equipment and Tools for BMP Implementation (Section X.H.4.a.vi)	Frequency for BMP Inspection (Section X.H.4.a.vii)	Individual and/or Position Responsible for Implementing the BMP (Section X.H.4.a.iv)
		10. Vactor material wash water area enclosed in berm and connected to sanitary sewer and under cover (Exposure Minimization BMP). 11. No significant repairs that could lead to a source of pollutants are performed on vehicles or movable equipment in outdoor, uncovered areas 12. Post rain inspections are performed for the stormwater holding tank for discharge to sanitary sewer during dry days, Discharges to occur 24 hours after rain event, no discharges during rain events per EBMUD permit, discharge logs. 13. BMP logs are kept to document trench drain maintenance and mister status at the Tipping Floor.					Waste Workers 10. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 11. Equipment Maintenance Supervisor and Workers 12. Environmental Compliance Specialist 13. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers
Equipment/Vehicle Maintenance Building, Temporary Mobile Diesel Fueling Area, Truck Parking, Mattress Area	Suspended Solids, Oil and Grease, metals, solvents, pH	1. Industrial materials including oil, grease, solvents, transmission fluids, and antifreeze are stored indoors (Exposure Minimization BMP). 2. Mattresses are stored under cover (Exposure Minimization BMP). 3. Above ground tanks with petroleum products or anti-freeze are stored with secondary containment under cover (Exposure Minimization BMP). 4. Maintenance at the Equipment Maintenance Building is performed indoors. 5. Spill kits and temporary spill booms are available for deployment in case of spills. 6. CB-3A is protected with sediment filter fabric. 7. Staff do not to top-off during fueling. Operator awareness is required during fueling activities. 8. A spill kit is located near the fueling area. 9. Wattles are placed in the flow path of CB-3 and CB-3A. 10. A Contech Filter treats stormwater from CB-3, CB-3A, CB-4, A second Contech Filter treats stormwater from TD-2A and CB-2 (Treatment Control BMP, see Appendix D for description and specifications). 11. BioChar, sediment wattles, and oil absorbent socks are installed around CB-3, CB-3A, and CB-4 (other advanced BMP). 12. BMP logs are kept to document sweeping, trench drain, storm drain and Contech Filter maintenance, picking up trash. 13. Vendor contracted for regular maintenance of the catch basins, and Contech Filters. 14. Repairs in the Truck Parking Area are performed undercover to the extent feasible. Outdoor vehicle maintenance work is not performed during rain events.	BioChar, sediment wattles, and oil absorbent socks are maintained on a weekly basis during dry weather and as-needed during wet weather. Catch basins maintained by vendor and on-site personnel on a weekly basis, identified problem areas will be addressed with more frequency if necessary. Catch basin and trench drain filter change outs will be performed as a result of visual inspections Contech filter vaults are maintained twice a year.	Drainage Area 2 and 3 Filter fabric, and or sediment, hydrocarbon, BioChar wattles are implemented at catch basins (CB-2 and CB-3, CB-3A, CB-4).	Sediment filter fabric, sediment/hydrocarbon/wattles	Weekly inspections of catch basins and trench drains Daily inspections occur for operational equipment. Contech filters are inspected monthly. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP.	1. Equipment Maintenance Supervisor/Workers 2. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 3. Equipment Maintenance Supervisor/Workers 4. Equipment Maintenance Supervisor/Workers 5. Equipment Maintenance Supervisor/Workers 6. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 7. All staff fueling vehicles with supervision 8. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 9. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 10. Stormwater Vendor 11. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 12. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 13. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers

Potential Pollution Source	Potential Pollutant(s) Reduced by BMP (Section X.H.4.a.i)	BMPs	Frequency of BMP implementation (Section X.H.4.a.ii)	Location of BMP (Section X.H.4.a.iii)	Equipment and Tools for BMP Implementation (Section X.H.4.a.vi)	Frequency for BMP Inspection (Section X.H.4.a.vii)	Individual and/or Position Responsible for Implementing the BMP (Section X.H.4.a.iv)
							14. Equipment Maintenance Supervisor/Workers
Wash Rack, Drum and Diesel Exhaust Fuel Storage	Suspended Solids, Oil and Grease, Solvents, pH	1. The manhole in the wash rack has gate valves to switch between stormwater and sanitary sewer. The valve is in the sanitary sewer position. 2. During rain events the manhole is covered per EBMUD permit. 3. Berms are located on the northern and southern ends of the wash rack to contain non-stormwater for discharge to the sanitary sewer. 4. Empty 55-gallon drums and DEF area stored in this area under cover and within secondary containment. Drums containing liquids (motor oil and hydraulic oil) are stored on secondary containment structures under cover.	Ongoing during operational hours.	Drainage Area 2	Manhole covers.	Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Daily inspections occur for operational equipment.	1. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 2. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 3. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 4. Equipment Maintenance Supervisor/Workers
Used Oil Depot, General Storage	Suspended Solids, Oil and Grease, Metals	1. The Used Oil Depot and General Storage are located under cover (Exposure Minimization BMP) 2. Used oil is put into a double walled above ground tank and stored within a concrete containment area. 3. Spill pallets are used to contain drips or spills from customers.	Ongoing during operational hours.	Drainage Area 3	Chemical storage	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP.	1. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 2. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 3. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers
Container/Bin/Tire Storage Area and Container Maintenance Building	Suspended Solids, Metals	1. Catch basin (CB-1) and trench drain (TD-1) are protected with BioChar, sediment wattles and hydrocarbon absorbent socks (other Advanced BMP). 2. Vendor contracted for regular maintenance of the trench drains, catch basins, and Contech Filters. 3. BMP logs are kept to document trench drain, storm drain and Contech Filter maintenance. 4. Contech filter at DP-1A (Treatment Control BMP, see Appendix D for description and specifications). 5. Tires are stored under cover (Exposure Minimization BMP).	Catch basins maintained by vendor and on-site personnel on a weekly basis, identified problem areas will be addressed with more frequency if necessary. Catch basin and trench drain filter change outs will be performed as a result of visual inspections Contech filter vaults are maintained twice a year.	Drainage Area 1 Spill response materials are located on the island and within the equipment maintenance building at the Transfer Station. Catch basin CB-1, trench drain TD-1 and Contech filter at DP-1A.	Equipment used for spill response may include absorbent pads, granular absorbent, shovels and brooms.	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Contech filters are inspected monthly.	1. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 2. Solid Waste Senior Supervisor, Solid Waste Supervisors and Environmental Compliance Specialist 3. Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers 4. Stormwater Vendor 5. Solid Waste Senior Supervisor, Solid Waste Supervisors, Solid Waste Workers, Equipment Maintenance Supervisor and Workers
Truck Parking, Container/Bin Storage Area	Suspended Solids, Metals, Oil and Grease	1. Relocating the vehicles elsewhere on the site to better facilitate sweeping of paved areas when feasible 2. Drip pans are placed under equipment stored or parked for a week or longer	Weekly inspections for evidence of leaks from equipment, and prompt clean-up of spills, drips, or	Drainage Area 4 Spill response materials are located on the island and within the	Equipment used for spill response may include absorbent pads,	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in	1. I Solid Waste Senior Supervisor, Solid Waste Supervisors and Solid Waste Workers

Potential Pollution Source	Potential Pollutant(s) Reduced by BMP (Section X.H.4.a.i)	BMPs	Frequency of BMP implementation (Section X.H.4.a.ii)	Location of BMP (Section X.H.4.a.iii)	Equipment and Tools for BMP Implementation (Section X.H.4.a.vi)	Frequency for BMP Inspection (Section X.H.4.a.vii)	Individual and/or Position Responsible for Implementing the BMP (Section X.H.4.a.iv)
		3. Inspections for evidence of leaks from such equipment, and prompt clean-up of spills, drips, or leaks from such equipment. 4. Containers and bins are cleaned to remove materials and liquids prior to storage.	leaks from such equipment.	equipment maintenance building at the Transfer Station.	granular absorbent, shovels and brooms. Drip Pans	accordance with the MIP in this SWPPP.	2. Equipment Maintenance Supervisor/Workers 3. Operator, Equipment Maintenance and Workers 4. Operator
Recycling Center - Customer Recyclable Area	Suspended Solids, Oil and Grease, Solvents, pH, Metals, Chemical Oxygen Demand	1. Customers are asked to drain any containers into provided buckets prior to recycling to prevent spills and leaks. Liquid in bucket disposed into sanitary sewer. 2. Two new canopy structures and berms installed to prevent stormwater contact and run-on with materials within the recycle bin area (Exposure Minimization BMP). 3. BMP logs are kept to document storm drain maintenance. 4. BioChar, sediment wattles and oil absorbent socks are installed around the catch basin DP-8A and DP-6 (other Advanced BMP). 5. DP-8A catch basin is sealed/covered during dry-season. 6. Vendor contracted for regular maintenance of DP-8A, BioChar, sediment wattles and oil absorbent BMPs. 7. Drip pans are placed under equipment stored or parked for a week or longer. 8. A drip pan is placed under the bottle and can conveyor belt to catch liquids. Liquids in the drip pan are disposed into the sanitary sewer. 9. Inspections for evidence of leaks from such equipment, and prompt clean-up of spills, drips, or leaks from such equipment. 10. No significant repairs that can be a source of potential pollutants are performed to vehicles or movable equipment at the Facility in outdoor, uncovered areas. 11. Maintain pavement by repaving certain sections to facilitate productive cleaning by the mechanical sweeper.	Storm drain protection and perimeter control is ongoing and maintained daily during the rainy season and before storm events. Prior to rain events wind screen along fence line is evaluated	Drainage Area 6 and 8 Spill response materials located adjacent to Recycling Center office near DP-8A	Equipment used for spill response may include absorbent pads, granular absorbent, shovels and brooms 5-gallon Buckets BioChar, Sediment Wattles, and Oil Absorbent Socks Catch Basin Cover Drip Pans	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Spill cleanup inspections. Weekly and annual inspections of BMPs for DP-5A, DP-6, and DP-8A. facility perimeter along Second Street.	1. Recycling Supervisors 2. General Manager and Operations Manager 3. Operations Manager 4. Operations Manager and Recycling Supervisors, Recycling Center Workers 5. Operations Manager and Recycling Supervisors 6. Operations Manager 7. Recycling Supervisor 8. Recycling Supervisor 9. Recycling Supervisor, Equipment Operators 10. Operations Manager 11. General Manager, Operations Manager
Recycling Center – Commingled Container Storage Area	Suspended Solids, Oil and Grease, Solvents, Chemical Oxygen Demand, pH	1. The commingled container storage area is surrounded by a wall and six-inch berm. Stormwater from source area drains into valley gutters and is collected in a sump where it is pumped into two (2) 4,000 gallon storage tanks. An additional 5,400 gallon storage tank is used to store stormwater collected from the 4,000 gallon tanks. Collected stormwater is discharged to EBMUD sanitary sewer during dry days (24 hours after rain event) and discharge logs (Stormwater Containment and Discharge Reduction BMP). 2. Stormwater storage tanks are inspected after rain events to assure adequate freeboard. 3. Observed spills and leaks outside the containment area are cleaned up upon discovery. 4. Weather tracking is performed to evaluate the need for additional temporary storage capacity, which is provided via rental tanks as-needed. 5. Piping and pumps associated with stormwater storage tanks are inspected regularly. 6. Debris is removed from grate cover and sump 7. BMP logs are kept to document sweeping, valley gutters/screen/sump/pump) maintenance. 8. Maintain pavement by repaving certain sections to facilitate productive cleaning by the mechanical sweeper.	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Daily trash pickup throughout facility and boundaries. Spills cleaned up upon discovery. Weekly and annual inspections of stormwater holding tank and associated equipment and valley gutter drains and sump pumps with maintenance logs. Sump is cleaned weekly	Drainage Area 7 Spill response materials located adjacent to Recycling Center office near DP-8A.	Drip pans Stormwater holding tank, sewage pump and associated piping used to collect stormwater for discharge to EBMUD sanitary sewer.	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Daily trash pickup throughout facility and boundaries. Spills cleaned up inspections. Weekly and annual inspections of stormwater holding tank and associated equipment and valley gutter drains and sump pumps with maintenance logs.	1. Operations Manager 2. Operations Manager 3. Recycling Supervisor, Recycling Center Workers 4. General Manager and Operations Manager 5. Operations Manager 6. Operations Manager 7. Operations Manager 8. General Manager, Operations Manager
Recycling Center – Glass Sorting Area, Paper Recycling	Suspended Solids, Oil and Grease, Chemical Oxygen Demand, pH	1. Stormwater from Glass Sorting Area drains into a valley gutter which flows to a sump that pumps stormwater into two (2) 4,000 gallon storage tanks. An additional 5,400 gallon storage tank is used to store stormwater collected from the 4,000 gallon tanks. Collected stormwater is discharge	Post rain inspections are performed for the stormwater holding tank	Drainage Area 7 and 9 Spill response materials are located near DP-8A near the Recycling	Equipment used for spill response may include absorbent pads,	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in	1. Operations Manager 2. Recycling Supervisors and Recycling Center Workers

Potential Pollution Source	Potential Pollutant(s) Reduced by BMP (Section X.H.4.a.i)	BMPs	Frequency of BMP implementation (Section X.H.4.a.ii)	Location of BMP (Section X.H.4.a.iii)	Equipment and Tools for BMP Implementation (Section X.H.4.a.vi)	Frequency for BMP Inspection (Section X.H.4.a.vii)	Individual and/or Position Responsible for Implementing the BMP (Section X.H.4.a.iv)
Warehouse and Rear Alleyway		into EBMUD sanitary sewer after 24 hours dry day rain event) (Stormwater Containment and Discharge Reduction BMP). 2. Broken glass outside the berm is controlled by hand and mechanical sweeping. 3. Observed spills and leaks outside the containment area are cleaned up upon discovery. 4. BMP logs are kept to document valley gutters/screen/sump/pump maintenance. 5. Wind screen along eastern fence adjacent to driveway near the Paper Recycling Warehouse. 6. Debris is removed from grate cover and sump. 7. A berm is placed to contain non-stormwater within the glass sorting area. 8. Sediment, BioChar wattles, and oil absorbent socks are installed along the flow path to discharge point DP-9 (other Advanced BMP). 9. Maintain pavement by repaving certain sections to facilitate productive cleaning by the mechanical sweeper. 10. Paper recycling located under cover within the Warehouse (Exposure Minimization BMP).	for discharge to sanitary sewer Prior to rain events wind screen along fence line is evaluated Spills cleaned up upon discovery. Weekly and annual inspections of stormwater holding tank and associated equipment and valley gutter drains and sump pumps with maintenance logs. Wattles and oil absorbent socks are maintained daily during the rainy season. Sump is cleaned weekly	Center office and within the equipment maintenance building at the Transfer Station.	granular absorbent, shovels and brooms Wind Screen Sediment and BioChar wattles Oil absorbent socks	accordance with the MIP in this SWPPP. Spill cleaned up inspections. Weekly and annual inspections of stormwater holding tank and associated equipment and valley gutter drains and sump pumps with maintenance logs. Weekly and annual inspections of BMPs for DP-9 along Gilman Street.	3. Recycling Supervisors and Equipment Operators 4. Operations Manager 5. Operations Manager and Supervisors 6. Operations Manager and Recycling Supervisors 7. General Manager and Operations Manager 8. Operations Manager, Recycling Supervisor 9. General Manager and Operations Manager 10. Not Applicable
Ecology Center – Truck Parking Area	Suspended Solids, Oil and Grease, Solvents, Chemical Oxygen Demand, pH	1. Major maintenance activities of the EC trucks are not conducted outdoors during rain events. 2. A vehicle containment pad is used for any work done onsite. 3. BioChar sediment wattles and oil absorbent socks are installed along the fence line along Second Street and associated discharge point DP-4 and DP-5A (other Advanced BMP). 4. Drip containers are deployed under garbage trucks. Spill Response equipment is located in the storage shed at the Ecology Center, and in a barrel outside the Ecology Center Office Building for easy access. 5. Vehicles are relocated elsewhere on the site to better facilitate sweeping of paved areas when feasible. 6. BMP logs are kept to document sweeping, maintenance of wattles, picking up trash, and containing truck oil/hydraulic oil leaks. 7. Large absorbent sheets or drip pans are placed under equipment stored or parked for a week or longer. 8. No significant repairs that can be a source of potential pollutants are performed on vehicles or movable equipment at the Facility in outdoor, uncovered areas.	Plastic drip containers and absorbent sheets are maintained daily. Wattles and oil absorbent socks installed along the fence line of 2nd street and around DP-5A are maintained daily during the rainy season.	Drainage Area 5 Spill response materials are located in the cargo bin next to truck parking area.	Plastic drip containers, absorbent sheets, BioChar sediment wattles, and oil absorbent socks.	Daily inspections occur for operational equipment. Monthly inspections are documented and performed in accordance with the MIP in this SWPPP. Daily inspections of BMPs for DP-5A and perimeter fence line along Second Street. Daily inspection of drip containers for garbage trucks.	1. Ecology Center Operations Manager 2. Ecology Center Operations Manager 3. Ecology Center Operations Manager 4. Ecology Center Operations Manager, Curbside Drivers 5. Ecology Center Operations Manager 6. Ecology Center Operations Manager 7. Ecology Center Operations Manager, Curbside Drivers 8. Ecology Venter Operations Manager

M. MONITORING IMPLEMENTATION PLAN (MIP) (GENERAL PERMIT SECTION X.I)

This document presents the MIP for the Facility. This MIP was prepared to address the following objectives:

1. Identify team members assisting in implementing the MIP (General Permit Section X.I.1);
2. Describe discharge locations (General Permit Section X.I.2.a);
3. Provide justification for representative sampling reduction, sample frequency reduction, and/or qualified combined samples, as applicable (General Permit Section X.I.3);
4. Describe visual observation requirements (General Permit Section X.I.2.b)
5. Describe visual observation response procedures (General Permit Section X.I.2.c);
6. Describe sample collection and handling procedures (General Permit Section X.I.B;
7. Describe sample analysis;
8. Describe field instrument calibration procedures (General Permit Section X.I.4); and
9. Provide an example Chain of Custody form and procedures when handling and shipping water quality samples (Section X.I.5).

M.1 TEAM MEMBERS (GENERAL PERMIT SECTION X.I.1)

Trained Facility compliance personnel are responsible for the visual monitoring, site inspection activities, and stormwater sampling described herein. A list of Facility employees selected for these tasks is provided within SWPPP Section B. Training for the team members is included in Section H. The Facility has selected the following certified laboratory to perform the sample analysis.

Laboratory Name:	Enthalpy, Inc.
Street Address:	5900 Hollis Street, Suite L (Service Center)
City, State Zip:	Emeryville, CA
Telephone Number:	510-486-0900
Street Address:	931 W Barkley Ave (Laboratory Sample Analysis Location)
City, State Zip:	Orange, CA 92868
Telephone Number:	714-771-9917
Point of Contact:	Richard Villafania
ELAP Certification Number:	1338

M.2 DISCHARGE LOCATIONS (GENERAL PERMIT SECTION X.I.2.A)

A description of the Facility's stormwater discharge locations associated with industrial activities are described with Section C.3 of this SWPPP, summarized below, and are included on

the Site Map. If there are discharge locations at the Facility that are affected by run-on from surrounding areas as described in Section C.4 of this SWPPP, or that are difficult or unsafe to sample, an alternative monitoring location can be selected.

Discharge Location No.	Include Latitude/Longitude for Discharge Point	Description[1, 2]
DP-1A	37.88150/-122.30655	The sample collection point is located after stormwater exits the Contech filter at the vault and is at the point of discharge from the facility into the MS4 located on Second Street.
DP-2A	37.88024/-122.30617	An alternative discharge location was selected due to the run-on condition and flooding described in Section 4.2. Stormwater is collected as it exits the Contech filter through the underground piping system.
DP-3A	37.88046/-122.30569	Stormwater samples are collected as stormwater exits the Contech filter through the underground piping system. The location includes flow from catch basins CB-3A, CB-3 and CB-4 within Drainage Area 3.
DP-4	37.87960/-122.30604	A Representative Sampling Reduction point was selected for Drainage Area 4. DP-4 is collected from sheet flow as it enters Second Street.
DP-5A	37.879156/-122.305919	A Representative Sampling Reduction point was selected for Drainage Area 5. DP-5A is collected from sheet flow as it enters Second Street.
DP-6	37.87914/-122.30588	A Representative Sampling Reduction point was selected for Drainage Area 6. DP-6 is collected from sheet flow as it enters Second Street.
DP-8A	37.87870/-122.30559	The sample collection point is collected as flow enters a storm drain catch basin on-site near the southwest corner of the property.
DP-9	37.87878/-122.30485	A Representative Sampling Reduction point was selected for Drainage Area 9. DP-9 is collected from sheet flow that is concentrated along the curb and gutter as it enters Gilman Street.

M.3 Representative Sampling Reduction or Qualified Combined Samples (General Permit Sections XI.C.4 and XI.C.5)

Presented below is the justification for representative sampling reduction and/or qualified combined samples in accordance with General Permit Sections XI.C.4 and XI.C.5, respectively. The Facility may reduce the number of sample locations in each drainage area (e.g., roofs with multiple downspouts) if the drainage areas are substantially similar. Alternatively, the Facility may authorize the laboratory to combine as many as four (4) discharge locations. Drainage areas must be substantially similar to one another including industrial activities, BMPs, and

[1] Any alternative sampling locations selected should be included on the site map.

[2] Justification for Representative Sampling Reduction, Sample Frequency Reduction, and/or Qualified Combined Samples is included in Section M.3.

physical characteristics. This SWPPP includes descriptions of the industrial activities within Section F; descriptions of BMPs are included within Sections G, J, and L; and descriptions of physical characteristics of the drainage areas are included within Section C.3. The rationale that demonstrates that drainage areas are substantially similar is presented below. If areas are not substantially similar, indicate not applicable in the table below.

Discharge Location Selected for Reduction	Corresponding Discharge Location	Are Industrial Activities Similar? (Y/N)	Are BMPs Similar? (Y/N)	Are Physical Characteristics (i.e., Grade, Surface Materials) Similar? (Y/N)	Is the sampling location representative? (Y/N)	Provide justification for Combining Samples or Sample Reduction[1]
Drainage Area 4 Gate at DP-4	DP-4	Yes	Yes	Yes	Yes	The four listed drainage areas do not have well-defined stormwater conveyance systems for sample collection due to sheet flow as it discharges across the driveway to Second Street or Gilman Street. The designated sampling points were selected based on concentrated flow observed.
Drainage Area 5 Gate at DP-5A	DP-5A	Yes	Yes	Yes	Yes	
Drainage Area 6 Gate at DP-6	DP-6	Yes	Yes	Yes	Yes	
Drainage Area 9 Corner of Curb and Gutter at DP-9	DP-9	Yes	Yes	Yes	Yes	

1. This section was completed to provide justification that a QISP reviewed the designated sampling locations to identify the representative sampling locations at each gate. Additional discharge points may occur at each gate as sheet flow discharges down the driveways, but these discharge points should be considered representative of the designated sampling locations shown on this table.

M.4 Visual Observation Requirements (General Permit Sections XI.A.2.b and c)

The General Permit requires visual observations as described below.

M.4.1 Monthly Visual Observations (General Permit Section XI.A.2.b)

At least once per calendar month, the Facility will visually observe each drainage area for the following:

- Presence or indications of prior, current, or potential unauthorized NSWDs and their sources;
- Authorized NSWDs, sources, and associated BMPs;
- Outdoor industrial equipment and storage areas, outdoor industrial activities areas, BMPs, and all other potential source of industrial pollutants.

Monthly visual observations will be completed during daylight hours, under dry conditions, and during scheduled Facility operating hours in accordance with General Permit Section XI.A.1.b. Monthly visual observations will be documented on the Monthly Visual Observation Form in Appendix E. The observation form includes a section on response actions and additional SWPPP revisions necessary in response to the visual observations as required in General Permit Section XI.A.3. If response actions are necessary (as described on the forms), BMPs will be revised and addressed within this SWPPP in accordance with the General Permit (Section XI.A.4).

M.4.2 Sampling Event Visual Observations (General Permit Section XI.A.2.c)

At each discharge location where a sample is obtained, the Facility will observe the discharge of stormwater associated with industrial activity. As described in the General Permit (Section XI.A.2):

- The Discharger shall ensure that visual observations of stormwater discharged from containment sources (e.g. secondary containment or storage ponds) are conducted at the time that the discharge is sampled.
  - Note: This requirement does not apply when stormwater from a containment area is released but no off-site discharge occurs.
- Any Discharger employing volume-based or flow-based treatment BMPs shall sample any bypass that occurs while the visual observations and sampling of stormwater discharges are conducted.
- The Discharger shall visually observe and record the presence or absence of
  - floating and suspended materials,
  - oil and grease,
  - discolorations,
  - turbidity,
  - odors,
  - trash/debris,

- and source(s) of any discharged pollutants
- In the event that a discharge location is not visually observed during the sampling event, the location of the discharge and reasoning for not obtaining observations must be recorded.

Sampling event visual observations will be documented on the Sampling Event Visual Observation Form within Appendix F. The observation form includes a section on response actions and additional SWPPP revisions necessary in response to the visual observations as required in General Permit Section XI.A.3. If response actions are necessary (as described on the forms), BMPs will be revised and addressed within this SWPPP in accordance with the General Permit Section XI.A.4.

Additional Stormwater Observations Per Settlement Agreement and Release

Though not required to video discharges of QSE events per the IGP, video of QSE during sampling will be taken at discharge points (DP-4, DP-5A, DP-6, DP-8, and DP-9). It is impracticable to video discharges at DP-1A, DP-2A, and DP-3A because discharges are underground and flow into the City's storm drainage system after treatment by the Contech Filter. Instead, video of discharges entering the catch basin leading to the Contech Filter will be taken. Videos will be taken during the 2023-2024, 2024-2025, and 2025-2026 compliance years if there are QSEs and occur during normal business operating hours.

M.5 Sample Collection and Handling Requirements (General Permit Sections XI.B)

M.5.1 Qualifying Storm Event (QSE) (General Permit Section XI.B)

In accordance with General Permit Section XI.B, stormwater discharge samples will be collected from QSEs four times per year: twice during the first half of the year (July 1 through December 31) and twice during the second half of the year (January 1 through June 30).

Samples will be taken from the first QSE (first flush) as defined in the IGP during the 2023-2024, 2024-2025, and 2025-2026 compliance years per settlement agreement and release.

The General Permit describes a QSE as a precipitation event that:

- Produces a discharge from at least one drainage area, and
- Is preceded by 48-hours with no discharges from any of the drainage areas.

Stormwater discharge samples should be collected under the following conditions by properly trained personnel:

- Within 4 hours of the start of discharge.
- The start of operations if the QSE occurs within the previous 12-hour period.
- During regularly scheduled facility operating hours (see Section C.1 regarding regularly scheduled operating hours).
- When weather and site conditions are safe.

M.5.1.1 Weather Monitoring

Weather will be monitored to identify potential QSEs from the National Weather Service San Francisco Bay Area, CA (weather.gov). A Weather Station to track rain accumulation in real time is located at the Transfer Station. A precipitation log sheet will be used to record rain events from this weather station. If the Transfer Station Weather Station malfunctions, the nearest weather station closest to the Transfer Station will be used. KJ6WEG Berkeley, CA (APRSWXNET/CWOP – MTR. KJ6WEG Berkeley (weather.gov).

M.5.2 Sample Collection (General Permit Section XI.B.8 and Attachment H)

Samples should be collected at the designated sampling locations shown on the Site Map(s) and listed in Section M.2. To maintain sample integrity and prevent cross-contamination, sample collection personnel will follow the protocols below.

- Think safety first when collecting samples;
- Collect samples (for laboratory analysis) only in analytical laboratory-provided sample containers;
- Wear clean, powder-free nitrile gloves when collecting samples;
- Change gloves whenever something not known to be clean has been touched;
- Change gloves between sites;
- Decontaminate equipment that may be in contact with the sample water with the exception of laboratory provided containers (e.g. collection buckets, tubing). Decontaminate prior to sample collection using a trisodium phosphate water wash, distilled water rinse, and final rinse with distilled water. Dispose of wash and rinse water appropriately (do not discharge into storm drain);
- Do not dip sample containers with preservatives into the flow path. Use a collection container provided by the laboratory to fill these containers. Alternatively, use decontaminated equipment (as described above) to fill the containers;

- Do not overfill containers. Fill containers to the top or to the level indicated on the bottle;
- Do not test pH from sample bottles. If using a bottle, use a designated container without laboratory preservatives and rinse the container with two or three volumes of stormwater from the flow at the discharge point prior to filling;
- Do not scrape the ground or collect floating debris;
- Do not smoke during sampling events;
- Never sample near a running vehicle;
- Avoid parking vehicles in the immediate sample collection area;
- Do not eat or drink during sample collection;
- Do not breathe, sneeze, or cough in the direction of an open sample container;
- Secure sample container lids immediately after sample collection; and
- Place containers in an ice-chilled cooler (or use blue ice) immediately following sample collection.

The most important aspect of grab sampling is to collect a sample that represents the entire runoff stream. Typically, samples are collected by dipping the collection container in the runoff flow paths and streams as noted below:

- For small flow paths, simply dip the bottle facing upstream until full.
- For larger flow that can be safely accessed, collect a sample in the middle of the flow stream by directly dipping the mouth of the bottle. Once again making sure that the opening of the bottle is facing upstream as to avoid any contamination by the sampler.
- Avoid collecting samples from ponded, sluggish or stagnant water as this does not represent true sample homogeneity. The sampled water must be flowing.
- Avoid collecting samples directly downstream from a bridge as the samples can be affected by the bridge structure or runoff from the road surface.
- Do not stand upstream of the sampling point within the flow path.

With the exception of oil and grease analyses, sample containers with preservatives should be filled from a collection container, rather than by immersion in a runoff stream to avoid overfilling. Samples collected for oil and grease analysis should be collected directly into the sample container with care taken to not overfill the container.

M.5.3 Sample Analysis (General Permit Section XI.B.6)

Presented below are the parameters selected for analysis at the discharge points (DP-1A, DP-2A, DP-3A, DP-4, DP-5A, DP-6, DP-8A, and DP-9 [8 locations]) and reason for including the parameter.

Constituent	Total	Reason
Field pH	8	Basic required constituent
Oil and Grease (O&G)	8	Basic required constituent
Total Suspended Solids (TSS)	8	Basic required constituent
Iron	8	SIC Code constituents
Lead	8	SIC Code constituents
Aluminum	8	SIC Code constituents
Zinc	8	SIC Code constituents
Chemical Oxygen Demand (COD)	8	SIC Code constituents
Copper	8	Pollutant Source Assessment constituent

Presented below are the analytical methods, sample container requirements, preservation requirements, reporting limits and holding times for the selected parameters at the Facility. Additionally, the NAL and instantaneous maximum NAL are provided for reference in the IGP.

Constituent	Analytical Method	Minimum Sample Volume	Sample Containers	Sample Preservation	Reporting Limit	Maximum Holding Time	NAL	Instantaneous Maximum NAL
Field pH	Litmus paper or meters	Not Applicable	Not Applicable	None	1 SU	Measurements to be taken immediately	Not Applicable (NA)	Less than 6.0 Greater than 9.0
TSS	SM 2540D	1 L	1 L plastic	None	1.0 mg/L	7 days	100 mg/L	400 mg/L
O&G	EPA 1664A	1 L	1 L amber glass	Hydrochloric acid	5.0 mg/L	28 days	15 mg/L	25 mg/L
Iron (total)	EPA 200.7	250 mL	250 mL plastic	Nitric acid (HNO3)	0.1 mg/L	180 days	1.0 mg/L	NA
Lead (total)	EPA 200.8	250 mL	250 mL plastic	Nitric acid (HNO3)	0.03 mg/L	180 days	0.262 mg/L	NA
Aluminum (total)	EPA 200.8	250 mL	250 mL plastic	Nitric acid (HNO3)	0.10 mg/L	180 days	0.75 mg/L	NA
Zinc (total)	EPA 200.8	250 mL	250 mL plastic	Nitric acid (HNO3)	0.02 mg/L	180 days	0.26 mg/L	NA
Copper (total)	EPA 200.8	250 mL	250 mL plastic	Nitric acid (HNO3)	0.001 mg/L	180 days	0.0332 mg/L	NA
COD (total))	SM 5220C	250 mL	250 mL plastic	Sulfuric acid (H2SO3)	10 mg/L	28 days	120 mg/L	NA

Abbreviations

mg/L = milligrams per liter
mL = milliliter
L = liter

SU = standard unit

M.5.4 Field pH Equipment Calibration Procedures (General Permit Section XI.C.2 and Attachment H)

Field pH measurements will be conducted within fifteen minutes of sample collection. Do not store pH samples for later measurement. If using a designated sample container for field pH, rinse the container with two or three volumes of stormwater from the flow at the discharge point prior to filling. Field pH will be monitored using either:

- Calibrated portable instrument for pH; or
- Methods in accordance with 40 Code of Federal Regulations 136 for testing stormwater.

If a calibrated portable instrument for pH is used, field measurements should be conducted in accordance with the portable instrument accompanying manufacturer's instructions. It is recommended that an equipment calibration is performed prior to forecast rain events that the Facility plans to monitor. The pH measurement will be recorded on the Sampling Event Visual Observation Form within Appendix F.

Note that, if the Facility has two or more pH results outside of the pH range of 6 to 9 pH units, this SWPPP will be revised and a calibrated portable instrument will be required.

M.5.5 Sampling Handling Procedures (General Permit Section XI.B.8 and Attachment H)

Samples for laboratory analysis will be handled as follows:

- Each sample container will be labeled immediately following sample collection with a unique sample identification;
- Field logs sheets included within Attachment E will be completed;
- Chain of Custody record will be completed. An example is included in Appendix G and described below;
- Cooler will be packaged with sufficient ice or blue ice to maintain samples between 0 and 6 degrees Celsius;
- Sample bottles will be packaged to reduce the chance of breakage during transit; and
- Samples will be delivered to the analytical laboratory promptly. Analytical laboratory should receive samples within 48 hours of the physical sampling unless required to be delivered sooner based on hold times.

M.5.6 Chain of Custody Record (General Permit Section X.I.5)

The Chain of Custody Record is a legal document used to track the samples from collection through analysis. The Chain of Custody Record must be signed by the sampler and the person taking custody of the samples. An example chain of custody record is included in Appendix G.

M.5.7 Data Evaluation and Reporting (General Permit Section XI.B.11)

A designated member of the Pollution Prevention Team will complete the evaluation listed below. The sampling analysis reporting requirements may be found in General Permit Section XI.B.11:

- All sampling and analytical results for all individual samples will be submitted via SMARTS within 30 days of obtaining all results for each sampling event.
- The method detection limit will be provided when an analytical result from samples taken is reported by the laboratory as a “non-detect" or less than the method detection limit. A value of zero will not be reported.
- Analytical results that are reported by the laboratory as below the minimum level (often referred to as the reporting limit) but above the method detection limit will be provided.
- Reported analytical results will be averaged automatically by SMARTS at the end of the reporting year. For any calculations required by the General Permit, a value of zero shall be used when effluent sampling analytical results are reported by the laboratory as “non-detect" or reported as less than the Method Detection Limit (MDL).

In addition to the reporting requirements above, the designated member of the Pollution Prevention Team will:

- Compare each distinct sample (individual or combined) to the corresponding instantaneous maximum NAL included in Table 2 of the General Permit; and
- Calculate the average of the analytical results and compare the average to the NALs included in Table 2 of the General Permit.

If an NAL exceedance occurs in a given reporting year, a Level 1 Exceedance Response Action (ERA) Evaluation and a Level 1 ERA Report will be required in the following year, or, if in a subsequent year, a Level 2 ERA Action Plan and a Level 2 ERA Report will be required in accordance with the General Permit.

An instantaneous maximum NAL exceedance occurs when two (2) or more analytical results from samples taken for any single parameter within a reporting year exceed the instantaneous maximum NAL or are outside the range for pH. An annual NAL exceedance occurs when the average of all the analytical results for a parameter from samples taken within a reporting year exceeds the annual NAL value for that parameter listed in Table 2.

The results of either of the ERA reports may require that the SWPPP be amended. ERA related documents, if applicable, are maintained with the hardcopy SWPPP files.

If the Facility is in ERA Level 1 or Level 2, the following person has been designated as the QISP.

QISP Name	QISP Cert Number	Contact Information (phone/ email address)
James Honniball	0051	415-515-3182 Jim.honniball@wsp.com

M.6 Annual Comprehensive Facility Compliance Evaluation (ACFCE) (General Permit Section XV)

The General Permit (Section XV) requires the Facility to conduct one Annual Comprehensive Facility Compliance Evaluation (ACFCE) for each reporting year (July 1 to June 30). The ACFCE should be conducted at least eight (8) months and not more than sixteen (16) months after the previous ACFCE. The SWPPP should be revised, as appropriate based on the results of the ACFCE, and the revisions will be implemented within 90 days of the ACFCE.

Presented below includes the minimum requirements of the ACFCE and reference to the applicable sections of the SWPPP how the Facility meets the requirements.

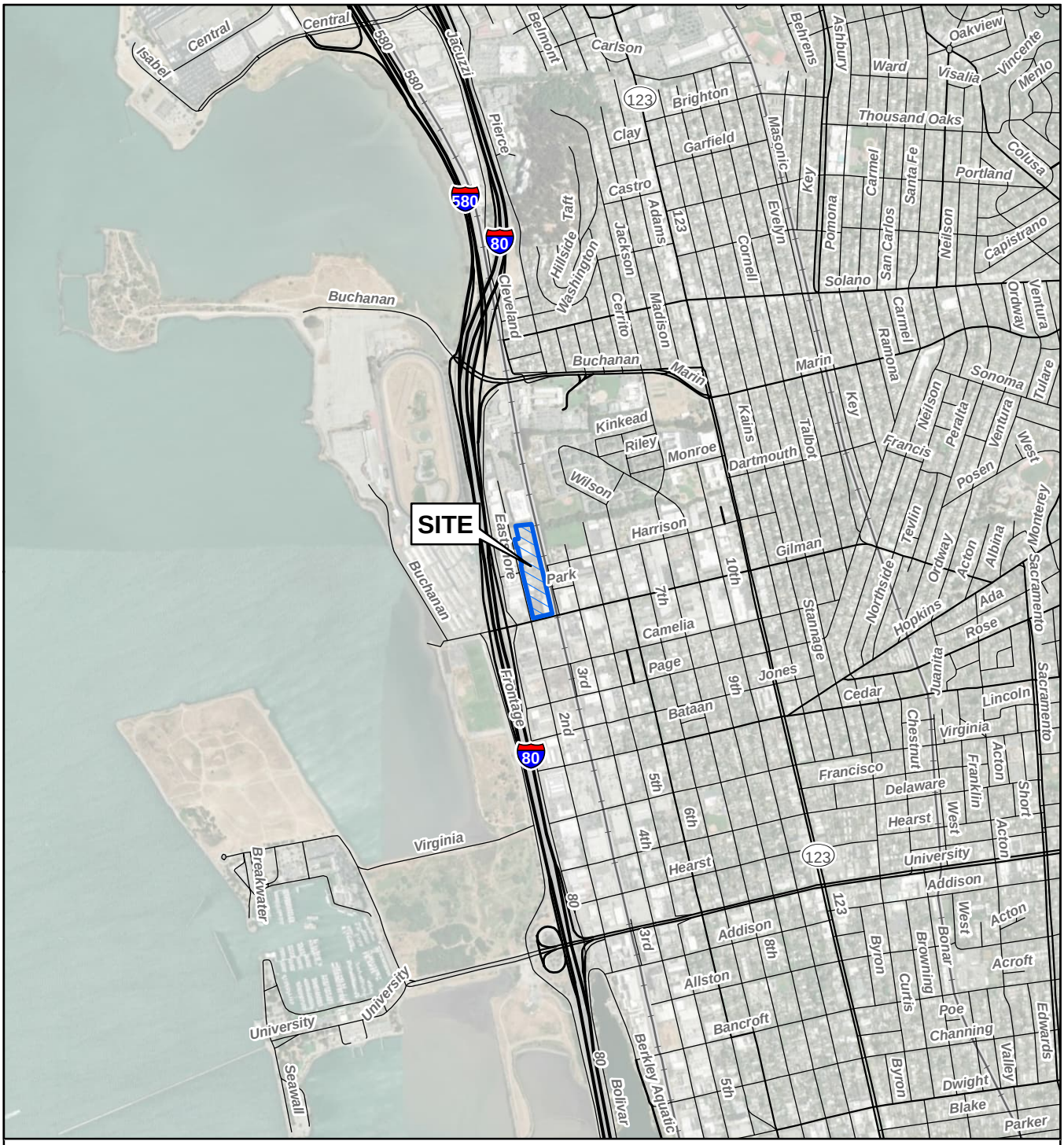
Annual Comprehensive Facility Compliance Evaluation Minimum Requirement	Reference to SWPPP Section
A review of all sampling, visual observation, and inspection and monitoring records and sampling and analysis results conducted during the previous reporting year.	Completed forms described within this SWPPP in Section M.4 and M.5 are reviewed by the designated team member. Analytical results are reviewed in accordance with Section M.5.G.
A visual inspection of all areas of industrial activity and associated potential pollutant sources for evidence of, or the potential for, pollutants entering the stormwater conveyance system.	This visual inspection is documented on the Monthly Visual Observation Form in Appendix E.

Annual Comprehensive Facility Compliance Evaluation Minimum Requirement	Reference to SWPPP Section
A visual inspection of all drainage areas previously identified as having no exposure to industrial activities and materials in accordance with the definitions in Section XVII.	Drainage Area 11 is inspected during the visual inspection.
A visual inspection of equipment needed to implement the BMPs.	This visual inspection is documented on the Monthly Visual Observation Form in Appendix E.
A visual inspection of BMPs.	This visual inspection is documented on the Monthly Visual Observation Form in Appendix E.
A review and effectiveness assessment of all BMPs for each industrial activity area and associated potential pollutant sources to determine if the BMPs are properly designed, implemented, and are effective in reducing and preventing pollutants in industrial stormwater discharges and authorized NSWDs.	Completed forms described within this SWPPP in Section M.4 and M.5 are reviewed by the designated team member. Analytical results are reviewed in accordance with Section M.5.G. These documents provide the basis for the effectiveness assessment. If revisions are required, the SWPPP will be amended as stated above.
An assessment of any other factors needed to comply with the Annual Reporting requirements in General Permit (Section XVI.B).	This item will be assessed by the Facility during the reporting year for any other factors to consider.

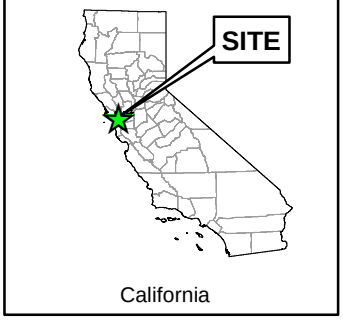
M.7 Annual Report (General Permit Section XVI)

The Annual Report should be prepared, certified, and electronically submitted no later than July 15th following each reporting year using the standardized format and checklists in SMARTS. The Annual Report (completed through SMARTS) should include:

- A Compliance Checklist that indicates whether a Discharger complies with, and has addressed all applicable requirements of this General Permit;
- An explanation for any non-compliance of requirements within the reporting year, as indicated in the Compliance Checklist;
- An identification, including page numbers and/or sections, of all revisions made to the SWPPP within the reporting year; and
- The date(s) of the ACFCE.



Street map from ESRI, 2007. Aerial image from Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community



N

0 2,000 Feet

SITE LOCATION MAP
Berkeley Transfer Station and
Recycling Center
1201 Second St.and 669 Gilman St.
Berkeley, California

wsp	By: TJH	Prj. No. 8620192840.6
	Date: 04/13/2026	Figure 1

C:\Users\USTH051727\WSP O365\WDS-Bay Area Graphics - Documents\City_of_Berkeley\22_0311_up_fig_01.mxd

C:\Users\USTH051727\WSP O365\WDS-Bay Area Graphics - Documents\City_of_Berkeley\22_0311_up_fig_02-FacilityImperviousAreas.mxd



Source: Esri, Maxar, Earthstar Geographics, and the GIS User Community

Explanation

- Facility Boundary
- Impervious Area
- Roof Area
- Permeable Area

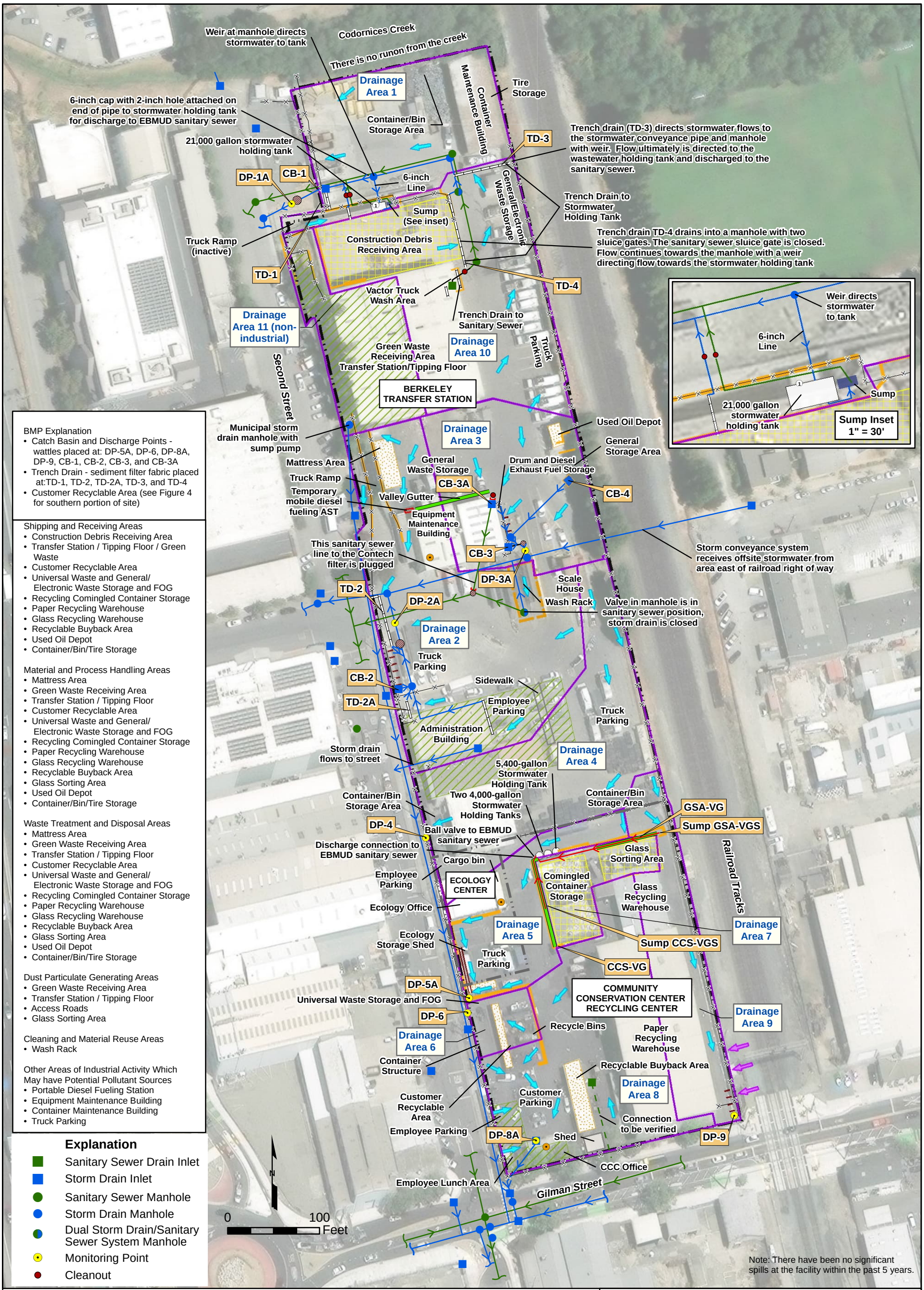
N

0 100 Feet

FACILITY IMPERVIOUS AREAS
Berkeley Transfer Station and
Recycling Center
1201 Second St.and 669 Gilman St.
Berkeley, California

wsp	By: TJH	Prj. No. 8620192840.6
	Date: 03/19/2026	Figure **2**

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BMP Explanation
- Catch Basin and Discharge Points - wattles placed at: DP-5A, DP-6, DP-8A, DP-9, CB-1, CB-2, CB-3, and CB-3A
- Trench Drain - sediment filter fabric placed at:TD-1, TD-2, TD-2A, TD-3, and TD-4
- Customer Recyclable Area (see Figure 4 for southern portion of site)

Shipping and Receiving Areas
- Construction Debris Receiving Area
- Transfer Station / Tipping Floor / Green Waste
- Customer Recyclable Area
- Universal Waste and General/ Electronic Waste Storage and FOG
- Recycling Comingled Container Storage
- Paper Recycling Warehouse
- Glass Recycling Warehouse
- Recyclable Buyback Area
- Used Oil Depot
- Container/Bin/Tire Storage

Material and Process Handling Areas
- Mattress Area
- Green Waste Receiving Area
- Transfer Station / Tipping Floor
- Customer Recyclable Area
- Universal Waste and General/ Electronic Waste Storage and FOG
- Recycling Comingled Container Storage
- Paper Recycling Warehouse
- Glass Recycling Warehouse
- Recyclable Buyback Area
- Glass Sorting Area
- Used Oil Depot
- Container/Bin/Tire Storage

Waste Treatment and Disposal Areas
- Mattress Area
- Green Waste Receiving Area
- Transfer Station / Tipping Floor
- Customer Recyclable Area
- Universal Waste and General/ Electronic Waste Storage and FOG
- Recycling Comingled Container Storage
- Paper Recycling Warehouse
- Glass Recycling Warehouse
- Recyclable Buyback Area
- Glass Sorting Area
- Used Oil Depot
- Container/Bin/Tire Storage

Dust Particulate Generating Areas
- Green Waste Receiving Area
- Transfer Station / Tipping Floor
- Access Roads
- Glass Sorting Area

Cleaning and Material Reuse Areas
- Wash Rack

Other Areas of Industrial Activity Which May have Potential Pollutant Sources
- Portable Diesel Fueling Station
- Equipment Maintenance Building
- Container Maintenance Building
- Truck Parking

Explanation

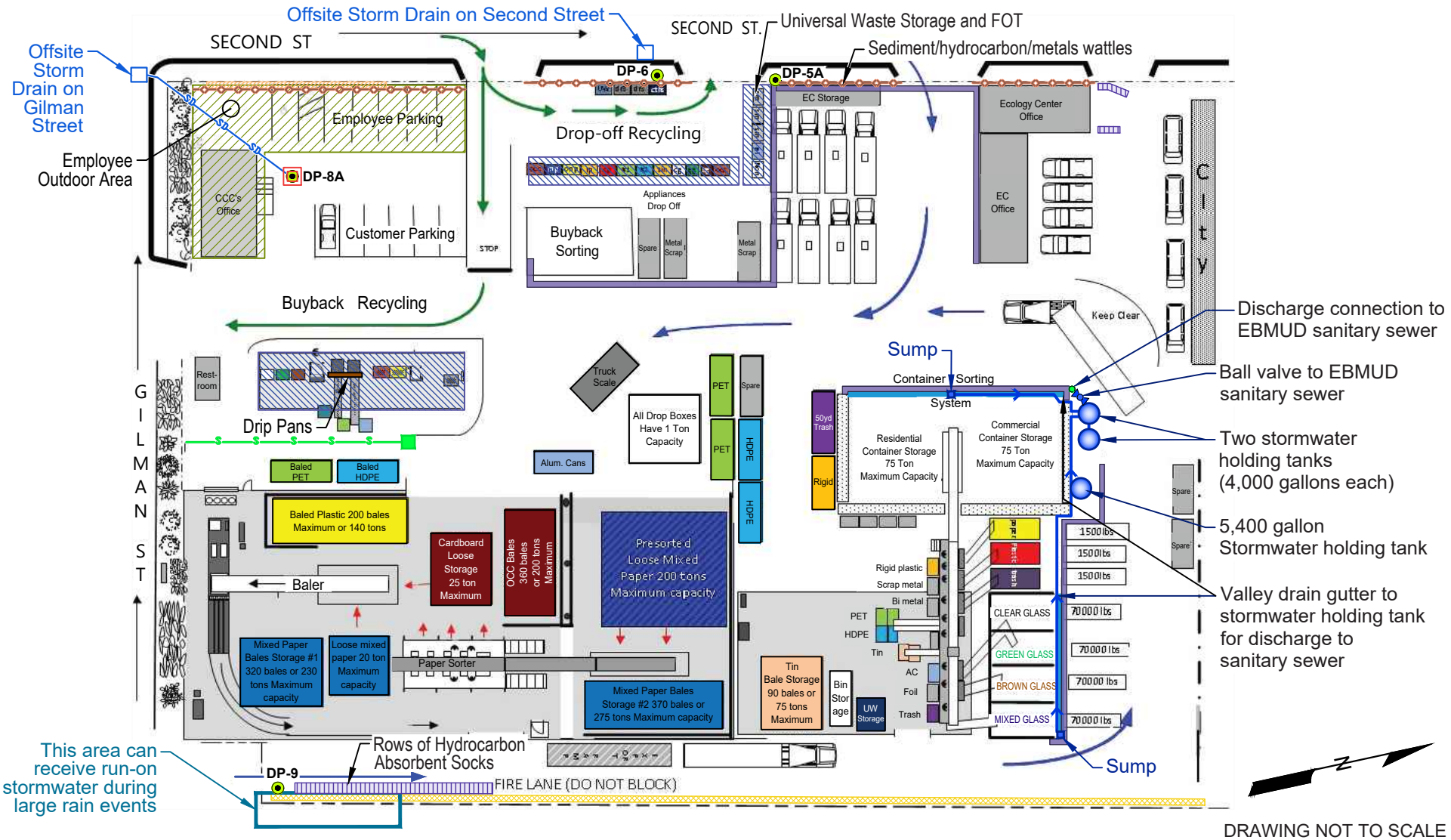
- Sanitary Sewer Drain Inlet
- Storm Drain Inlet
- Sanitary Sewer Manhole
- Storm Drain Manhole
- Dual Storm Drain/Sanitary Sewer System Manhole
- Monitoring Point
- Cleanout
- Spill Kit
- Non-industrial Area
- Materials directly exposed to rain
- Run-on Area
- Flow Arrow
- Wattle
- Property Boundary
- Curb/Berm
- Contech Filter Vault
- Sanitary Sewer Line with Flow Direction
- Storm Drain Line with Flow Direction
- Concrete Plug
- Underground flow to the 4,000-gallon stormwater holding tanks
- Trench Drain
- Valley Drain Gutter
- Canopy Cover
- Drainage area
- Fence

SITE PLAN
Berkeley Transfer Station and Recycling Center
1201 Second St.and 669 Gilman St.
Berkeley, California

wsp

By: TJH	Prj. No. 8620192840.6
Date: 04/16/2026	Figure 3

Plot Date: 4/13/2026 11:46:49 AM, Plotted by: USJW720210 Drawing Path: WDS-Bay Area Graphics – Documents\City of Berkeley\22_0311_up_Site Plan.dwg, Site Plan (5025-220024)



DRAWING NOT TO SCALE

BMP Explanation

- Debris grate and filter fabric insert
- Fence windscreen
- Fine mesh fence screen
- Hydrocarbon absorbent socks
- Sediment wattles
- Sump
- DP-9 Monitoring point

- Sanitary drain connection; not verified
- Valley drain gutter
- Berm
- Public traffic route
- Truck traffic route
- Canopy cover
- Non-industrial area

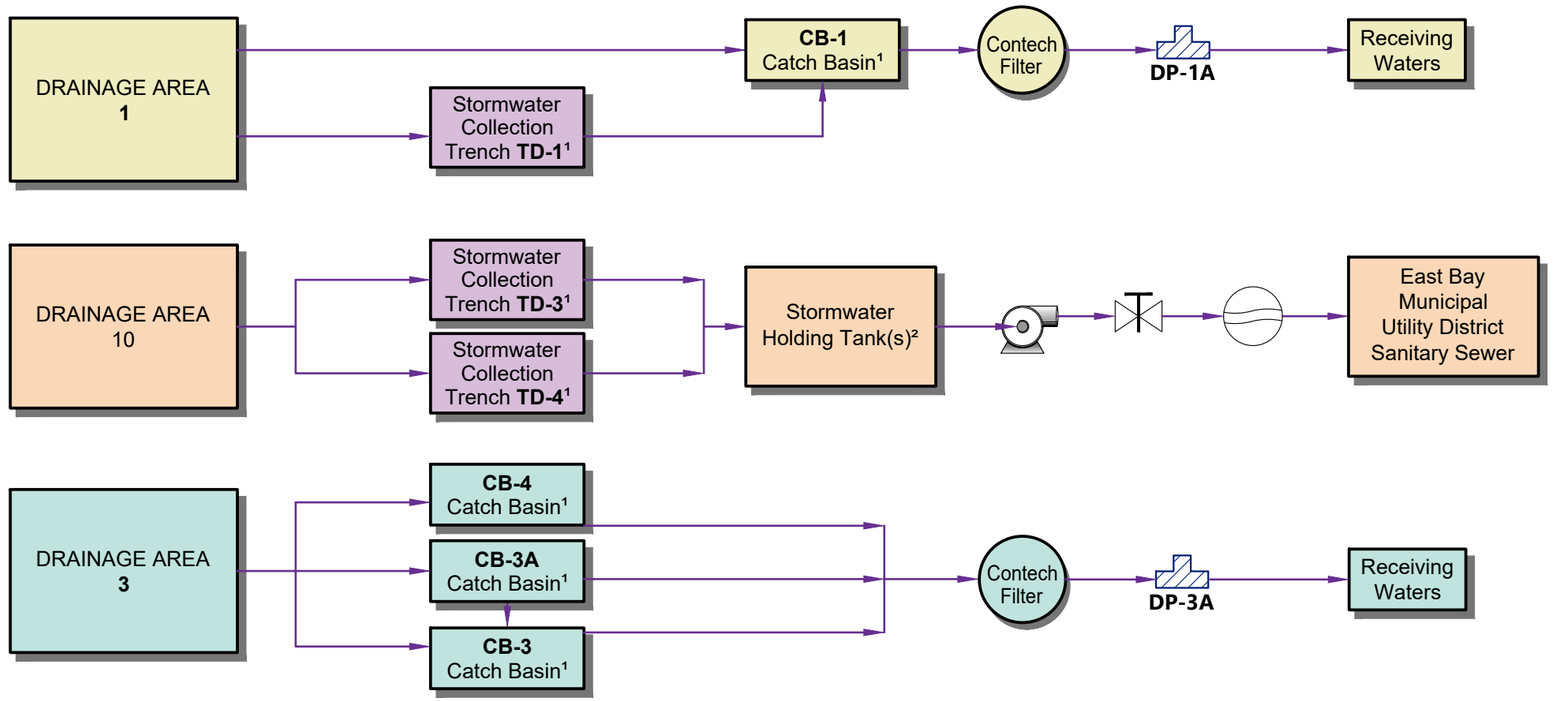
Note:
Site Plan provided by the Community Conservation Centers, Inc.

RECYCLING CENTER SITE PLAN
Berkeley Transfer Station and Recycling Center
1201 Second St. and 669 Gilman St.
Berkeley, California

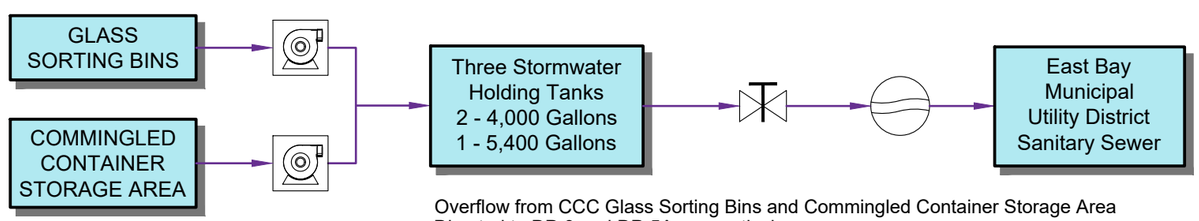
wsp

By: jrw
Date: 04/13/2026
Proj. No. 8620192840.6
Figure 4

BERKELEY TRANSFER STATION



COMMUNITY CONSERVATION CENTER (CCC)



Overflow from CCC Glass Sorting Bins and Commingled Container Storage Area Directed to DP-9 and DP-5A, respectively.

Explanation

- Effluent sample port
- Flow totalizer
- Pump
- Effluent sample location
- Drainage inlet with pump

Notes:

1. Sediment filter fabric inserts.
2. 21,000 gallon storage tank is used to capture stormwater. Additional 21,000-gallon storage tank is added as needed.

STORMWATER CAPTURE AND TREATMENT FLOW PROCESS DIAGRAM
Berkeley Transfer Station and Recycling Center
1201 Second St. and 669 Gilman St.
Berkeley, California

wsp

By: jrw
Date: 04/14/2026
Proj. No. 8620192840.6
Figure 5

Plot Date: 4/14/2026 1:47:41 PM, Plotted by: USJW720210 Drawing Path: WDS-Bay Area Graphics – Documents\City of Berkeley\22_0311_up_Trtmnt Flow Process.dwg, Flow Process (5025-220024)

APPENDIX A

Permit Registration Documents

CALIFORNIA Water Boards

State Water Resources Control Board

NOTICE OF INTENT

GENERAL PERMIT TO DISCHARGE STORM WATER
ASSOCIATED WITH INDUSTRIAL ACTIVITY (WQ ORDER No. 2014-0057-DWQ)
(Excluding Construction Activities)

EDMUND G. BROWN JR.
GOVERNOR

MATTHEW RODRIQUEZ
SECRETARY FOR
ENVIRONMENTAL PROTECTION

WDID: 2 01I009237 Status: Active

Operator Information

Type:

Name: City of Berkeley
Address: 1201 2nd St
Address 2:
City/State/Zip: Berkeley CA 94710
Federal Tax ID:

Contact Name: Elizabeth Brown
Title: Env. Comp. Specialist
Phone Number: 510-981-6629
Email Address: ejbrown@ci.berkeley.ca.us

Facility Information

Level:

Contact Name: Manuel Hector
Title: Acting Zero Wate Manager
Site Name: Berkeley City Transfer Station
Address: 1201 2nd St
City/State/Zip: Berkeley CA 94710
County: Alameda
Latitude: 37.88015 Longitude: -122.30636

Site Phone #: 510-981-2489
Email Address: mhector@cityofberkeley.info
Site Size: 7.4 Acres
Industrial Area Exposed to Storm Water: 2 Acres
Percent of Site Impervious (Including Rooftops): 98 %

SIC Code Information

1. 4212 Local Trucking Without Storage
2. 5093 Scrap and Waste Materials
3.

Additional Information

Receiving Water: San Francisco Bay
Flow: Indirectly
Storm Drain System: City of Berkeley
Compliance Group:

RWQCB Jurisdiction: Region 2 - San Francisco Bay
Phone: 510-622-2300
Email: r2_stormwater@waterboards.ca.gov

Certification

Name: Elizabeth Brown
Date: June 11, 2015
Title: Environmental Compliance Specialist

APPENDIX B

Example Training Log

TRAINED TEAM MEMBER LOG

Storm Water Management Training Log and Documentation

The following employees have received training in the subjects described below as required by NPDES Industrial Permit WDID #201I009237. Refresher training on these subjects must be conducted annually.

- Good Housekeeping
- Spill and Leak Prevention and Response Management
- Erosion and Sediment Controls
- Advanced BMPs
- Storm Water Sampling and Analysis
- Preventative Maintenance
- Material handling and waste
- Visual Monitoring
- Quality Assurance and Record Keeping

Training Objective: Annual Training on Industrial General Permit

Location: _____

Instructor: _____

Date: _____

Employee Name	Signature	Job Title	Date

Employee Name	Signature	Job Title	Date

Questions?
Contact Environmental Compliance (510) 981-6629

APPENDIX C

East Bay Municipal District Sanitary Sewer Discharge Permit

EAST BAY
MUNICIPAL UTILITY DISTRICT

BENNETT K. HORENSTEIN
DIRECTOR OF WASTEWATER

CERTIFIED MAIL
(Return Receipt Requested)
Certified Mail No. 7014 2870 0001 5084 5497

October 26, 2016

Ms. Joy Brown
City of Berkeley Municipal Transfer Station
1326 Allston Way
Berkeley, CA 94702

Dear Ms. Joy Brown:

Re: Wastewater Discharge Permit No. 51407600

Enclosed is the Wastewater Discharge Permit (Permit) for the Berkeley Transfer Station located at 1201 Second Street and the Community Conservation Center located at 669 Gilman Street in Berkeley. Please read the Permit and the enclosed Wastewater Discharge Permit Standard Terms and Conditions. As a permit holder, you are legally responsible for complying with all permit conditions and requirements.

Berkeley Transfer Station shall report to the EBMUD Environmental Services Division any changes, permanent or temporary, to the premises or operations that either significantly affects the quality or volume of wastewater discharge, or deviate from the terms and conditions under which the Permit was granted.

If you have any questions regarding this Permit, please contact Marie Kulka of the Environmental Services Division at (510) 287-1727.

Sincerely,

JACQUELINE T. ZIPKIN
Manager of Environmental Services

JTZ:MK:mk

W:\NAB\IDS\Permits\Stormwater Discharge Permits\Berkeley Municipal Transfer Station\Berkeley Transfer Station permit 2016.doc

Enclosures

Recycled Paper

WASTEWATER DISCHARGE PERMIT
Terms and Conditions
APPLICANT INFORMATION

APPLICANT BUSINESS NAME			PERMIT NUMBER
Berekely Transfer Station			51407600
ADDRESS OF SITE DISCHARGING WASTEWATER			
1201 Second Street and 669 Gilman Street	Berkeley		94710
STREET ADDRESS	CITY		ZIP CODE

PERSON TO BE CONTACTED REGARDING THIS APPLICATION

NAME	EMAIL ADDRESS	PHONE NUMBER	FAX NUMBER
Joy Brown	ejbrown@ci.berkeley.ca.us	(510) 981-6629	

PERSON(S) TO RECEIVE PERMIT AND CORRESPONDENCE IF DIFFERENT THAN PERSON SIGNING APPLICATION

NAME	MAILING ADDRESS
NAME	MAILING ADDRESS

PERSON TO BE CONTACTED IN THE EVENT OF AN EMERGENCY

NAME	DAYTIME TELEPHONE NUMBER	EVENING TELEPHONE NUMBER
Joy Brown	(510) 981-6629	(510) 774-5039

AUTHORIZATION

Joy Brown, Environmental Compliance Specialist — NAME & TITLE

is authorized to sign reports, documents, and other correspondence required by this Permit.

CERTIFICATION

I understand that I am legally responsible for discharge of wastewater from the facility and for complying with the Terms and Conditions of this Wastewater Discharge Permit.

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

NAME	TITLE
Joy Brown	Environmental Compliance Specialist
E. Joy Brown — Digitally signed by E. Joy Brown DN: cn=E. Joy Brown, o=City of Berkeley, ou=Public Works, email=ejbrown@cityofberkeley.info, c=US Date: 2016.08.15 11:54:12 -07'00'	8/15/16
SIGNATURE	DATE

(TO BE SIGNED BY CHIEF EXECUTIVE OFFICER OR DULY AUTHORIZED REPRESENTATIVE. SEE CERTIFICATION REQUIREMENTS ON REVERSE)

MAILING ADDRESS	PHONE NUMBER
1326 Allston Way, Berkeley, CA 94702	(510) 981-6629

EBMUD

WASTEWATER DISCHARGE PERMIT
Terms and Conditions
PROCESS DESCRIPTION

APPLICANT BUSINESS NAME Berkeley Transfer Station

The information on this form provides a description of wastewater generating processes, characteristics of the wastewater, and waste management activities. Instructions are on the back of this form.

Permit Number: 51407600

BUSINESS ACTIVITY
Solid waste transfer facility

Standard Industrial Classification: 5093, 4212

Business Classification Code:

PROCESSES

Process Description	Wastewater Characteristics	Schematic Process Number
solid construction debris	TSS, oil and grease, metals	1
general waste storage	TSS, oil and grease, COD, metals	2
commingled container storage	COD	3
glass sorting	COD	4

POLLUTION PREVENTION TECHNIQUES / BEST MANAGEMENT PRACTICES (BMPs)

Housekeeping (routine sweeping, spill cleanup), routine maintenance of equipment and vehicles, spill prevention (use of secondary containment and drip pans), covering materials with canopies, roofs, and tarps to prevent contact with potential pollutants, filter fabric, and absorbent booms.

For details in BMPs, see the attached Stormwater Pollution Prevention Plan for the Transfer Station.

PRETREATMENT

Pretreatment System	Design Capacity	Loading Rate	Size	Side Sewer Number
☐ filtration				
☐ grease trap/oil and water separator				
☐ granular activated carbon				
☐ sedimentation				
☐ pH adjustment				
☐ chlorination				
☐ chemical precipitation				
☐ other (describe)				
☒ none				

PROCESS GENERATED WASTE

Waste / Disposal Method	Annual Waste Generation Quantity	Annual Waste Generation Unit
None	--	--

EBMUD

WASTEWATER DISCHARGE PERMIT

APPLICANT BUSINESS NAME Berkeley Transfer Station

TERMS AND CONDITIONS

WATER BALANCE/STRENGTH SUMMARY

The information on this form describes the volume, source, and strength of wastewater discharged to the community sewer. Instructions are on the back of this form.

Permit Number 51407600

WATER USE AND WASTEWATER DISCHARGE BALANCE

Units expressed in: [✓] gallons per calendar day or [ ] gallons per working day (Number of working days per year____)

	Source			Wastewater Discharge to each Side Sewer					Water Diverted	Code
	EBMUD	Other	Code[1]	No.	No.	No.	No.	No.		
Sanitary										
Processes			B	21,000	4,000	-	-	-	0 gallons	-
Product										
Boiler										
Cooling										
Washing										
Irrigation										
Sub-total				21,000	4,000					
Total	All Sources	25,000		All Side Sewers		25,000	All Side Sewers + Water Diverted		25,000	
Maximum Daily Discharge (gallons)				21,000	4,000					

METERED WATER

Water Meter Number	Code[3]	Percent Discharge to each Side Sewer					Total % Discharge
None							

[1]Other / Code: Compute the average gallon per day water use from non-EBMUD sources and enter the value in the Other "Sub-total" box. Do not include sources that discharge only to the stormdrain. Allocate the subtotal value to each type of water use. Enter the code(s) that identifies the source water:

A= Well Water / Groundwater B= Stormwater C= Reclaimed Water D= Other (describe)

[2]Water Diverted/Code: Enter the diverted volume for each type of water use. Enter the code(s) that identifies the diversion:

A= Product B= Evaporation C= Irrigation D= Creek/Bay E= Rail, Truck, Vessel F= Other (describe)

[3]Metered Water Code(s): E= EBMUD Meter P= Private Meter

EBMUD

WASTEWATER DISCHARGE PERMIT

APPLICANT BUSINESS NAME Berkeley Transfer Station

TERMS AND CONDITIONS

#51407600

WATER BALANCE/STRENGTH SUMMARY

WASTEWATER STRENGTH ESTIMATES		Wastewater Discharge to each Side Sewer				
		No.	No.	No.	No.	No.
Total Suspended Solids mg/L (TSS)	Average					
	Maximum	See Table 1				
Chemical Oxygen Demand (mg/L)	Average					
	Maximum					

DISCHARGE FREQUENCY

Days of Week	Monday - Friday *				
Time of Day (Start & Stop Time)	8AM to 4PM				
Volume, if Batch Discharge	21,000 max	4,000 max			

*** Discharge only occurs Monday through Friday after storm events**

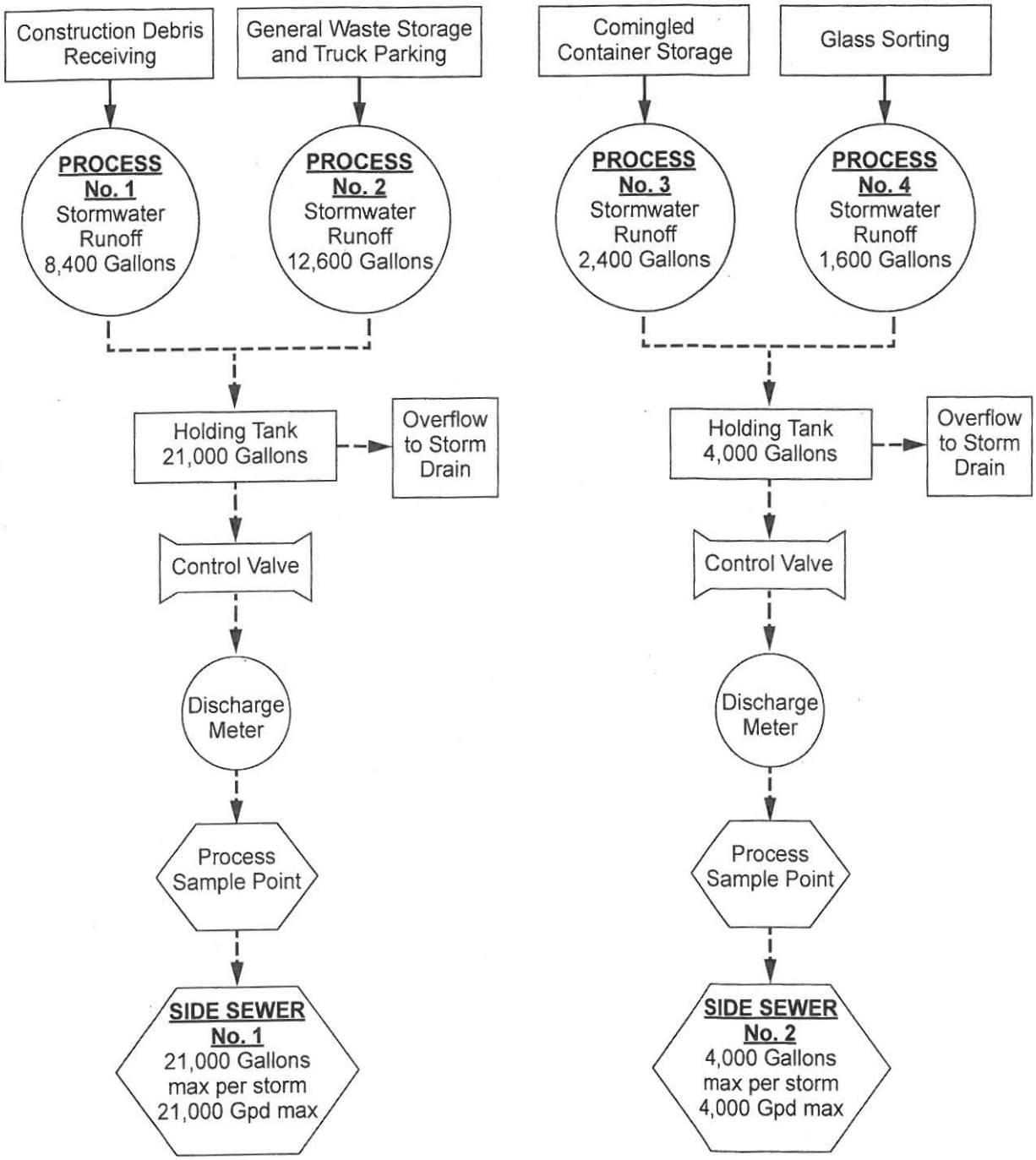
SIDE SEWER LOCATION

No.
No.
No.
No.
No.

STORMWATER AREA

Total square-foot area exposed to stormwater that drains to the sanitary sewer: 4,500 sq. ft.

Permit #51407600



LEGEND

Product Flow

Wastewater Flow

SCHEMATIC FLOW DIAGRAM
Berkeley Transfer Station
1201 Second St. and 669 Gilman St.
Berkeley, California

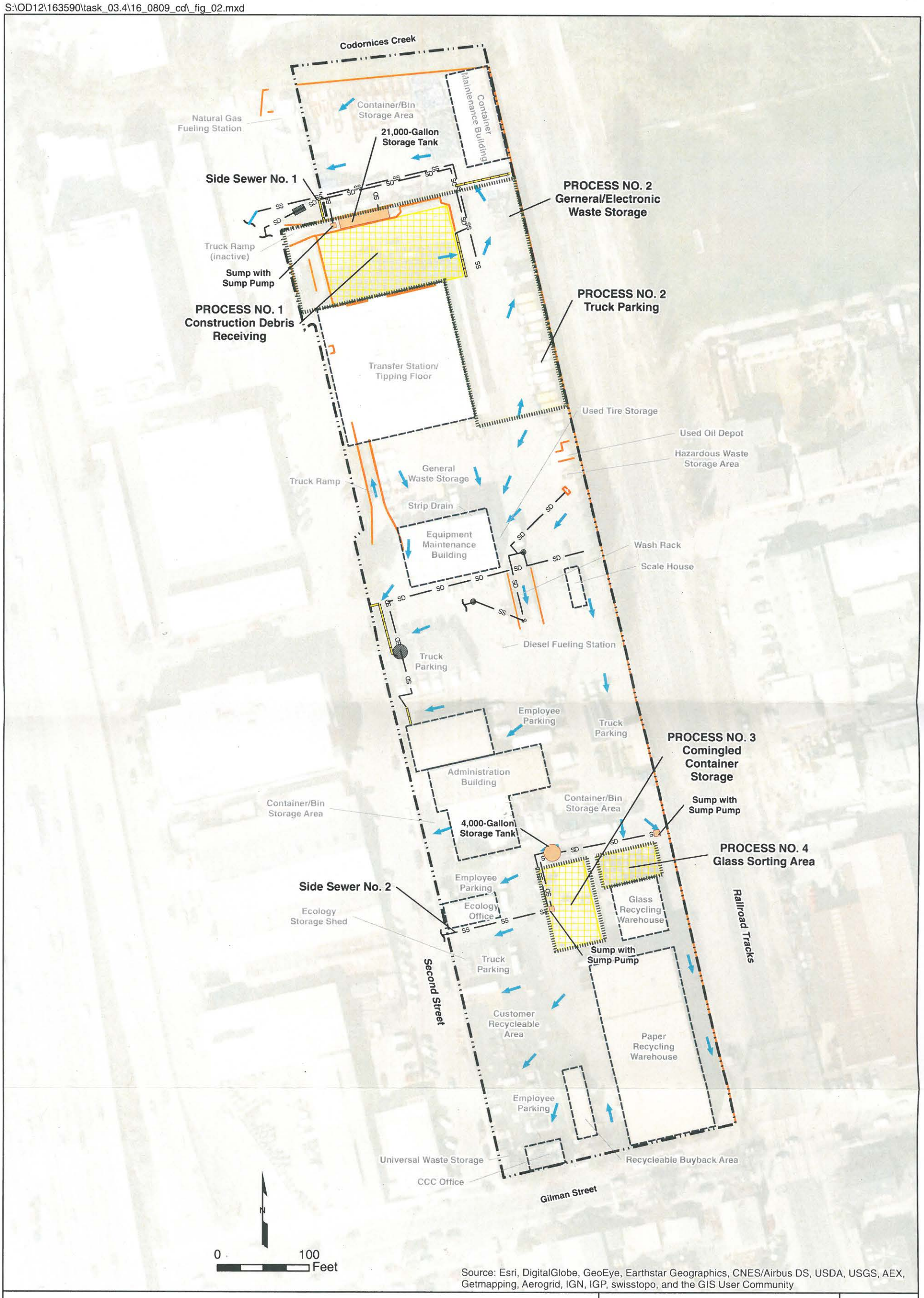
amec foster wheeler

Date: 08/09/2016 | Project No. 163590.025

Figure 1

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S:\OD12\163590\task_03.4\16_0809_cd_fig_02.mxd



Source: Esri, DigitalGlobe, GeoEye, Earthstar Geographics, CNES/Airbus DS, USDA, USGS, AEX, Getmapping, Aerogrid, IGN, IGP, swisstopo, and the GIS User Community

Explanation

- Flow Arrow
- Uncovered Activity Area
- Property Boundary
- Curb/Berm
- Storm Filter Vault
- SS — Sanitary Sewer
- SD — Storm Drain
- Trench Drain
- Building
- Area Discharges to Sanitary Sewer

FACILITY LAYOUT
Berkeley Transfer Station and Recycling Center
1201 Second St.and 669 Gilman St.
Berkeley, California

amec foster wheeler

Date: 08/12/2016 | Project No. OD12163590.03.4

Figure 2

GENERAL CONDITIONS

I. Berkeley Transfer Station shall comply with all items of the East Bay Municipal Utility District Wastewater Discharge Permit Standard Terms and Conditions, most recent edition.

II. Berkeley Transfer Station shall comply with East Bay Municipal Utility District Wastewater Control Ordinance (Ordinance).

This permit waives Title I, Section 5 (a) storm water prohibition to allow the discharge of treated storm water to the sanitary sewer at 1201 Second Street and 669 Gilman Street in Berkeley.

III. Berkeley Transfer Station shall practice pollution prevention techniques to reduce or eliminate pollutants released from the facility. Wastes remaining should be recycled whenever possible.

IV. Berkeley Transfer Station shall:

a. Seal all floor drains in the equipment/container service bays to prevent contaminated or polluted wastes from entering the sanitary sewers.
b. Post a sign in the equipment and container maintenance building and the wash rack work areas indicating that the discharge of automotive materials and waste to the sanitary sewer is prohibited.
c. Provide District access to hazardous waste manifest records upon request.

COMPLIANCE REQUIREMENTS

I. Berkeley Transfer Station shall treat collected storm water according to *Schematic Flow Diagram and Facility Layout* plans.

II. Berkeley Transfer Station shall immediately cease discharge of treated storm water if not in compliance with any of the terms and conditions of this wastewater discharge permit.

III. Berkeley Transfer Station shall not discharge storm water to the sanitary sewer during a rain event or within 24 hours after a rain event, which is defined as any precipitation greater than a drizzle.

IV. Berkeley Transfer Station shall not discharge storm water to the sanitary sewer at a flow rate greater than 100 gallons per minute.

V. Berkeley Transfer Station shall maintain a discharge logbook recording the date, time, and total volume of all storm water discharged to the sanitary sewer.

VI. Berkeley Transfer Station shall implement the Best Management Practices outlined in the current Storm Water Pollution Prevention Plan associated with the Industrial Storm Water General Permit to minimize storm water contamination and maintain the storm water collection system in proper operation condition.

REPORTING REQUIREMENTS

I. Berkeley Transfer Station shall submit a semi-annual Discharge Log Report, including:

- A copy of all entries recorded in the discharge logbook, described under *Compliance Requirements,* Paragraph V., for the six-month reporting period.
- The authorized signature and certification statement.

All semi-annual reports are due on the 15th of the month subsequent to the end of the six-month reporting period. The initial report is due by January 15, 2017, for reporting period July 2016 through December 2016. The following report is due by July 15, 2017 for reporting period January 2017 through June 2017, and so on.

II. Berkeley Transfer Station shall report to the Environmental Services Division any changes, permanent or temporary, to the premise or operations that change the quality or volume of the wastewater discharge or deviate from the terms and conditions under which this Permit is granted.

WASTEWATER DISCHARGE LIMITATIONS

Berkeley Transfer Station shall not discharge wastewater to the sanitary sewer if the strength of the wastewater exceeds Ordinance wastewater discharge limits:

SELF-MONITORING REPORTING REQUIREMENTS

I. Berkeley Transfer Station shall collect a representative sample **of the first batch of effluent discharge,** as depicted in the *Schematic Flow Diagram.*

II. Berkeley Transfer Station shall submit analytical data to EBMUD for approval prior to discharge. **Discharge to the sanitary sewer is prohibited until EBMUD reviews the self-monitoring report and grants approval.**

III. The self-monitoring report shall include:

a. A signed analytical report
b. The chain of custody documentation
c. The authorized signature and certification statement

IV. The parameters to be monitored, sample type, and analytical test methods shall be in accordance with the following table:

Parameter	Sample Type	Method
Total Suspend Solids	Grab	SM 2540D
Chemical Oxygen Demand	Grab	EPA 410.4
Total Metals[1]	Grab	EPA 200.7
Oil & Grease (HC)	Grab	EPA 1664 HEM-SGT

[1] cadmium; chromium; copper; lead; nickel; silver; zinc

V. The District may require monitoring of the treated storm water if deemed necessary. Berkeley monitored the treated storm water collected from the March 10, 2016 rain event. The treatment charges included in this wastewater discharge permit are based on the storm water's measured strength. The strengths may be revised based on new sampling data received and reviewed.

VI. The designated sampling location for compliance determination shall be the final effluent designated as the Process Sample Point on the *Schematic Flow Diagram* at the side sewer number 1 (SS No. 1) and side sewer number 2 (SS No. 2) locations as designated on the *Facility Layout* plan.

INSPECTIONS

The District shall conduct random, unannounced inspections to verify Berkeley Transfer Station compliance with the terms and conditions of this wastewater discharge permit. Berkeley Transfer Station shall grant District personnel access to the facility to conduct inspections and collect samples, in addition to reviewing the discharge logbook.

ENFORCEMENT AND PENALTIES

Failure to comply with the terms and conditions of this wastewater discharge permit may result in enforcement actions, including violation follow-up fees, civil enforcement penalties, and administrative fines of up to $5,000 per day.

WASTEWATER DISCHARGE PERMIT
Terms and Conditions

Rates Effective July 1, 2016, or
Revised Rates Approved by EBMUD Board of Directors thereafter

CALCULATION OF WASTEWATER DISPOSAL SERVICE CHARGES

Metered Water Consumption:

Wastewater discharged off metered water consumption will be charged at the Business Classification Code unit rate. Charges are determined by multiplying the metered consumption by the percent discharged and then by the treatment rate.

Account Number	Meter Number	BCC Code	BCC Description	Percent Discharged	Fixed Volume Ccf / month	Meter Rate $/Ccf
51407600	40474028	6800	OFFICES	100.0%	0	$2.48

Storm water Discharges:

Unit Treatment Rates

Total Suspended Solids (TSS) =	$0.469	/pound
Chemical Oxygen Demand-Filtered (CODF) =	$0.321	/pound
Volume =	$1.085	/hundred cubic feet ($/Ccf)

Step 1: Convert concentrations of TSS and CODF to $/Ccf for each side sewer (SS).
The concentration in mg/L, multiplied by 0.00624 and the rate/pound, equals $/Ccf.
The side sewer rate is the sum of TSS + CODF + Volume.

	Side sewer #1		Side sewer #2	
	mg/L	$/Ccf	mg/L	$/Ccf
TSS	470	1.375	300	0.878
CODF	350	0.701	1600	3.205
Volume		1.085		1.085
SS rate =		$3.16		$5.17

Step 2: Determine the rate to be applied to each water meter. The meter rate is the sum of the individual side sewer contributions.

	Side sewer #1		Side sewer #2		Total
SS rate =	3.16	$/Ccf	5.17	$/Ccf	for meter
Meter No.	%	$/Ccf	%	$/Ccf	$/Ccf
Stormwater	87.0%	2.75	13.0%	0.67	3.42

Step 3: Wastewater Charges are determined by multiplying the metered volume by the percent discharged, plus any fixed volume, all multiplied by the meter rate.

Account Number	Meter Number	Units	Percent Discharged	Fixed Volume Ccf/month	Meter Rate $/Ccf
51407600	Stormwater	Ccf	0.0%	0	3.42

WASTEWATER DISCHARGE PERMIT

Terms and Conditions

CALCULATION OF WASTEWATER CAPACITY FEE FOR STORMWATER DISCHARGES

The wastewater capacity fee is calculated by multiplying the reported maximum monthly stormwater discharge volume by applicable fee in effect at start-up of the discharge. Each month, 1/36 of the capacity fee will be assessed, until the entire fee has been paid in 3 years.

The wastewater capacity fee is calculated as follows:

Discharge volume = 154 Ccf maximum monthly (115,561 gallons)

Unit Treatment Rates

Total Suspended Solids (TSS) =	$63.10	/pound
Chemical Oxygen Demand-Filtered (CODF) =	$46.88	/pound
Volume =	$159.07	/hundred cubic feet ($/Ccf)

SS#1

lbs/mth					
CODF (mg/l)		C.Factor		Ccf/mo	lbs/mo
350	*	0.00624	*	134.00 =	292.66
TSS (mg/l)					
470	*	0.00624	*	134.00 =	393.00

WCF

Flow	134.00 *	$159.07 =	$21,315.38
CODF	292.66 *	$46.88 =	$13,719.71
	393.00 *	$63.10 =	$24,798.00
SS#1 WCF Total:			**$59,833.09**

SS#2

lbs/mth					
CODF (mg/l)		C.Factor		Ccf/mo	lbs/mo
1600	*	0.00624	*	20.00 =	199.68
TSS (mg/l)					
300	*	0.00624	*	20.00 =	37.44

WCF

Flow	20.00 *	$159.07 =	$3,181.40
CODF	199.68 *	$46.88 =	$9,361.00
	37.44 *	$63.10 =	$2,362.46
SS#2 WCF Total:			**$14,904.86**

Total Wastewater Capacity Fee: $74,737.95

Monthly Capacity Fee applied to the first 36 months of the permit term: $2,076.05

SD-30.7 2/91

WASTEWATER DISCHARGE PERMIT

Terms and Conditions

Rates Effective July 1, 2016, or
Revised Rates Approved by EBMUD Board of Directors thereafter

Berkeley Transfer Station
Permit No. 51407600

FEES AND WASTEWATER CHARGES

The following fees and charges are due when billed by the District:

Annual Permit Fee:	$2,450.00
Inspecton Fee*:	$1,300.00
Monthly Wastewater Capacity Fee (36 Mths):	$2,076.05

*Inspection every five years. Fee assessed after inspection is conducted. The District may revised the permit to increase inspection frequency as necessary.

WASTEWATER DISPOSAL SERVICE CHARGE

Account Number	Meter Number	BCC Code	BCC Description	Percent Discharged	Fixed Volume Ccf / month	Meter Rate $/Ccf
51407600	40474028	6800	OFFICES	100%	0	$2.48
51407600	Stormwater				*	$3.42

* Volume based discharge values reported in the semi-annual reports. Fee will be assessed upon receipt of the Discharge Report.

The District may change the terms and conditions of a Wastewater Discharge Permit, including changing the average limits on the elements of wastewater strength and rates and charges, from time to time as circumstances may require. Treatment rate updates shall be provided by means of a rate revision letter. The District shall allow a discharger reasonable time to comply with any District required changes in the permit except that a change in average limits of wastewater strength shall immediately affect calculation of the wastewater disposal charge.

Authorization

Permit Holder shall report to EBMUD, Wastewater Department any changes, permanent or temporary, to the premises or operations that significantly change the quality or volume of the wastewater discharge or deviation from the terms and conditions under which this permit is granted.

Permit Holder is hereby authorized to discharge wastewater to the community sewer, subject to said Applicant's compliance with *East Bay Municipal Utility District Wastewater Control Ordinance* as well as the permit terms and conditions.

Effective: 10/24/16 — Director, Wastewater Department — Date: 10/24/16

WASTEWATER DISCHARGE PERMIT
Terms and Conditions
APPLICANT INFORMATION

APPLICANT BUSINESS NAME			PERMIT NUMBER
Berekely Transfer Station			51407600
ADDRESS OF SITE DISCHARGING WASTEWATER			
1201 Second Street and 669 Gilman Street	Berkeley		94710
STREET ADDRESS	CITY		ZIP CODE

PERSON TO BE CONTACTED REGARDING THIS APPLICATION

NAME	EMAIL ADDRESS	PHONE NUMBER	FAX NUMBER
Joy Brown	ejbrown@ci.berkeley.ca.us	(510) 981-6629	

PERSON(S) TO RECEIVE PERMIT AND CORRESPONDENCE IF DIFFERENT THAN PERSON SIGNING APPLICATION

NAME	MAILING ADDRESS
NAME	MAILING ADDRESS

PERSON TO BE CONTACTED IN THE EVENT OF AN EMERGENCY

NAME	DAYTIME TELEPHONE NUMBER	EVENING TELEPHONE NUMBER
Joy Brown	(510) 981-6629	(510) 774-5039

AUTHORIZATION

Joy Brown, Environmental Compliance Specialist — NAME & TITLE

is authorized to sign reports, documents, and other correspondence required by this Permit.

CERTIFICATION

I understand that I am legally responsible for discharge of wastewater from the facility and for complying with the Terms and Conditions of this Wastewater Discharge Permit.

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

NAME	TITLE
Joy Brown	Environmental Compliance Specialist
E. Joy Brown — Digitally signed by E. Joy Brown DN: cn=E. Joy Brown, o=City of Berkeley, ou=Public Works, email=ejbrown@cityofberkeley.info, c=US Date: 2016.08.15 11:54:12 -07'00'	8/15/16
SIGNATURE	DATE

(TO BE SIGNED BY CHIEF EXECUTIVE OFFICER OR DULY AUTHORIZED REPRESENTATIVE. SEE CERTIFICATION REQUIREMENTS ON REVERSE)

MAILING ADDRESS	PHONE NUMBER
1326 Allston Way, Berkeley, CA 94702	(510) 981-6629

EBMUD

WASTEWATER DISCHARGE PERMIT
Terms and Conditions
PROCESS DESCRIPTION

APPLICANT BUSINESS NAME Berkeley Transfer Station

The information on this form provides a description of wastewater generating processes, characteristics of the wastewater, and waste management activities. Instructions are on the back of this form.

Permit Number: 51407600

BUSINESS ACTIVITY
Solid waste transfer facility

Standard Industrial Classification: 5093, 4212

Business Classification Code:

PROCESSES

Process Description	Wastewater Characteristics	Schematic Process Number
solid construction debris	TSS, oil and grease, metals	1
general waste storage	TSS, oil and grease, COD, metals	2
commingled container storage	COD	3
glass sorting	COD	4

POLLUTION PREVENTION TECHNIQUES / BEST MANAGEMENT PRACTICES (BMPs)

Housekeeping (routine sweeping, spill cleanup), routine maintenance of equipment and vehicles, spill prevention (use of secondary containment and drip pans), covering materials with canopies, roofs, and tarps to prevent contact with potential pollutants, filter fabric, and absorbent booms.

For details in BMPs, see the attached Stormwater Pollution Prevention Plan for the Transfer Station.

PRETREATMENT

Pretreatment System	Design Capacity	Loading Rate	Size	Side Sewer Number
☐ filtration				
☐ grease trap/oil and water separator				
☐ granular activated carbon				
☐ sedimentation				
☐ pH adjustment				
☐ chlorination				
☐ chemical precipitation				
☐ other (describe)				
☒ none				

PROCESS GENERATED WASTE

Waste / Disposal Method	Annual Waste Generation Quantity	Annual Waste Generation Unit
None	--	--

EBMUD

WASTEWATER DISCHARGE PERMIT

APPLICANT BUSINESS NAME Berkeley Transfer Station

TERMS AND CONDITIONS

WATER BALANCE/STRENGTH SUMMARY

The information on this form describes the volume, source, and strength of wastewater discharged to the community sewer. Instructions are on the back of this form.

Permit Number 51407600

WATER USE AND WASTEWATER DISCHARGE BALANCE

Units expressed in: [✓] gallons per calendar day or [ ] gallons per working day (Number of working days per year_____)

	Source			Wastewater Discharge to each Side Sewer					Water Diverted	Code
	EBMUD	Other	Code[1]	No.	No.	No.	No.	No.		
Sanitary										
Processes			B	21,000	4,000 8,000		-	-	0 gallons	-
Product										
Boiler										
Cooling										
Washing										
Irrigation										
Sub-total	29,000			21,000	4,000 8,000					
Total	All Sources ~~25,000~~			All Side Sewers		25,000	All Side Sewers + Water Diverted 29,000		~~25,000~~	
Maximum Daily Discharge (gallons)				21,000	4,000 8,000					

METERED WATER

Water Meter Number	Code[3]	Percent Discharge to each Side Sewer					Total % Discharge
None							

[1]Other / Code: Compute the average gallon per day water use from non-EBMUD sources and enter the value in the Other "Sub-total" box. Do not include sources that discharge only to the stormdrain. Allocate the subtotal value to each type of water use. Enter the code(s) that identifies the source water:

A= Well Water / Groundwater *B= Stormwater* *C= Reclaimed Water* *D= Other (describe)*

[2]Water Diverted/Code: Enter the diverted volume for each type of water use. Enter the code(s) that identifies the diversion:

A= Product *B= Evaporation* *C= Irrigation* *D= Creek/Bay* *E= Rail, Truck, Vessel* *F= Other (describe)*

[3]Metered Water Code(s): E= EBMUD Meter P= Private Meter

EBMUD

WASTEWATER DISCHARGE PERMIT

APPLICANT BUSINESS NAME Berkeley Transfer Station

TERMS AND CONDITIONS

#51407600

WATER BALANCE/STRENGTH SUMMARY

WASTEWATER STRENGTH ESTIMATES		Wastewater Discharge to each Side Sewer				
		No.	No.	No.	No.	No.
Total Suspended Solids mg/L (TSS)	Average					
	Maximum	See Table 1				
Chemical Oxygen Demand (mg/L)	Average					
	Maximum					

DISCHARGE FREQUENCY

Days of Week	Monday - Friday *				
Time of Day (Start & Stop Time)	8AM to 4PM				
Volume, if Batch Discharge	21,000 max	~~4,000~~ max 8,000			

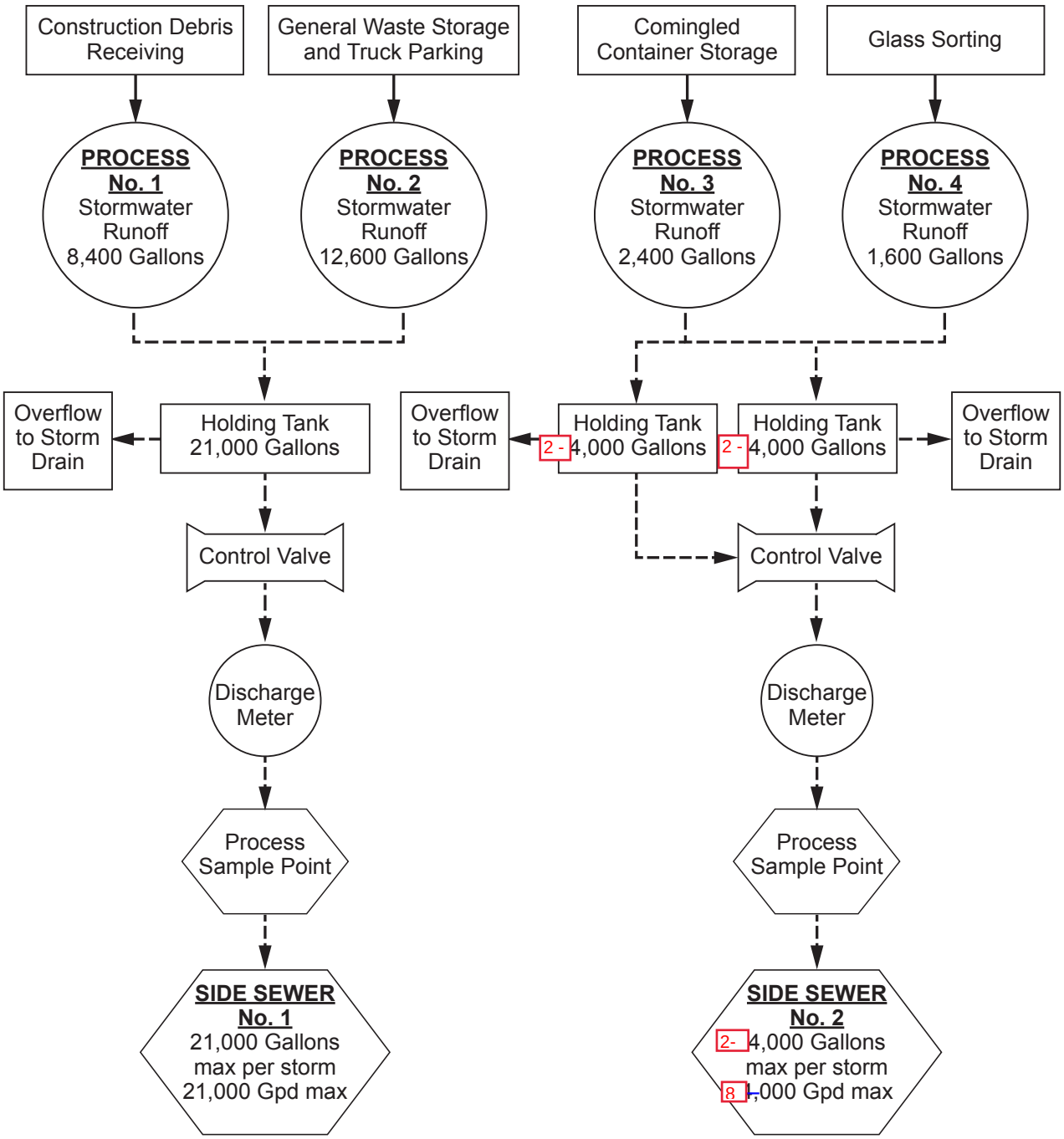
*** Discharge only occurs Monday through Friday after storm events**

SIDE SEWER LOCATION

No.
No.
No.
No.
No.

STORMWATER AREA

Total square-foot area exposed to stormwater that drains to the sanitary sewer: 4,500 sq. ft.



LEGEND

Product Flow

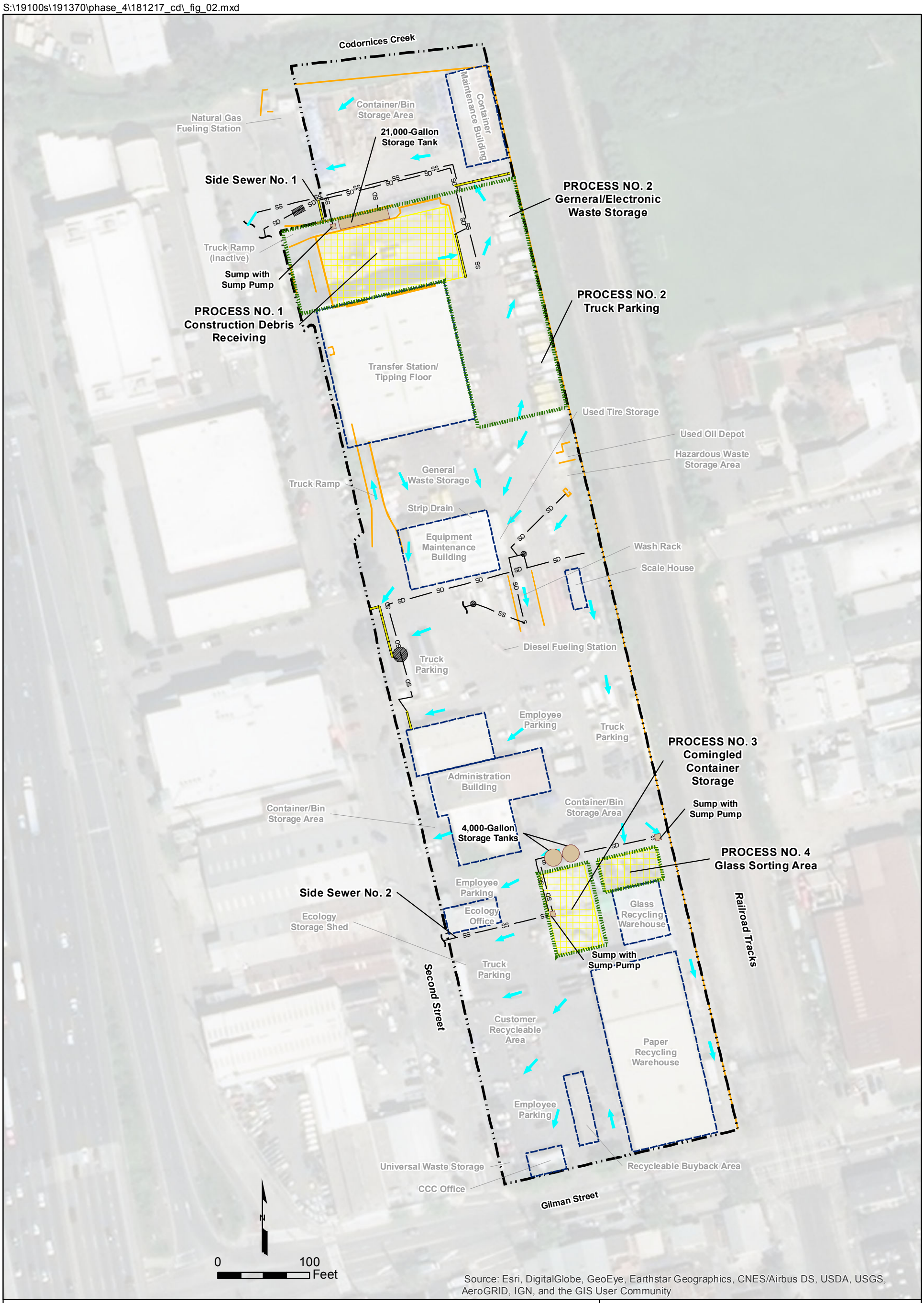
Wastewater Flow

SCHEMATIC FLOW DIAGRAM
Berkeley Transfer Station
1201 Second St. and 669 Gilman St.
Berkeley, California

wood.	By: DMO	Proj. No. 8618191370.4
	Date: 12/18/2018	Figure 1

S:\19100s\191370\phase_4\181217_cd_fig_01.ai

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Explanation

- Flow Arrow
- Uncovered Activity Area
- Property Boundary
- Curb/Berm
- Storm Filter Vault
- SS — Sanitary Sewer
- SD — Storm Drain
- Trench Drain
- Building
- Area Discharges to Sanitary Sewer

FACILITY LAYOUT
Berkeley Transfer Station and Recycling Center
1201 Second St.and 669 Gilman St.
Berkeley, California

wood.

By: DMO
Date: 12/18/2018
Project No. 8618191370.4
Figure 2

SPECIAL DISCHARGE PERMIT
Terms and Conditions

PERMIT NUMBER: 51407600

GENERAL CONDITIONS

Berkeley Transfer Station shall comply with the provisions of the following two documents:

- EBMUD Wastewater Control Ordinance (Wastewater Control Ordinance)
- EBMUD Special Discharge Permit Standard Terms and Conditions, most recent edition

II. This Special Discharge Permit is a waiver of Wastewater Control Ordinance, Title I, Section 5, which prohibits the discharge of stormwater, drainage water, and groundwater to the community sewer.

III. Berkeley Transfer Station shall discharge Special Discharge Wastewater only from the site described in the Special Discharge Permit Applicant Form.

IV. Berkeley Transfer Station shall immediately cease discharge of treated or managed Special Discharge Wastewater if not in compliance with any of the terms and conditions of this Special Discharge Permit.

V. Berkeley Transfer Station shall not discharge Special Discharge Wastewater authorized by this Special Discharge Permit after the expiration date.

COMPLIANCE REQUIREMENTS

I. Berkeley Transfer Station shall pretreat/manage, including sediment control, all Special Discharge Wastewater prior to discharge to the community sewer. Pretreatment or management shall be sufficient to achieve compliance with the discharge limits established in this Special Discharge Permit.

II. Berkeley Transfer Station shall post a sign in the work area stating, "All Wastewater Discharge must comply with the Special Discharge Permit."

III. Berkeley Transfer Station shall not discharge Special Discharge Wastewater to the community sewer at a flow rate greater than 100 gallons per minute. The flow rate for discharges made directly to the EBMUD interceptor is under the discretion of the EBMUD Construction Inspector.

IV. Berkeley Transfer Station shall not discharge Special Discharge Wastewater to the community sewer during a rain event or within 24 hours after a rain event, which is defined as any precipitation greater than a drizzle.

V. Berkeley Transfer Station shall obtain permission from the applicable local agency to discharge Special Discharge Wastewater to the community sewer. Note that local sewer authorities may have different or more restrictive discharge requirements than EBMUD.

VI. Berkeley Transfer Station shall discharge all Special Discharge Wastewater to the community sewer through a totalizing flow meter.

VII. Berkeley Transfer Station shall maintain a discharge logbook. Each entry shall include the date, time, source, and total volume of all Special Discharge Wastewater discharged to the community sewer.

REPORTING REQUIREMENTS

I. Berkeley Transfer Station shall submit quarterly discharge log reports, including:

- A copy of discharge logs which include dates, times, volumes, flow totalizer readings, and the total volume of Special Discharge Wastewater discharged to the sanitary sewer to date.
- The authorized signature and certification statement.

DRAFT

II. The discharge log report is due according to the following schedule:

Discharge Period	Discharge Log Due
July 1 – December 31, 2021, 2022, 2023, 2024, 2025	January 31, 2022, 2023, 2024, 2025
January 1 – June 30, 2022, 2023, 2024, 2025, 2026	July 31, 2022, 2023, 2024, 2025, 2026

III. Submit all reports to EBMUD by email to adam.kern@ebmud.com.

SELF-MONITORING REQUIREMENTS

I. This permit requires Berkeley Transfer Station to submit analytical data to EBMUD due by January 31, 2022 for samples at side sewer no. 1 (transfer station), and side sewer no. 2 recycling center. The parameters to be monitored, sample type, and analytical test methods shall be in accordance with the following table:

Parameter	Sample Type	Method
Metals *(arsenic, cadmium, chromium, copper, iron, lead, nickel, silver, zinc)*	Grab	EPA 200.8 or 200.7
Oil & Grease (HC)	Grab	EPA 1664 HEM-SGT
Total Suspended Solids	Grab	SM2540D
Chemical Oxygen Demand	Grab	EPA 410.4

EBMUD may prohibit the discharge of the Special Discharge Wastewater and require additional treatment if any constituents exceed Wastewater Discharge Limits set forth by the permit. Subsequent testing may be required depending on the initial test results.

II. The self-monitoring report shall be submitted by email to adam.kern@ebmud.com. The self-monitoring report shall include:

- A signed analytical report
- The chain of custody documentation
- The authorized signature and certification statement

WASTEWATER DISCHARGE LIMITS

I. Berkeley Transfer Station shall not discharge Special Discharge Wastewater to the community sewer if the strength of the wastewater exceeds Wastewater Control Ordinance Discharge Limits.

DRAFT

INSPECTIONS

The District may conduct random, unannounced inspections to verify compliance with the terms and conditions of this Special Discharge Permit. Berkeley Transfer Station shall grant District personnel site access to conduct inspections and collect Special Discharge Wastewater samples.

ENFORCEMENT AND PENALTIES

Failure to comply with the terms and conditions of this Special Discharge Permit may result in enforcement actions, including violation follow-up fees, civil enforcement penalties, and administrative fines of up to $5,000 per day.

RATES AND CHARGES

This Special Discharge Permit may be amended to include changes to rates and charges that may be established by the District during the term of this Special Discharge Permit.

DRAFT

AUTHORIZATION

Berkeley Transfer Station is hereby authorized to discharge Special Discharge Wastewater to the community sewer, subject to compliance with *EBMUD Wastewater Control Ordinance, Special Discharge Permit Standard Terms and Conditions*, and established billing conditions.

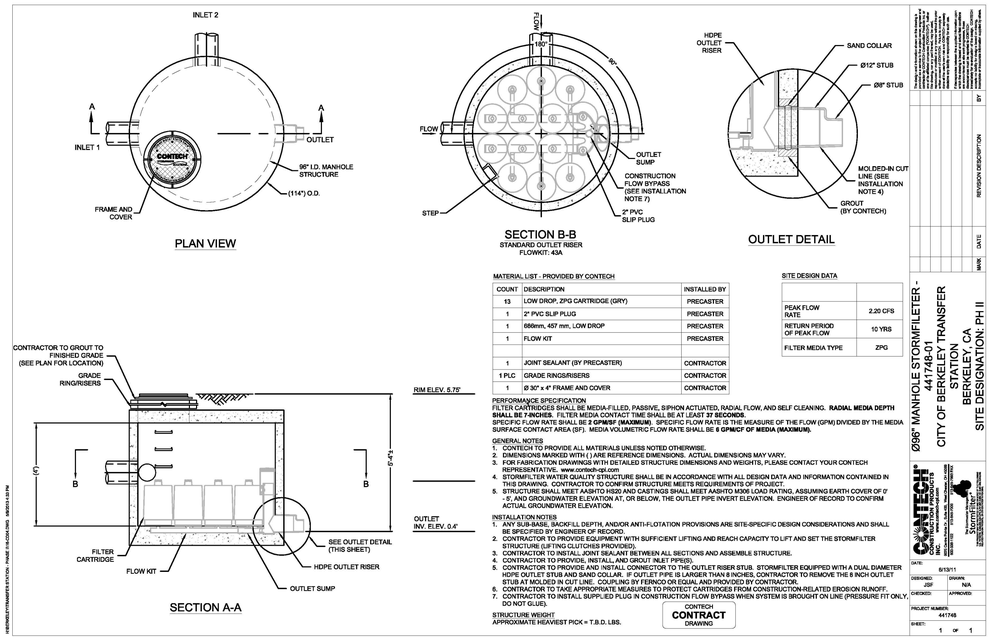
Expiration: October 25, 2026

_____ _____

Director, Wastewater Department — Effective Date

APPENDIX D

Contect Filter Descriptions and Specifications



MATERIAL LIST - PROVIDED BY CONTECH

COUNT	DESCRIPTION	INSTALLED BY
13	LOW DROP, ZPG CARTRIDGE (GRY)	PRECASTER
1	2" PVC SLIP PLUG	PRECASTER
1	660mm, 45° mm, LOW DROP	PRECASTER
1	FLOW KIT	PRECASTER
1	JOINT SEALANT (BY PRECASTER)	CONTRACTOR
1 PLC	GRADE RINGS/RISERS	CONTRACTOR
1	Ø 30" x 4" FRAME AND COVER	CONTRACTOR

SITE DESIGN DATA

PEAK FLOW RATE	2.20 CFS
RETURN PERIOD OF PEAK FLOW	10 YRS
FILTER MEDIA TYPE	ZPG

PERFORMANCE SPECIFICATION
FILTER CARTRIDGES SHALL BE MEDIA-FILLED, PASSIVE, SIPHON ACTUATED, RADIAL FLOW, AND SELF CLEANING. **RADIAL MEDIA DEPTH SHALL BE 7-INCHES**. FILTER MEDIA CONTACT TIME SHALL BE AT LEAST **37 SECONDS**.
SPECIFIC FLOW RATE SHALL BE **2 GPM/SF (MAXIMUM)**. SPECIFIC FLOW RATE IS THE MEASURE OF THE FLOW (GPM) DIVIDED BY THE MEDIA SURFACE CONTACT AREA (SF). MEDIA VOLUMETRIC FLOW RATE SHALL BE **6 GPM/CF OF MEDIA (MAXIMUM).**

GENERAL NOTES
1. CONTECH TO PROVIDE ALL MATERIALS UNLESS NOTED OTHERWISE.
2. DIMENSIONS MARKED WITH ( ) ARE REFERENCE DIMENSIONS. ACTUAL DIMENSIONS MAY VARY.
3. FOR FABRICATION DRAWINGS WITH DETAILED STRUCTURE DIMENSIONS AND WEIGHTS, PLEASE CONTACT YOUR CONTECH REPRESENTATIVE. www.contech-cpi.com
4. STORMFILTER WATER QUALITY STRUCTURE SHALL BE IN ACCORDANCE WITH ALL DESIGN DATA AND INFORMATION CONTAINED IN THIS DRAWING. CONTRACTOR TO CONFIRM STRUCTURE MEETS REQUIREMENTS OF PROJECT.
5. STRUCTURE SHALL MEET AASHTO H20 AND CASTINGS SHALL MEET AASHTO M306 LOAD RATING, ASSUMING EARTH COVER OF 0' - 5', AND GROUNDWATER ELEVATION AT, OR BELOW, THE OUTLET PIPE INVERT ELEVATION. ENGINEER OF RECORD TO CONFIRM ACTUAL GROUNDWATER ELEVATION.

INSTALLATION NOTES
1. ANY SUB-BASE, BACKFILL DEPTH, AND/OR ANTI-FLOTATION PROVISIONS ARE SITE-SPECIFIC DESIGN CONSIDERATIONS AND SHALL BE SPECIFIED BY ENGINEER OF RECORD.
2. CONTRACTOR TO PROVIDE EQUIPMENT WITH SUFFICIENT LIFTING AND REACH CAPACITY TO LIFT AND SET THE STORMFILTER STRUCTURE (LIFTING CLUTCHES PROVIDED).
3. CONTRACTOR TO INSTALL JOINT SEALANT BETWEEN ALL SECTIONS AND ASSEMBLE STRUCTURE.
4. CONTRACTOR TO PROVIDE, INSTALL, AND GROUT INLET PIPE(S).
5. CONTRACTOR TO PROVIDE AND INSTALL CONNECTOR TO THE OUTLET RISER STUB. STORMFILTER EQUIPPED WITH A DUAL DIAMETER HDPE OUTLET STUB AND SAND COLLAR. IF OUTLET PIPE IS LARGER THAN 8 INCHES, CONTRACTOR TO REMOVE THE 8 INCH OUTLET STUB AT MOLDED IN CUT LINE. COUPLING BY FERNCO OR EQUAL AND PROVIDED BY CONTRACTOR.
6. CONTRACTOR TO TAKE APPROPRIATE MEASURES TO PROTECT CARTRIDGES FROM CONSTRUCTION-RELATED EROSION RUNOFF.
7. CONTRACTOR TO INSTALL SUPPLIED PLUG IN CONSTRUCTION FLOW BYPASS WHEN SYSTEM IS BROUGHT ON LINE (PRESSURE FIT ONLY, DO NOT GLUE).

STRUCTURE WEIGHT
APPROXIMATE HEAVIEST PICK = T.B.D. LBS.

CONTECH
CONTRACT
DRAWING

Ø96" MANHOLE STORMFILETER - 441748-01
CITY OF BERKELEY TRANSFER STATION
BERKELEY, CA
SITE DESIGNATION: PH II

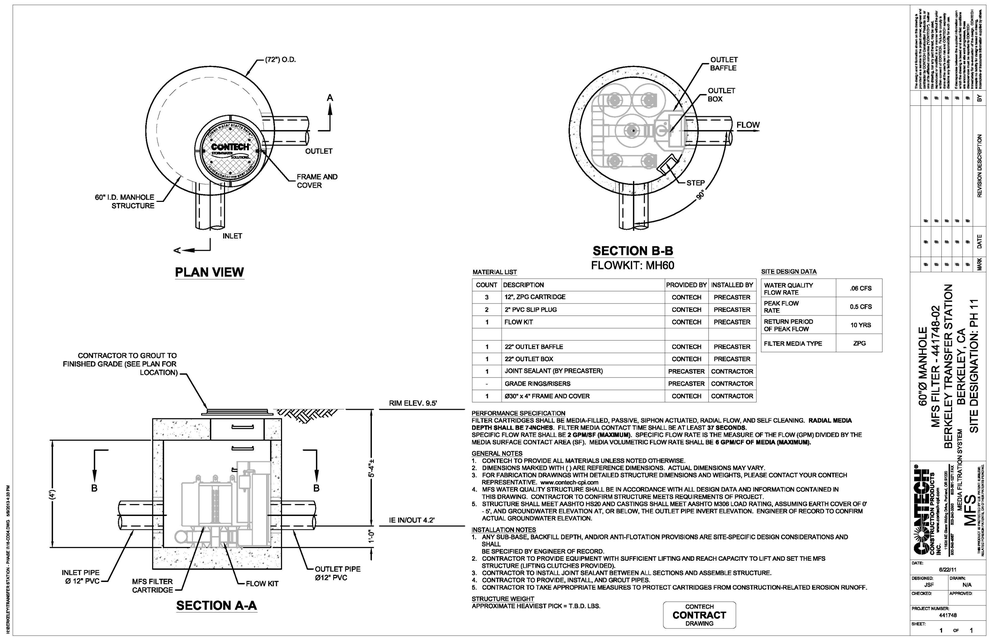
CONTECH Construction Products Inc.
Stormwater Management
StormFilter

DATE: 5/13/11
DESIGNED: JSF
DRAWN: N/A
CHECKED:
APPROVED:
PROJECT NUMBER: 441748
SHEET: 1 OF 1

1 STORM FILTER AT MAIN GATE

SCALE: NOT TO SCALE

NOTE: THIS DRAWING IS FOR INFORMATIONAL PURPOSES ONLY



MATERIAL LIST

COUNT	DESCRIPTION	PROVIDED BY	INSTALLED BY
3	12", ZPG CARTRIDGE	CONTECH	PRECASTER
2	2" PVC SLIP PLUG	CONTECH	PRECASTER
1	FLOW KIT	CONTECH	PRECASTER
1	22" OUTLET BAFFLE	CONTECH	PRECASTER
1	22" OUTLET BOX	CONTECH	PRECASTER
1	JOINT SEALANT (BY PRECASTER)	PRECASTER	CONTRACTOR
-	GRADE RINGS/RISERS	PRECASTER	CONTRACTOR
1	Ø30" x 4" FRAME AND COVER	CONTECH	CONTRACTOR

SITE DESIGN DATA

WATER QUALITY FLOW RATE	.06 CFS
PEAK FLOW RATE	0.5 CFS
RETURN PERIOD OF PEAK FLOW	10 YRS
FILTER MEDIA TYPE	ZPG

PERFORMANCE SPECIFICATION
FILTER CARTRIDGES SHALL BE MEDIA-FILLED, PASSIVE, SIPHON ACTUATED, RADIAL FLOW, AND SELF CLEANING. **RADIAL MEDIA DEPTH SHALL BE 7-INCHES**. FILTER MEDIA CONTACT TIME SHALL BE AT LEAST **37 SECONDS**.
SPECIFIC FLOW RATE SHALL BE **2 GPM/SF (MAXIMUM)**. SPECIFIC FLOW RATE IS THE MEASURE OF THE FLOW (GPM) DIVIDED BY THE MEDIA SURFACE CONTACT AREA (SF). MEDIA VOLUMETRIC FLOW RATE SHALL BE **6 GPM/CF OF MEDIA (MAXIMUM).**

GENERAL NOTES
1. CONTECH TO PROVIDE ALL MATERIALS UNLESS NOTED OTHERWISE.
2. DIMENSIONS MARKED WITH ( ) ARE REFERENCE DIMENSIONS. ACTUAL DIMENSIONS MAY VARY.
3. FOR FABRICATION DRAWINGS WITH DETAILED STRUCTURE DIMENSIONS AND WEIGHTS, PLEASE CONTACT YOUR CONTECH REPRESENTATIVE. www.contech-cpi.com
4. MFS WATER QUALITY STRUCTURE SHALL BE IN ACCORDANCE WITH ALL DESIGN DATA AND INFORMATION CONTAINED IN THIS DRAWING. CONTRACTOR TO CONFIRM STRUCTURE MEETS REQUIREMENTS OF PROJECT.
5. STRUCTURE SHALL MEET AASHTO H20 AND CASTINGS SHALL MEET AASHTO M306 LOAD RATING, ASSUMING EARTH COVER OF 0' - 5', AND GROUNDWATER ELEVATION AT, OR BELOW, THE OUTLET PIPE INVERT ELEVATION. ENGINEER OF RECORD TO CONFIRM ACTUAL GROUNDWATER ELEVATION.

INSTALLATION NOTES
1. ANY SUB-BASE, BACKFILL DEPTH, AND/OR ANTI-FLOTATION PROVISIONS ARE SITE-SPECIFIC DESIGN CONSIDERATIONS AND SHALL BE SPECIFIED BY ENGINEER OF RECORD.
2. CONTRACTOR TO PROVIDE EQUIPMENT WITH SUFFICIENT LIFTING AND REACH CAPACITY TO LIFT AND SET THE MFS STRUCTURE (LIFTING CLUTCHES PROVIDED).
3. CONTRACTOR TO INSTALL JOINT SEALANT BETWEEN ALL SECTIONS AND ASSEMBLE STRUCTURE.
4. CONTRACTOR TO PROVIDE, INSTALL, AND GROUT PIPES.
5. CONTRACTOR TO TAKE APPROPRIATE MEASURES TO PROTECT CARTRIDGES FROM CONSTRUCTION-RELATED EROSION RUNOFF.

STRUCTURE WEIGHT
APPROXIMATE HEAVIEST PICK = T.B.D. LBS.

CONTECH
CONTRACT
DRAWING

60"Ø MANHOLE
MFS FILTER - 441748-02
BERKELEY TRANSFER STATION
BERKELEY, CA
SITE DESIGNATION: PH 11

CONTECH Construction Products Inc.
Stormwater Management
MFS
MEDIA FILTRATION SYSTEM

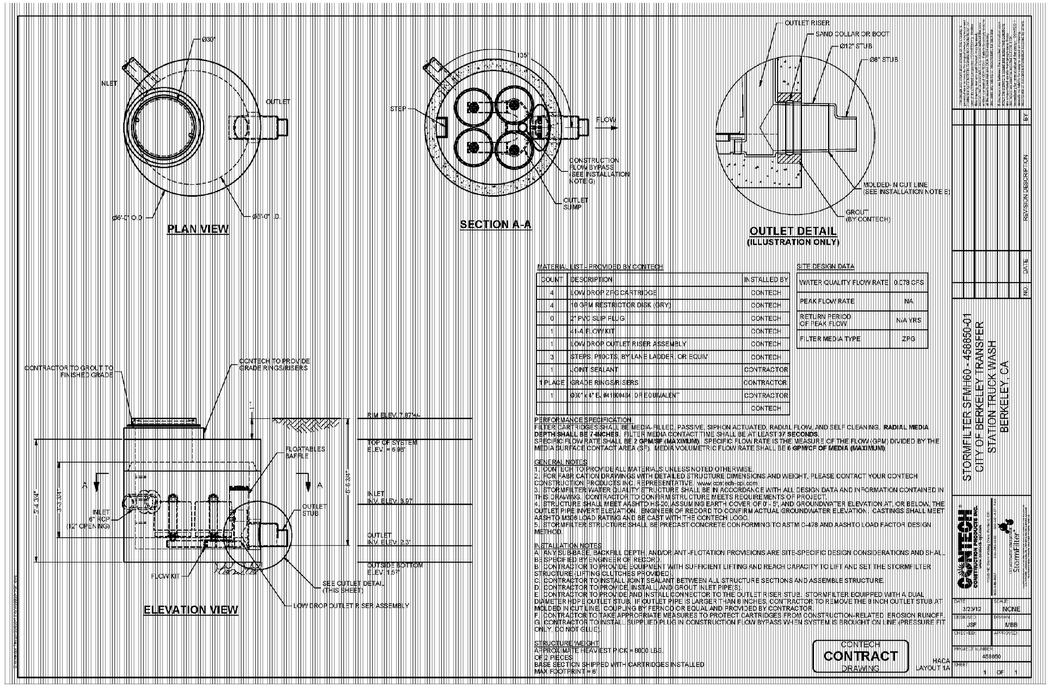
DATE: 6/22/11
DESIGNED: JSF
DRAWN: N/A
CHECKED:
APPROVED:
PROJECT NUMBER: 441748
SHEET: 1 OF 1

2 STORM FILTER AT WASH RACK

SCALE: NOT TO SCALE

NOTE: THIS DRAWING IS FOR INFORMATIONAL PURPOSES ONLY

DP-3A



MATERIAL LIST - PROVIDED BY CONTECH

COUNT	DESCRIPTION	INSTALLED BY
4	LOW DROP ZPG CARTRIDGE	CONTECH
4	18" SIM RESTRICTOR DISK (GRY)	CONTECH
6	2" PVC SLIP PLUG	CONTECH
1	4'-6" FLOW KIT	CONTECH
1	LOW DROP OUTLET RISER ASSEMBLY	CONTECH
3	STEPS: PROTI, BYLAN E LADDER, OR EQUIV.	CONTECH
1	JOINT SEALANT	CONTRACTOR
1 PLACE	GRADE RINGS/RISERS	CONTRACTOR
1	6'-6" x 4'-6" ELLIPTICAL OR EQUIVALENT	CONTRACTOR
		CONTECH

SITE DESIGN DATA

WATER QUALITY FLOW RATE	0.675 CFS
PEAK FLOW RATE	NA
RETURN PERIOD OF PEAK FLOW	N/A YRS
FILTER MEDIA TYPE	ZPG

PERFORMANCE SPECIFICATION
FILTER CARTRIDGES SHALL BE MEDIA-FILLED, PASSIVE, SIPHON ACTUATED, RADIAL FLOW, AND SELF CLEANING. **RADIAL MEDIA DEPTH SHALL BE 7-INCHES**. FILTER MEDIA CONTACT TIME SHALL BE AT LEAST **37 SECONDS**.
SPECIFIC FLOW RATE SHALL BE **2 GPM/SF (MAXIMUM)**. SPECIFIC FLOW RATE IS THE MEASURE OF THE FLOW (GPM) DIVIDED BY THE MEDIA SURFACE CONTACT AREA (SF). MEDIA VOLUMETRIC FLOW RATE SHALL BE **6 GPM/CF OF MEDIA (MAXIMUM)**.

GENERAL NOTES
1. CONTECH TO PROVIDE ALL MATERIALS UNLESS NOTED OTHERWISE.
2. DIMENSIONS MARKED WITH ( ) ARE REFERENCE DIMENSIONS. ACTUAL DIMENSIONS MAY VARY.
3. FOR FABRICATION DRAWINGS WITH DETAILED STRUCTURE DIMENSIONS AND WEIGHTS, PLEASE CONTACT YOUR CONTECH REPRESENTATIVE. www.contech-cpi.com
4. STORMFILTER WATER QUALITY STRUCTURE SHALL BE IN ACCORDANCE WITH ALL DESIGN DATA AND INFORMATION CONTAINED IN THIS DRAWING. CONTRACTOR TO CONFIRM STRUCTURE MEETS REQUIREMENTS OF PROJECT.
5. STRUCTURE SHALL MEET AASHTO H-20 (ASSUMING EARTH COVER OF 0' - 5', AND GROUNDWATER ELEVATION AT, OR BELOW, THE OUTLET PIPE INVERT ELEVATION. ENGINEER OF RECORD TO CONFIRM ACTUAL GROUNDWATER ELEVATION. CASTINGS SHALL MEET AASHTO M306 LOAD RATING AND BE CAST WITH THE CONTECH LOGO.
6. STORMFILTER STRUCTURE SHALL BE PRECAST CONCRETE CONFORMING TO ASTM C-478 AND AASHTO LOAD FACTOR DESIGN METHOD.

INSTALLATION NOTES
A. ANY SUB-BASE, BACKFILL DEPTH, AND/OR ANTI-FLOTATION PROVISIONS ARE SITE-SPECIFIC DESIGN CONSIDERATIONS AND SHALL BE SPECIFIED BY ENGINEER OF RECORD.
B. CONTRACTOR TO PROVIDE EQUIPMENT WITH SUFFICIENT LIFTING AND REACH CAPACITY TO LIFT AND SET THE STORMFILTER STRUCTURE (LIFTING CLUTCHES PROVIDED).
C. CONTRACTOR TO INSTALL JOINT SEALANT BETWEEN ALL STRUCTURE SECTIONS AND ASSEMBLE STRUCTURE.
D. CONTRACTOR TO PROVIDE, INSTALL, AND GROUT INLET PIPE(S).
E. CONTRACTOR TO PROVIDE AND INSTALL CONNECTOR TO THE OUTLET RISER STUB. STORMFILTER EQUIPPED WITH A DUAL DIAMETER HDPE OUTLET STUB. IF OUTLET PIPE IS LARGER THAN 8 INCHES, CONTRACTOR TO REMOVE THE 8 INCH OUTLET STUB AT MOLDED IN CUT LINE. COUPLING BY FERNCO OR EQUAL AND PROVIDED BY CONTRACTOR.
F. CONTRACTOR TO TAKE APPROPRIATE MEASURES TO PROTECT CARTRIDGES FROM CONSTRUCTION-RELATED EROSION RUNOFF.
G. CONTRACTOR TO INSTALL SUPPLIED PLUG IN CONSTRUCTION FLOW BYPASS WHEN SYSTEM IS BROUGHT ON LINE (PRESSURE FIT ONLY, DO NOT GLUE).

STRUCTURE WEIGHT
APPROXIMATE HEAVIEST PICK = 9600 LBS.
OF 2 PIECES
BASE SECTION SHIPPED WITH CARTRIDGES INSTALLED
MAX FOOTPRINT = 8'

CONTECH
CONTRACT
DRAWING

STORMFILTER SFM1H20 - 456850-01
CITY OF BERKELEY TRANSFER STATION TRUCK WASH
BERKELEY, CA

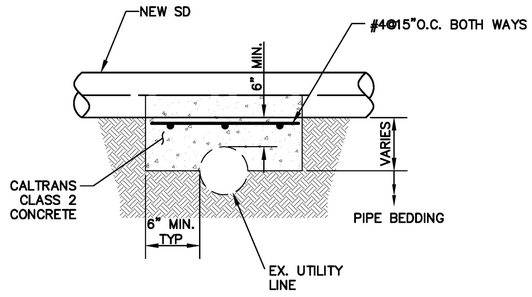
CONTECH Construction Products Inc.
Stormwater Management
StormFilter

DATE: 5/23/12
DESIGNED: JSF
DRAWN: KRB
CHECKED:
APPROVED:
PROJECT NUMBER: 456850
SHEET: 1 OF 1
HACA LAYOUT 1A

3 STORMFILTER IN FRONT OF VEHICLE MAINTENANCE BUILDING

SCALE: NOT TO SCALE

NOTE: THIS DRAWING IS FOR INFORMATIONAL PURPOSES ONLY



NOTE:

1. INSTALL ½" STYROFOAM WHERE CONCRETE COMES IN CONTACT WITH PIPE.

4 CONCRETE SADDLE

SCALE: NOT TO SCALE

REVISION

NO.	DATE	BY	REVISION DESCRIPTION	DATE	BY

THESE DRAWINGS AND SPECIFICATIONS ARE THE PROPERTY AND COPYRIGHT OF THE ENGINEER AND SHALL NOT BE USED ON ANY PROJECT WITHOUT THE WRITTEN AUTHORIZATION OF THE ENGINEER. WRITTEN DIMENSIONS SHALL TAKE PRECEDENCE OVER SCALED DIMENSIONS. CONTRACTOR SHALL VERIFY AND BE RESPONSIBLE FOR ALL DIMENSIONS AND CONDITIONS ON THE JOB SITE. ANY DISCREPANCY SHALL BE BROUGHT TO THE ATTENTION OF THE ENGINEER PRIOR TO THE START OF ANY WORK.

Harris & Associates
1401 Willow Pass Road, Suite 500, Concord CA, 94520
harris-assoc.com (925) 827-4900

CITY OF BERKELEY
SECOND STREET TRANSFER STATION
STORMWATER MANAGEMENT IMPROVEMENTS - PHASE II

DRAWN BY: MCL
DESIGNED BY: KI
CHECKED BY: VP

DATE ISSUED: 05/01/13
JOB NO.: 091-0684.02

REGISTERED PROFESSIONAL ENGINEER
KOUROSH IRANPOUR
No. C 53336
Exp. 06-30-13
CIVIL
STATE OF CALIFORNIA

CONSTRUCTION DETAILS

DWG. CD-4

SHEET 16 OF 16

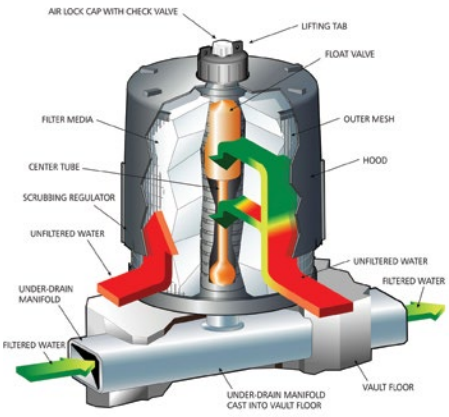
H:\Berkeley\Transfer Station - Phase II\ 16-CD04 05/09/13 14:50 lgabriel XREFS: X-TB

Appendix D

Contech Filter Information and Specifications

The Contech® stormwater treatment vaults are located at DP-1A, DP-2A and DP-3A. The fourth vault is located downstream from the truck wash rack that pretreats wash water in the sanitary line to East Bay Municipal Utility District (EBMUD). DP-1A, DP-2A and the downstream wash rack filter are composed of ZPG™ filtering media. According to Contech® ZPG™ multipurpose media, a proprietary blend of organic and inorganic media is effective in the removal of solids, metals and organic chemicals. DP-3A is composed of Perlite filter media.
The vaults are composed of a series of circular filters. The mode of operation of the filters are described below.

- During a storm, runoff passes through the filtration media and starts filling the cartridge center tube. The air inside the hood is purged through a one-way check valve as the water rises.
- When water reaches the top of the float, buoyant forces pull the float free and allow filtered water to exit the cartridge. A siphon is established within each cartridge that draws water uniformly across the full height of the media bed ensuring even distribution of pollutants and prolonged media longevity.
- After the storm, the water level in the structure starts falling. A hanging water column remains under the cartridge hood until the water level reaches the scrubbing regulators at the bottom of the hood.
- Air then rushes through the regulators, breaking the siphon and creating air bubbles that agitate the surface of the filter media, causing accumulated sediment to settle on the treatment bay floor. This unique surface-cleaning mechanism prevents surface blinding and further extends cartridge life.



Appendix D

The website link below has a video explaining operation of the filter.
https://www.conteches.com/stormwater-management/treatment/stormwater-management-stormfilter

The Contech filter information for the BTS are presented in table below. Manufacturer informational engineering drawings are provided in attachment 4. A record of inspection and maintenance of the Contech filters is provided in attachment 5.

Discharge Location	Cartridge Size (inches)	Number of Cartridges	Media Type
DP-1A	18	13	ZPG
DP-2A	18	13	ZPG
DP-3A	12	3	Perlite
EBMUD Filter from Truck Wash Rack	18	4	ZPG

APPENDIX E

Monthly Visual Observation and BMP Inspection Forms

MONTHLY VISUAL OBSERVATION LOG	
Date of Inspection:	**Inspection Time:**
Site Information	
Facility Name: Berkeley Transfer Station and Recycling Center	
Facility Address: 1201 Second Street and 669 Gilman Street Berkeley, California	
Inspector Information	
Inspector Name:	**Inspector Title:**
Signature:	**Date:**
Reviewer Name:	**Inspector Title:**
Signature:	**Date:**
Weather	
Antecedent Conditions:	**Current Weather:**
Part I. NSWD Observations	
Were any authorized non-stormwater discharges observed?	**No**
Were any **unauthorized** non-stormwater discharges observed?	**No**

Part II. BMP Observations.		
Industrial Activity Areas (Inspect all BMPs Implemented)	**Action Required (yes/no)**	**Notes:**
DP-1A – Catchment Area A		
Container/Bin Storage Area		
Container Maintenance Building		
Cargo Cab and Truck Parking		
Catch Basins CB-1 and Trench Drain TD-1		
Stormwater Inspection Area Summary:		**Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs** **General Maintenance Recommendations:** • Cover open bins/containers • Use drip pins when possible; clean up oil spills and leaks immediately • Sweep area regularly • Maintain proper wattle placement • Clean sediment and debris • Ensure tires are kept under roof cover **Actions Required:**
Stormwater Collection Tank Discharge to EBMUD Sanitary Sewer - Catchment Area B		
General/Electronic Waste Storage		
Trench Drains TD-3 and TD-4		
Construction Debris Receiving Area		

Garbage Truck Parking		
Transfer Station/Tipping Floor		
Stormwater Holding Tank		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Store general/electronic waste in sheltered or closed containers • Maintain dust control and sweeping measures • Clean sediment and debris
DP-2A – Catchment Area C		
Equipment Maintenance Building		
Truck Parking		
Diesel Fueling Station		
Catch Basins CB-2 and Trench Drain TD-2		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Use drip pans when possible; clean up oil spills and leaks immediately • Sweep area regularly • Maintain proper wattle placement • Clean sediment and debris **Actions Required:**
DP-3A – Catchment Area C		

Transfer Station/Tipping Floor		
General Waste Storage		
Hazardous Waste Storage Area **To comply with:** **40 CFR Part 262 - Standards Applicable To Generators Of Hazardous Waste** **40 CFR § 262.11 - Hazardous waste determination and recordkeeping** **40 CFR § 262.15 - Satellite accumulation area regulations for small and large quantity generators**		
Used Oil Depot		
Used Tire Storage		
Truck Wash Rack		
Drum Storage Area Adjacent to Catch Basin CB-3		
Catch Basin CB-3 (large catch basin)		
Catch Basin CB-3A (small catch basin)		
Mattress Area		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Maintain dust, sediment, and trash control measures • Keep drums closed and in proper storage • Maintain proper wattle placement **Actions Required:** •

DP-4 – Catchment Area D		
Truck Parking		
Container/Bin Storage Areas		
Discharge point DP-4		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Clean leaks/spills immediately • Maintain sweeping and debris control **Actions Required:**
DP-5A – Catchment Area F		
Ecology center building		
Ecology Truck Parking and Maintenance Area		
Comingled Container Storage		
Stormwater Storage Tank		
Stormwater Sump/Pump to Storage Tank		
Glass Sorting Area		

Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Maintain proper wattle and filter placement • Ensure regular sweeping and cleaning for spills, sediment, debris, and trash • Pickup trash **Actions Required:**
DP-6 – Catchment Area F		
Customer Recyclable Area		
Berm Behind Canopy Structure (recyclable buyback area)		
Universal Waste (Battery, fluorescent tube and fats, oil and grease, etc. recycling area)		
Cardboard collection Bins		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Maintain proper wattle placement • Ensure regular sweeping and cleaning • Keep materials in proper containment and under canopies • Use drip pans **Actions Required:**
DP-7 – Catchment Area F		
Customer Parking		

DP-8A – Catchment Areas F		
Recyclable Buyback Area		
Storm Drain Catch Basin		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Maintain proper wattle placement • Ensure regular sweeping and cleaning • Keep materials in proper containment and under canopies **Actions Required:**
DP-9 – Catchment Area E		
Paper Recycling Warehouse		
Glass Recycling Warehouse		
Wattle filters		
Alley way behind Warehouses		
Stormwater Inspection Area Summary:		**<u>Per Industrial General Permit Section X. Stormwater Pollution Prevention Plan, H Best Management Practices (BMPs), 1. Minimum BMPs</u>** **General Maintenance Recommendations:** • Maintain proper wattle placement and windscreens • Ensure regular sweeping and cleaning **Actions Required:**

Part III. Summary of Deficiencies and Actions Required		
Deficiency (list corresponding discharge location, include photo)	**Date Observed**	**Corrective Action Recommended**

Part IV. Additional or General Observations/Notes Requiring Corrective Action. Identify additional corrective actions not included with BMP Deficiencies (Part III) above. Identify BMPs that need more frequent inspection. Note if SWPPP change is required.		
Observation (include photo)	**Date Observed**	**Corrective Action Recommended**

WEEKLY COMMUNITY CONSERVATION CENTER STORMWATER BMP INSPECTION CHECKLIST

GENERAL INFORMATION

Inspection Week (Start – End)
Inspector Name / Title
Weather Conditions

DAILY BMP INSPECTION & MAINTENANCE TASKS

Verify tasks were completed each operating day during the inspection week.

BMP / Activity	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Notes
Mechanical sweeping completed by CCC	☐	☐	☐	☐	☐	☐	☐	
Vacuum/mechanical sweeping completed by City	☐	☐	☐	☐	☐	☐	☐	
Trash pickup and litter control	☐	☐	☐	☐	☐	☐	☐	
Leak and spill inspection and cleanup	☐	☐	☐	☐	☐	☐	☐	
Post-Rain Inspection of Stormwater holding tanks	☐	☐	☐	☐	☐	☐	☐	
Pre-Rain Inspection of wind screen along fence line	☐	☐	☐	☐	☐	☐	☐	
Daily maintenance/ emptying of drip pans to the sanitary sewer	☐	☐	☐	☐	☐	☐	☐	
Daily inspection of operational equipment in the commingled container storage area	☐	☐	☐	☐	☐	☐	☐	
Daily inspection of operational equipment in the glass sorting area	☐	☐	☐	☐	☐	☐	☐	

WEEKLY COMMUNITY CONSERVATION CENTER STORMWATER BMP INSPECTION CHECKLIST

BMP / Activity	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Notes
Daily inspection/ maintenance of storm drain protection/ perimeter control in customer recyclable area	☐	☐	☐	☐	☐	☐	☐	
Wattles and oil absorbent socks inspected/maintained along 2nd Street fence line and around DP-5A	☐	☐	☐	☐	☐	☐	☐	
End of shift hand sweeping completed	☐	☐	☐	☐	☐	☐	☐	

WEEKLY BMP INSPECTION & MAINTENANCE TASKS

BMP / Activity	Date Completed	Notes
Stormwater BMP visual inspection		
Inspection of stormwater holding tanks, valley gutter drains, and associated equipment (testing float switches, sump pumps and totalizer)		
Cleaning of comingled container storage area and glass sorting area sumps		

MONTHLY INSPECTION & MAINTENANCE TASKS

BMP / Activity	Last Date Completed	Notes
Inspection and repair of bins		

WEEKLY COMMUNITY CONSERVATION CENTER STORMWATER BMP INSPECTION CHECKLIST

ANNUAL BMP TASKS

BMP / Activity	Frequency	Last Completed Date	Notes
Pavement cleaning / pressure washing	Annual		

WEEKLY COMMUNITY CONSERVATION CENTER STORMWATER BMP INSPECTION CHECKLIST

CORRECTIVE ACTIONS & FOLLOW-UP

Issue / Deficiency	Corrective Action Taken	Date Completed

INSPECTOR CERTIFICATION

I certify that this inspection was conducted in accordance with the facility SWPPP and that noted deficiencies have been addressed or scheduled for correction.

Inspector Signature	Date

WEEKLY ECOLOGY CENTER STORMWATER BMP INSPECTION CHECKLIST

GENERAL INFORMATION

Inspection Week (Start – End)
Inspector Name / Title
Weather Conditions

DAILY BMP INSPECTION & MAINTENANCE TASKS

Verify tasks were completed each operating day during the inspection week.

BMP / Activity	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Notes
Mechanical sweeping completed	☐	☐	☐	☐	☐	☐	☐	
Trash pickup and litter control	☐	☐	☐	☐	☐	☐	☐	
Leak and spill inspection	☐	☐	☐	☐	☐	☐	☐	
Drip containers and absorbent sheets maintained	☐	☐	☐	☐	☐	☐	☐	
Wattles and oil absorbent socks inspected/maintained along 2nd Street fence line and around DP-5A	☐	☐	☐	☐	☐	☐	☐	
End of shift hand sweeping completed	☐	☐	☐	☐	☐	☐	☐	

WEEKLY BMP INSPECTION & MAINTENANCE TASKS

BMP / Activity	Date Completed	Notes
Stormwater BMP visual inspection		

WEEKLY ECOLOGY CENTER STORMWATER BMP INSPECTION CHECKLIST

MONTHLY INSPECTION & MAINTENANCE TASKS

BMP / Activity	Last Date Completed	Notes
Confirm vendor utilizes BMPs to capture wash water during bi-monthly truck washing		

ANNUAL / SEMI-ANNUAL BMP TASKS

BMP / Activity	Frequency	Last Completed Date	Notes
Pavement cleaning / pressure washing	Annual		

CORRECTIVE ACTIONS & FOLLOW-UP

Issue / Deficiency	Corrective Action Taken	Date Completed

INSPECTOR CERTIFICATION

I certify that this inspection was conducted in accordance with the facility SWPPP and that noted deficiencies have been addressed or scheduled for correction.

Inspector Signature	Date

WEEKLY TRANSFER STATION STORMWATER BMP INSPECTION CHECKLIST

GENERAL INFORMATION

Inspection Week (Start – End)
Inspector Name / Title
Weather Conditions

DAILY BMP INSPECTION & MAINTENANCE TASKS

Verify tasks were completed each operating day during the inspection week.

BMP / Activity	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Notes
Mechanical sweeping completed AM	☐	☐	☐	☐	☐	☐	☐	
Mechanical sweeping completed PM	☐	☐	☐	☐	☐	☐	☐	
Trash pickup and litter control	☐	☐	☐	☐	☐	☐	☐	
Confirm Truck Wash drain inlet is sealed prior to rain events	☐	☐	☐	☐	☐	☐	☐	
Leak and spill inspection and cleanup	☐	☐	☐	☐	☐	☐	☐	
Dust control misters inspected	☐	☐	☐	☐	☐	☐	☐	

WEEKLY TRANSFER STATION STORMWATER BMP INSPECTION CHECKLIST

WEEKLY BMP INSPECTION & MAINTENANCE TASKS

BMP / Activity	Date Completed	Notes
Stormwater BMP visual inspection		
Trench drains TD-1, TD-2, TD-2A, TD-3, and TD-4 inspected and maintained		
Sluice gate manhole in front of construction debris area from trench drain TD-4 inspected and cleaned. Coordinate with public works for cleaning.		
Weir manhole to stormwater holding tank inspected and cleaned. Coordinate with public works for cleaning.		
BioChar, sediment, and oil absorbent wattles maintained at trench drain TD-1, and catch basins CB-1, CB-2, CB-3, CB-3A, and CB-4		
Catch basins inspected and maintained CB-1, CB-2, CB-3, CB-3A and CB-4.Coordinate with public works for cleaning.		

WEEKLY TRANSFER STATION STORMWATER BMP INSPECTION CHECKLIST

MONTHLY INSPECTION & MAINTENANCE TASKS

BMP / Activity	Last Date Completed	Notes
Hand sweeping in areas inaccessible areas by mechanical/vacuum sweeper		
Inspection and repair of bins		
Contech filter and catch basin (CB-1, CB-2, CB-3, CB-3A, and CB-4) inspection/debris removal. Coordinate with public works for cleaning as needed.		

ANNUAL / SEMI-ANNUAL BMP TASKS

BMP / Activity	Frequency	Last Completed Date	Notes
Pavement cleaning / pressure washing	Annual		
Contech filter maintenance by vendor	Twice Annually		
Exercising of sluice gates in manhole in front of construction debris area from trench drain TD-4	Annual		

WEEKLY TRANSFER STATION STORMWATER BMP INSPECTION CHECKLIST

CORRECTIVE ACTIONS & FOLLOW-UP

Issue / Deficiency	Corrective Action Taken	Date Completed

INSPECTOR CERTIFICATION

I certify that this inspection was conducted in accordance with the facility SWPPP and that noted deficiencies have been addressed or scheduled for correction.

Inspector Signature	Date

APPENDIX F

Sampling Event Visual Observation Form

SAMPLING EVENTS VISUAL OBSERVATION LOG

Date and Time of Inspection:	Report Date:
Facility Name:	

Weather

Antecedent Conditions (last 48 hours):	Weather:
Precipitation Total:	Predicted % chance of rain:

Estimate storm beginning: _____ (date and time)	Estimate storm duration:_____ (hours)	Estimate time since last storm: _____ (days or hours)	Rain gauge reading: _____ (inches)

Sampling Event Observations

Observations: If yes identify location and observe drainage area to identify probable cause		
Odors	Yes □	No □
Floating material	Yes □	No □
Suspended Material	Yes □	No □
Sheen	Yes □	No □
Discolorations	Yes □	No □
Turbidity	Yes □	No □

NSWD Observations

Were any authorized non-stormwater discharges observed?	Yes □	No □
Were any **unauthorized** non-stormwater discharges observed?	Yes □	No □
If yes to either, identify source		

Drainage Area Observations

Drainage Area	Deficiencies Noted

Exception Documentation (explanation required if inspection could not be conducted).

Inspector Information

Inspector Name:	Inspector Title:
Signature:	Date:

APPENDIX G

Example Chain of Custody and Field Log Form

CHAIN-OF-CUSTODY **DATE:** **Lab ID:**

DESTINATION LAB: ATTN: **ADDRESS:** **Office Phone:** **Cell Phone:**							REQUESTED ANALYSIS				Notes:
SAMPLED BY:											
Contact:											
Facility Name											
Client Sample ID	**Sample Date**	**Sample Time**	**Sample Matrix**	**Container #**	**Container Type**	**Container Pres.**					

SENDER COMMENTS:	**RELINQUISHED BY**		
	Signature:		
	Print:		
	Company:		
	Date:	TIME:	

LABORATORY COMMENTS:	**RECEIVED BY**		
	Signature:		
	Print:		
	Company:		
	Date:	TIME:	

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